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교재명 : Introduction to Mathematical Statistics
저자 : Hogg, McKean, Craig
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2019년 1월 1일
부산대학교 통계학과 교수 김충락

# 1 Probability and Distributions

## 1.1 Introduction

- *Statistical(random) experiment*: the outcome cannot be predicted with certainty prior to the performance of the experiment.
- *Sample space*: collection of every possible outcome from the random experiment, and denoted by $\mathscr{C}$.
- *Event*: subset of sample, and denoted by $A, B, C$.

**Example 1.1.1.** Consider tossing a coin, then $\mathscr{C} = \{H, T\}$.

**Example 1.1.2.** Consider tossing two die (one red, the other white), then $\mathscr{C} = \{(1,1), \cdots, (1,6), (2,1), \cdots, (6,6)\}$.

**Example 1.1.3.** Let $C$ denote an event of sum seven when tossing two die, then $\mathscr{C} = \{(1,6), (2,5), \cdots, (6,1)\}$.

**Remark 1.1.1.** Two types of probability

(i) Relative frequancy approach.

(ii) Personal or subjective approach.

## 1.2 Set Theory

**Definition 1.2.1.** If each element of set $C_1$ is also an element of set $C_2$, then $C_1$ is called *subset* of $C_2$, and denoted by $C_1 \subset C_2$.

**Definition 1.2.2.** If a set $C$ has no elements, $C$ is called the *null(empty)* set, and denoted by $C = \phi$.

**Definition 1.2.3.** The set of all elements that belong to at least one of $C_1$ and $C_2$ is called the *union* of $C_1$ and $C_2$, and denoted by $C_1 \cup C_2$ and it can be generalized to any number of sets. For example, $C_1 \cup C_2 \cup \cdots \cup C_n = \cup_{k=1}^{\infty} C_k$.

**Example 1.2.1.** Let $C_k = \left\{x : \frac{1}{k+1} \leq x \leq 1\right\}$, then $\bigcup_{k=1}^{\infty} C_k = \{x : 0 < x \leq 1\}$.

**Definition 1.2.4.** The set of all elements that belong to each of $C_1$ and $C_2$ is called the *intersection* of $C_1$ and $C_2$. and denoted by $C_1 \cap C_2$, and it can be generalized to any number of sets $C_1 \cap C_2 \cap \cdots := \cap_{k=1}^{\infty} C_k$

**Example 1.2.2.** Let $C_k = \left\{x : 0 < x < \frac{1}{k}\right\}$, then $\bigcap_{k=1}^{\infty} C_k = \phi$.

**Definition 1.2.5.** Let $C$ be a subset of $\mathscr{C}$, then the set that consists of all elements of $\mathscr{C}$ that are not elements of $C$ is called *complement* of $C$, and denoted by $C^c$ or $\overline{C}$.

- A function is called *point* or *set function* if its domain is point or set, respectively.

**Example 1.2.3.** point function: $f(x) = 2x$, $f(1) = 2$
set function: $Q(A) =$ # of positive integers in $A$
$A = \{x : -\infty < x < 6\} \Rightarrow Q(A) = 5$

- The symbol
$$\int_C f(x)dx$$
means the ordinary Riemann integral of $f(x)$ over a one-dimensional set $C$, the symbol
$$\int\int_C g(x,y)dxdy$$

means the Riemann integral of $g(x,y)$ over a two-dimensional set $C$. Similarly, one or two-dimensional sum is

$$\sum_C f(x), \sum\sum_C g(x,y).$$

**Example 1.2.4.** Let $Q(C) = \int_C \cdots \int dx_1 dx_2 \cdots dx_n$. If $C = \{(x_1, x_2, \cdots, x_n) : 0 \le x_1 \le x_2 \le \cdots \le x_n \le 1\}$, then

$$Q(C) = \int_0^1 \int_0^{x_n} \cdots \int_0^{x_3} \int_0^{x_2} dx_1 dx_2 \cdots dx_{n-1} dx_n = \frac{1}{n!}.$$

## 1.3 The Probability Set Function

**Definition 1.3.1.** *($\sigma$-field)* Let $\mathscr{B}$ be a collection of subsets of $\mathscr{C}$. We say $\mathscr{B}$ is a *$\sigma$-field* if

(i) $\phi \in \mathscr{B}$.

(ii) $C \in \mathscr{B} \Rightarrow C^c \in \mathscr{B}$ (closed under complement).

(iii) $C_1, C_2, \cdots \in \mathscr{B} \Rightarrow \bigcup_{i=1}^{\infty} C_i \in (B)$ (closed under countable union).

- Example of $\sigma$-field

1. $\mathscr{B} = \{\phi, C, C^c, \mathscr{C}\}$.
2. $\mathscr{B}$ is the power set of $\mathscr{C}$, i.e. the collection of all subsets of $\mathscr{C}$.
3. $\mathscr{B} = \bigcap_{i=1}^{\infty} \{\varepsilon_i : \mathscr{D} \subset \varepsilon_i, \varepsilon_i \text{ is a } \sigma\text{-field } \}$. This is the smallest $\sigma$-field which containing $\mathscr{D}$, and it is called the $\sigma$-field generated by $\mathscr{D}$.
4. Let $\mathscr{I}$ be the set of all open intevals in $\mathbb{R}$ (set of real numbers), then the $\sigma$-field generated by $\mathscr{I}$ is called the Borel $\sigma$-field.

**Definition 1.3.2.** (probability) Let $\mathscr{C}$ be a sample space, $\mathscr{B}$ be a $\sigma$-field on $\mathscr{C}$. Let $P$ be a real-valued function defined on $\mathscr{B}$. Then $P$ is called a *probability set function* if it satisfies the following three conditions

(i) $P(C) \geq 0$, $^{\forall} C \in \mathscr{B}$ (*non-negativity*).

(ii) $P(\mathscr{C}) = 1$ (*normality*).

(iii) $C_1, C_2, \cdots \in \mathscr{B}$ s.t. $C_m \cup C_n = \phi$, $^{\forall} m \neq n$,
then $P\left(\bigcap_{i=1}^{\infty} C_n\right) = \sum_{i=1}^{\infty} P(C_i)$ (*countable additivity*).

**Theorem 1.3.1.** $P(C) = 1 - P(C^c)$, $^{\forall} C \in \mathscr{B}$.

(pf) Since $C \cup C^c = \mathscr{C}$ and $C \cap C^c = \phi$,

$$\begin{aligned} 1 &= P(\mathscr{C}) \\ &= P(C) + P(C^c). \end{aligned}$$

**Theorem 1.3.2.** $P(\phi) = 0$.

(pf) By taking $C = \phi$, we have $C^c = \mathscr{C}$, then by Thm.1.3.1,

$$P(\phi) = 1 - P(\mathscr{C}) = 0.$$

**Theorem 1.3.3.** $C_1 \subset C_2 \Rightarrow P(C_1) \leq P(C_2)$.

(pf) $C_2 = C_1 \cup (C_1^c \cap C_2) \Rightarrow P(C_2) = P(C_1) + P(C_1^c \cap C_2) \geq P(C_1)$.

**Theorem 1.3.4.** $0 \leq P(C) \leq 1$, $^{\forall} C \in \mathscr{B}$.

(pf) $\phi \subset C \subset \mathscr{C} \Rightarrow P(\phi) \leq P(C) \leq P(\mathscr{C}) \Rightarrow 0 \leq P(C) \leq 1$.

**Theorem 1.3.5.** $P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$.

(pf) $C_1 \cup C_2 = C_1 \cup (C_1^c \cap C_2) \Rightarrow P(C_1 \cup C_2) = P(C_1) + P(C_2 \cap C_1^c)$

$C_2 = (C_1 \cap C_2) \cup (C_1^c \cap C_2) \Rightarrow P(C_2) = P(C_1 \cap C_2) + P(C_2 \cap C_1^c)$

Hence, we have

$P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$.

**Remark 1.3.1.** (inclusion-exclusion formula)
For 3 sets $C_1, C_2, C_3$, it is not difficult to show that

$$P(C_1 \cup C_2 \cup C_3) = p_1 - p_2 + p_3$$

where $p_1 = P(C_1) + P(C_2) + P(C_3)$,
$p_2 = P(C_1 \cap C_2) + P(C_1 \cap C_3) + P(C_2 \cap C_3)$,
$p_3 = P(C_1 \cap C_2 \cap C_3)$.
In general,

$$P(C_1 \cup C_2 \cup \cdots \cup C_k) = p_1 - p_2 + p_3 - \cdots + (-1)^{k-1} p_k$$

where $p_i$ is sum of probability of all possible intersections of $i$ sets.

- $C_1, C_2, \cdots$ are called mutually exclusive if $C_i \cap C_j = \phi$, $^{\forall} i \neq j$
- Mutually exclusive sets $C_1, C_2, \cdots$ are called exhaustive if $\bigcup_{i=1}^{\infty} C_i = \mathscr{C}$
- Notation:

$$\lim_{n\to\infty} C_n = \begin{cases} \bigcup_{n=1}^{\infty} C_n & \text{for increasing sequence} \\ \bigcap_{n=1}^{\infty} C_n & \text{for decreasing sequence} \end{cases}$$

**Theorem 1.3.6.** Let $\{C_n\}$ be a increasing sequence of events. Then

$$\lim_{n\to\infty} P(C_n) = P(\lim_{n\to\infty} C_n) = P\left(\bigcup_{n=1}^{\infty} C_n\right).$$

Let $\{C_n\}$ be a decreasing sequence of events. Then

$$\lim_{n\to\infty} P(C_n) = P(\lim_{n\to\infty} C_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right).$$

(pf) Assume $\{C_n\}$ is increasing sequence, and let $R_1 = C_1$, $R_n = C_n \cap C_{n-1}^c$

$$\begin{aligned}
P(\lim_{n\to\infty} C_n) &= P\left(\bigcup_{n=1}^{\infty} C_n\right) \\
&= P\left(\bigcup_{n=1}^{\infty} R_n\right) \\
&= \sum_{n=1}^{\infty} P(R_n) \\
&= \lim_{n\to\infty} \sum_{j=1}^{n} P(R_j) \\
&= \lim_{n\to\infty} \left\{ P(R_1) + \sum_{j=2}^{n} P(R_j) \right\} \\
&= \lim_{n\to\infty} \left[ P(C_1) + \sum_{j=2}^{n} \{P(C_j) - P(C_{j-1})\} \right] \\
&= \lim_{n\to\infty} [P(C_1) + \{P(C_2) - P(C_1)\} + \{P(C_3) - P(C_2)\} + \cdots + \{P(C_n) - P(C_{n-1})\} \\
&= \lim_{n\to\infty} P(C_n).
\end{aligned}$$

**Theorem 1.3.7.** (*Boole's Inequality*) Let $\{C_n\}$ be an arbitrary sequence of events. Then

$$P\left(\bigcup_{n=1}^{\infty} C_n\right) \leq \sum_{n=1}^{\infty} P(C_n).$$

(pf) Let $D_n = \bigcup_{i=1}^{n} C_i$, then $\{D_n\}$ is increasing sequence of sets.
Since $D_j = D_{j-1} \cup C_j$

$$\begin{aligned}
P(D_j) &= P(D_{j-1}) + P(C_j) - P(D_{j-1} \cap C_j) \\
&\leq P(D_{j-1}) + P(C_j)
\end{aligned}$$

i.e. $P(D_j) - P(D_{j-1}) \leq P(C_j)$.

Now,

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} C_n\right) &= P\left(\bigcup_{n=1}^{\infty} D_n\right) \\ &= \lim_{n\to\infty} P(D_n) \\ &= \lim_{n\to\infty}\left[P(D_1) + \sum_{j=2}^{n}\left\{P(D_j) - P(D_{j-1})\right\}\right] \\ &\leq \lim_{n\to\infty}\left\{P(D_1) + \sum_{j=2}^{n} P(D_j)\right\} \\ &= \lim_{n\to\infty}\sum_{j=1}^{n} P(C_j) \\ &= \sum_{n=1}^{\infty} P(C_n). \end{aligned}$$

## 1.4 Conditional Probability and Independence

Let $C_1, C_2 \subset \mathscr{C}$, then the *conditional probability of* $C_2$ *given* $C_1$ is defined as

$$P = (C_2 \mid C_1) = \frac{P(C_2 \cap C_1)}{P(C_1)}, \text{ if } P(C_1) > 0$$

Note that the conditional probability satisfies 3 conditions of probability

(i) $P(C_2 \mid C_1) \geq 0$ (non-negativity).

(ii) $P\left(\bigcup_{i=2}^{\infty} C_i \mid C_1\right) = \sum_{i=2}^{\infty} P(C_i \mid C_1)$ if $C_2, C_3, \cdots$ are mutually disjoint (countable additivity).

(iii) $P(C_1 \mid C_1) = 1$ (normality).

**Example 1.4.1.** Consider drawing cards successively from a deck, at random and without replacement. Find thd probability that the third spade

appears on the sixth draw. (♠ : *spade*, ♢ : *diamond*, ♣ : *clover*, ♡ : *heart*)

(sol) $C_1$ : two spades in the first five draws.
$C_2$ : a spade on the sixth draw.
We need to compute $P(C_1 \cap C_2)$, and use $P(C_1 \cap C_2) = P(C_2 \mid C_1)P(C_1)$

$$P(C_1) = \frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}} = 0.2743,\ P(C_2 \mid C_1) = 11/47 = 0.234 \Rightarrow\ P(C_1 \cap C_2) = 0.064$$

From the definition of conditional probability, we have $P(C_1 \cap C_2) = P(C_2 \mid C_1)P(C_1)$ which is called the multiplication rule. For 3 events,

$$P(C_2 \mid C_1 \cap C_2) = P(C_3 \cap C_1 \cap C_2)/P(C_1 \cap C_2)$$

$$\Rightarrow\ P(C_1 \cap C_2 \cap C_3) = P(C_3 \mid C_1 \cap C_2)P(C_1 \cap C_2) = P(C_3 \mid C_1 \cap C_2)P(C_2 \mid C_1)P(C_1).$$

In general,

$$P(C_1 \cap C_2 \cap C_3 \cap \cdots) = P(C_1)P(C_2 \mid C_1)P(C_3 \mid C_1 \cap C_2)P(C_4 \mid C_1 \cap C_2 \cap C_3)\cdots.$$

Bayes Theorem: Let $C_1, C_2, \cdots, C_k$ be mutually exclusive and exhaustive events, s.t. $P(C_i) > 0$, $i = 1, \cdots, k$. Then,

$$P(C_j \mid C) = \frac{P(C_j)P(C \mid C_j)}{\sum_{i=1}^{k} P(C_i)P(C \mid C_i)},\ j = 1, \cdots, k.$$

(pf) Since $C = (C \cap C_1) \cup (C \cap C_2) \cup \cdots \cup (C \cap C_k)$

$$\begin{aligned} \Rightarrow P(C) &= P(C \cap C_1) + \cdots + P(C \cap C_k) \\ &= P(C_1)P(C \mid C_1) + \cdots + P(C_k)P(C \mid C_k) \\ &= \sum_{i=1}^{k} P(C_i)P(C \mid C_i) : \text{law of total probability} \end{aligned}$$

Now,

$$P(C_j \mid C) = \frac{P(C_j \cap C)}{P(C)} = \frac{P(C_j)P(C \mid C_j)}{\sum_{i=1}^{k} P(C_i)P(C \mid C_i)}.$$

**Remark 1.4.1.** $P(C_j)$: *prior* probability, $P(C_j \mid C)$: *posterior* probability.

**Definition 1.4.1.** Two events $C_1$ and $C_2$ are independent if $P(C_1 \mid C_2) = P(C_1)$, i.e.

$$P(C_1 \mid C_2) = \frac{P(C_1 \cap C_2)}{P(C_2)} = P(C_1) \Rightarrow P(C_1 \cap C_2) = P(C_1)P(C_2).$$

In general, $C_1, \cdots, C_n$ are called independent iff for every collection of $k$ events $(2 \leq k \leq n)$,

$$P(C_{i_1} \cap C_{i_2} \cap \cdots \cap C_{i_k}) = P(C_{i_1}) \cdots P(C_{i_k}).$$

## 1.5 Random Variables

**Definition 1.5.1.** A function $X$ is called a *random variable*(r.v.) if it assigns to each element $c \in \mathscr{C}$ one and only one number $X(c) = x$. The space or range of $X$ is $\mathscr{D} = \{x : x = X(c), c \in \mathscr{C}\}$.

The r.v. $X$ is called *discrete* r.v. or *continuous* r.v. if $\mathscr{D}$ is countable set or an interval of real numbers, respectively.

The probability function $P$ is defined on $\mathscr{B}$. Now, we define a probability function $P_X$ defined on $\mathscr{F}$, and $P_X$ is often called induced probability function by r.v. $X$. i.e.

$$P(C),\ C \in \mathscr{B},\ P_X(B),\ B \in \mathscr{F}.$$

i.e.

$$P_X(B) = P[c \in \mathscr{C} : X(c) \in B],\ B \in \mathscr{F}.$$

Let $X$ is discrete r.v. with $\mathscr{D} = \{d_1, \cdots, d_m\}$, then

$$P_X(d_i) = P(X = d_i),\ i = 1, \cdots, m$$

is called probability mass function(pmf) of $X$.

**Example 1.5.1.** Consider tossing two fair die and let $X$ be the sum of upfaces. Then, $\mathscr{C} = \{(1,1), (1,2), \cdots, (6,6)\}$ and $\mathscr{D} = \{2, 3, \cdots, 12\}$. The probability of sum 4 is

$$P\left((1,3) \cup (2,3) \cup (3,1)\right) = P_X(4) = 3/36.$$

**Definition 1.5.2.** (Cumulative Distribution Function) The cumulative distribution function(cdf) of r.v. $X$ is defined as

$$F_X(x) = P_X((-\infty, x]) = P(X \leq x).$$

**Example 1.5.2.** Let $X$ be the upface of tossing a fair dice, then the cdf of $X$ is

**Example 1.5.3.** Let $X$ be a real number chosen at random from the interval $(0, 1)$. Then, it is reasonable to assign

$$P_X[(a, b)] = b - a \text{ for}, 0 < a < b < 1.$$

Want to obtain cdf of r.v. $X$. Let $x < 0$, then $P(X \leq x) = 0$. Let $x > 1$, then $P(X \leq x) = 1$. Let $0 < x < 1$, then $P(X \leq x) = P(0 < X \leq x) = x - 0 = x$. Hence, the cdf of $X$ is

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases}$$

**Theorem 1.5.1.** (Properties of cdf)

(a) $F(a) \leq F(b), \forall\, a < b$ (nondecreasing).

(b) $\lim_{x \to -\infty} F(x) = 0$.

(c) $\lim_{x \to \infty} F(x) = 1$.

(d) $\lim_{x \downarrow x_0} F(x) = F(x_0)$ (right continuous).

(pf) (a) $\{X \leq a\} \subset \{X \leq b\} \Rightarrow P(X \leq a) \leq P(X \leq b)$ by Thm.1.3.3
(b) $\lim_{x \to -\infty} \{X \leq x\} = \phi \Rightarrow \lim_{x \to -\infty} P(X \leq x) = 0$ by Thm.1.3.2
(c) $\lim_{x \to \infty} \{X \leq x\} = \mathscr{C} \Rightarrow \lim_{x \to \infty} P(X \leq x) = 1$
(d) Let $\{X_n\}$ be nay sequence s.t. $x_n \downarrow x_0$, and let $C_n = \{X \leq x_n\}$.

Then, $\{C_n\}$ is decreasing and $\bigcap_{n=1}^{\infty} C_n = \{X \leq x_0\}$. Hence, by Thm.1.3.6, $\lim_{n\to\infty} F(x_n) = P\left(\bigcap_{n=1}^{\infty} C_n\right) = F(x_0)$.

**Theorem 1.5.2.** $P(a < X \leq b) = F_X(b) - F_X(a)$, $^{\forall}a < b$.

(pf) $\{-\infty < X \leq b\} = \{-\infty < X \leq a\} \cup \{a < X \leq b\}$.

**Theorem 1.5.3.** $P(X = x) = F_X(x) - F_X(x-)$, $F_X(x-) = \lim_{z\uparrow x} F_X(z)$, i.e. left limit.

(pf) $^{\forall}x \in R$, $\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x\right]$, therefore by Thm.1.3.6,

$$\begin{aligned} P(X = x) &= P\left[\bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x\right]\right] = \lim_{n\to\infty} P\left(x - \frac{1}{n} < X \leq x\right) \\ &= \lim_{n\to\infty} P\left[F_X(x) - F_X\left(x - \frac{1}{n}\right)\right] = F_X(x) - F_X(x-). \end{aligned}$$

## 1.6 Discrete Random Variables

**Definition 1.6.1.** (Discrete Random Variable) A r.v. is called *discrete* if its space is either finite or countable.

**Definition 1.6.2.** (Probability Mass Function) The *probability mass function*(pmf) of a discrete r.v. $X$ with space $\mathscr{D}$ is given by

$$P_X(x) = P(X = x),\ x \in \mathscr{D}$$

- The support of a discrete r.v. $X$ is the points where $P_X(x) > 0$.

**Example 1.6.1.** Consider tossing a fair coin. Let $X$ be the number of flips need to obtain the first head. Find the the pmf of $X$.

(sol) We must have a string of $x-1$ tails followed by a head, i.e. $T\cdots TH$. Hence, by independence of each flip,

$$P(X=x)=\left(\frac{1}{2}\right)^{x-1}\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^{x},\ x=1,2,3,\cdots.$$

**Example 1.6.2.** An urn contains 100 balls, 20 white and 80 black. Let $X$ be the number of white balls when we draw 5 ball. Find the pmf of $X$.

(sol)

$$P_X(x)=\begin{cases}\frac{\binom{20}{x}\binom{80}{5-x}}{\binom{100}{5}}, & x=0,1,2,3,4,5\\ 0, & \text{otherwise.}\end{cases}$$

We are interested in computing the pmf of $Y=g(X)$ where the pmf of $X$ is known and $g$ is 1-1.

$$P_Y(y)=P(Y=y)=P[g(X)=y]=P(X=g^{-1}(y))=P_X(g^{-1}(y)).$$

**Example 1.6.3.** Find pmf of $Y=X-1$ when $P_X(x)=\left(\frac{1}{2}\right)^{x}$, $x=1,2,\cdots$.

(sol) $g(x)=x-1\ \Rightarrow\ g^{-1}(y)=y+1$

$$\therefore P_Y(y)=P_X(y+1)=\left(\frac{1}{2}\right)^{y+1},\ y=0,1,2,\cdots:\textit{geometric distribution}.$$

## 1.7 Continuous Random Variables

**Definition 1.7.1.** (Continuous Random Variables) A r.v. $X$ is called *continuous* if its cdf $F_X(x)$ is continuous, $^{\forall}x\in R$.

When we write cdf as

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$$

then $f_X(t)$ is called the probability density function(pdf) of a continuous r.v. X.

$$f_X(x) = \frac{d}{dx}F_X(x)$$

Note that

$$P(X = x) = F_X(x) - F_X(x-) = 0$$

for conti. r.v. also,

$$P(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(t)dt$$

and

$$P(a < X \leq b) = P(a \leq X \leq b) = P(a \leq X < b) = P(a < X < b).$$

By the properties of $F_X(x)$, we have

(i) $f_X(x) \geq 0 \leftarrow F_X(x)$ is nondecreasing

(ii) $\int_{-\infty}^{\infty} f_X(t)dt = 1 \leftarrow F_X(\infty) = 1$

**Example 1.7.1.** Consider selecting a point at random in the interior of a circle of radius 1. Find the pdf of $X$, where $X$ denote the distance of the selected point from the origin.

(sol) Note that $0 \leq x \leq 1$

$$F_X(x) = P(X \leq x) = \frac{\pi x^2}{\pi 1^2} = x^2$$

and $P(X \leq 0) = 0, P(X \leq 1) = 1$. Therefore,

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x \geq 1 \end{cases}$$

$$\therefore f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

**Example 1.7.2.** Find the pdf of $Y = X^2$ in Ex.1.7.1.

(sol)

$$\begin{aligned} F_Y(y) &= P(Y \le y) \\ &= P(X^2 \le y),\ y > 0 \\ &= P(-\sqrt{y} \le X \le \sqrt{y}) \\ &= P(0 \le X \le \sqrt{y}) \\ &= F_X(\sqrt{y}) \\ &= (\sqrt{y})^2 \\ &= y,\ 0 < y < 1 \end{aligned}$$

$$\therefore f_Y(y) = I(0 < y < 1)$$

**Example 1.7.3.** Find the pdf of $Y = X^2$ when $f_X(x) = \frac{1}{2} I(-1 < x < 1)$.

(sol)

$$\begin{aligned} F_Y(y) &= P(Y \le y) \\ &= P(X^2 \le y),\ y > 0 \\ &= P(-\sqrt{y} \le X \le \sqrt{y}) \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} f_X(x)dx \\ &= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx \\ &= \sqrt{y} \end{aligned}$$

$$\therefore F_Y(y) = \begin{cases} 0, & y < 0 \\ \sqrt{y}, & 0 \le y \le 1 \\ 1, & y \ge 1 \end{cases}$$

$$\therefore f_Y(y) = \frac{1}{2\sqrt{y}} I(0 \le y \le 1)$$

**Theorem 1.7.1.** Let $X$ be a continuous random variable with pdf $f_X(x)$ and support $S_X$. Let $Y = g(X)$, where $g(x)$ is a one-to-one differentiable function, on the support of $X$, $S_X$. Then the pdf of $Y$ is given by

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|, \, y \in S_Y,$$

where the support of $Y$ is the set $S_Y = \{y = g(x) : x \in S_X\}$.

(pf) Since $g$ is one-to-one and continuous, it is either increasing or decreasing. First, assume it is increasing.

$$F_Y(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$$
$$\therefore f_Y(y) = \frac{d}{dy} F_Y(y) = f_X(g^{-1}(y)) \frac{dx}{dy}$$

$g$ is decreasing, then

$$F_Y(y) = P(g(X) \leq y) = P(X > g^{-1}(y)) = 1 - F_X(g^{-1}(y))$$
$$\therefore f_Y(y) = -f_X(g^{-1}(y)) \frac{dx}{dy}$$

Therefore,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

**Example 1.7.4.** Find pdf of $Y = -2 \log X$, where $f_X(x) = I(0 < x < 1)$

(sol) $g^{-1}(y) = e^{-y/2}$, $dx/dy = -\frac{1}{2} e^{-y/2}$

$$\therefore f_Y(y) = \frac{1}{2} e^{-y/2}, \, y > 0$$

**Example 1.7.5.**

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}(x+1), & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

$$P\left(-3 < X \leq \frac{1}{2}\right) = F\left(\frac{1}{2}\right) - F(-3) = \frac{3}{4} - 0 = \frac{3}{4}$$

$$P(X = 0) = F(0) - F(0-) = \frac{1}{2} - 0 = \frac{1}{2}$$

## 1.8 Expectation of a Random Variable

**Definition 1.8.1.** The expectation of r.v. $X$ is defined as

$$E(x) = \begin{cases} \int_{-\infty}^{\infty} x f_X(x)dx & \text{if } \int_{-\infty}^{\infty} |x| f(x)dx < \infty \text{ (conti.)} \\ \sum_{x \in S_X} x p_X(x) & \text{if } \sum |x| p(x) < \infty \text{ (discrete)} \end{cases}$$

**Theorem 1.8.1.** The expectation of $Y = g(X)$ is given by

$$E[g(X)] = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x)dx & \text{if } \int_{-\infty}^{\infty} |g(x)| f(x)dx < \infty \text{ (conti.)} \\ \sum_{x \in S_X} g(x) p_X(x) & \text{if } \sum |g(x)| p(x) < \infty \text{ (discrete)} \end{cases}$$

(pf) discrete case only

$$\begin{aligned} \sum_{x \in S_X} g(x) p_X(x) &= \sum_{g \in S_Y} \sum_{\{x \in S_X, g(x) = y\}} g(x) p_X(x) \\ &= \sum_{y \in S_Y} y \sum_{\{x \in S_X, g(x) = y\}} p_X(x) \\ &= \sum_{y \in S_Y} y p_Y(y) \\ &= E(Y) \end{aligned}$$

**Theorem 1.8.2.** $E[k_1 g_1(X) + k_2 g_2(X)] = k_1 E[g_1(X)] + k_2 E[g_2(X)]$ if $E[g_1(X)]$ and $E[g_2(X)]$ exist.

(pf) We are only to show $\int |k_1 g_1(x) + k_2 g_2(x)| f_X(x)dx < \infty$. By triangular inequality($|\boldsymbol{a} + \boldsymbol{b}| \leq |\boldsymbol{a}| + |\boldsymbol{b}|$)

$$\int |k_1 g_1(x) + k_2 g_2(x)| f_X(x)dx \leq |k_1| \int |g_1(x)| f_X(x)dx + |k_2| \int |g_2(x)| f_X(x)dx < \infty$$

## 1.9 Some Special Expectations

**Definition 1.9.1.** $\mu = E(X)$: *mean* of r.v. $X$

**Definition 1.9.2.** $\sigma^2 = E(X-\mu)^2$: *variance* of r.v. $X$ and $\sigma = \sqrt{(}\sigma^2)$: standard deviation

**Definition 1.9.3.** $X$: r.v. s.t. $E(e^{tX}) < \infty$, $|t| < h$ for some $h > 0$. Then, $M(t) = E(e^{tX})$ is called the *moment generating function*(mgf) of r.v. $X$.

**Theorem 1.9.1.** $X$, $Y$: r.v. with mgf $M_X(t)$ and $M_Y(t)$, respectively. Then, $F_X(z) = F_Y(z)$, $^\forall z \in R$ iff $M_X(t) = M_Y(t)$, $^\forall t \in (-h, h)$, $h > 0$.(uniqueness of mgf)

**Remark 1.9.1.** (1) mgf may not exist. For example, let $X$ be r.v. with pdf $f(x) = \frac{1}{x^2} I(x > 1)$, then

$$\begin{aligned} M_X(t) &= \int_1^\infty e^{tX} \frac{1}{x^2} dx \\ &= \lim_{b\to\infty} \int_1^b \left(1 + tx + \frac{t^2x^2}{2} + \cdots\right) \frac{1}{x^2} dx \\ &= \lim_{b\to\infty} \int_1^b \left[-\frac{1}{x} + t\log x + \frac{t^2x^2}{2} + \cdots\right]_1^b = \infty \end{aligned}$$

(2) Sometimes can find the pdf from the mgf. Let

$$M_x(t) = \frac{1}{10}e^t + \frac{2}{10}e^{2t} + \frac{3}{10}e^{3t} + \frac{4}{10}e^{4t}.$$

Now,

$$M_X(t) = \sum e^{tx} p(x) = p(1)e^t + p(2)e^{2t} + p(3)e^{3t} + p(4)e^{4t}.$$

By the uniqueness of polynomial coeff., we must have

$$p(x) = \frac{x}{10},\ x = 1, 2, 3, 4$$

(3) Can compute $E(X^m)$, $m = 1, 2, \cdots$ using the mgf. By Taylor expansion,

$$\begin{aligned} M_X(t) &= E(e^{tX}) \\ &= E\left[1 + tX + \frac{t^2X^2}{2!} + \frac{t^3X^3}{3!} + \cdots\right] \\ &= 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \cdots \\ &\therefore M_X^{(m)}(0) = E(X^m). \end{aligned}$$

(4) *characteristic function(ch.f)*

$$\varphi(t) = E(e^{itX}) : \text{ch.f of r.v.} X.$$

Note that ch.f always exist. why?

$$|\varphi(t)| = \left|\int e^{itx} f(x)dx\right| \leq \int \left|e^{itx} f(x)\right| dx$$

Now,

$$|e^{itx}| = |\cos tx + i \sin tx| = \sqrt{\cos^2 tx + \sin^2 tx} = 1$$

$$\therefore |\varphi(t)| \leq 1.$$

Also, can show $E(X) = -i\varphi'(0)$, $E(X^2) = -\varphi''(0)$

(5) *cumulant generating function*(cgf)

$$\psi(t) = \log M(t) : \text{cgf of r.v. } X.$$

Relation between moment and cumulant. Recall that

$$M(t) = 1 + \mu_1 t + \frac{\mu_2 t^2}{2!} + \frac{\mu_3 t^3}{3!} + \cdots, \ \mu_m = E(X^m),$$

assume that

$$\begin{aligned}
\psi(t) &= \kappa_0 + \kappa_1 t + \frac{\kappa_2 t^2}{2!} + \frac{\kappa_3 t^3}{3!} + \cdots, \ \kappa_m : \text{m-th cumulant} \\
&= \log\left(1 + \mu_1 t + \frac{\mu_2 t^2}{2!} + \cdots\right) \\
&= \left(\mu_1 t + \frac{\mu_2 t^2}{2!} + \cdots\right) - \frac{1}{2}\left(\mu_1 t + \frac{\mu_2 t^2}{2!} + \cdots\right)^2 + \frac{1}{3}\left(\mu_1 t + \frac{\mu_2 t^2}{2!} + \cdots\right)^3 - \cdots \\
&= \mu_1 t + \frac{1}{2}(\mu_2 - \mu_1^2)t^2 + \frac{1}{6}(\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3) + \cdots
\end{aligned}$$

$\therefore \kappa_0 \equiv 0$, $\kappa_1 = \mu_1$, $\kappa_2 = \mu_2 - \mu_1^2 \equiv \sigma^2$, $\kappa_3 = \mu_3 - 3\mu_1\mu_2 + 2\mu_1^3 = E(X-\mu)^3 = \mu_3'$.

(6) skewness and kurtosis

$$\rho_3 = E[(X-\mu)^3]/\sigma^3 : \textit{skewness}.$$

$$\rho_4 = E[(X-\mu)^4]/\sigma^4 : \textit{kurtosis}.$$

## 1.10 Important Inequalities

**Theorem 1.10.1.** If $E(X^m)$ exists then $E(X^k)$ exists for $k \leq m$.

(pf) We are only to prove $\int |x|^k f(x)dx < \infty$

$$\begin{aligned}
\int_{-\infty}^{\infty} |x|^k f(x)dx &= \int_{|x|\leq 1} |x|^k f(x)dx + \int_{|x|>1} |x|^k f(x)dx \\
&\leq \int_{|x|\leq 1} f(x)dx + \int_{|x|>1} |x|^m f(x)dx \\
&\leq \int_{-\infty}^{\infty} f(x)dx + \int_{-\infty}^{\infty} |x|^m f(x)dx \\
&= 1 + E|X|^m < \infty
\end{aligned}$$

**Theorem 1.10.2.** (*Markov's Inequality*). $u(X)$: nonnegative function of r.v. $X$. Assume $E[u(X)]$ exists. Then, $^\forall c > 0$, $P[u(X) \geq c] \leq E[u(X)]/c$.

(pf)Let $A = \{x : u(x) \geq c\}$. Then,

$$\begin{aligned} E[u(X)] &= \int u(x)f(x)dx \\ &= \int_A u(x)f(x)dx + \int_{A^c} u(x)f(x)dx \\ &\geq \int_A u(x)f(x)dx \\ &\geq c\int_A f(x)dx \\ &= cP(u(X) \geq c). \end{aligned}$$

**Theorem 1.10.3.** (*Chebyshev Inequality*). $P(|X - \mu| \geq k\sigma) \leq 1/k^2$, $^\forall k > 0$.

(pf) Let $u(X) = (X - \mu)^2$ and $c = k^2\sigma^2$, then

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq E[(X - \mu)^2]/k^2\sigma^2 \Rightarrow P(|X - \mu| \geq k\sigma) \leq 1/k^2.$$

**Definition 1.10.1.** $\phi$: function defined on $(a, b)$, $-\infty \leq a < b \leq \infty$. $\phi$ is said to be *convex* if for all $x, y$ in $(a, b)$ and $0 < \gamma < 1$,

$$\phi[\gamma x + (1 - \gamma)y] \leq \gamma\phi(x) + (1 - \gamma)\phi(y).$$

$\phi$ is said to be strictly convex if the inequality is strict.

**Theorem 1.10.4.** Assume $\phi$ is differentiable on $(a, b)$, then

(a) $\phi$: convex iff $\phi'(x) \leq \phi'(y)$, $^\forall a < x < y < b$

(b) $\phi$: strictly convex iff $\phi'(x) < \phi'(y)$, $^\forall a < x < y < b$.
If $\phi$ is twice differentiale on $(a, b)$, then

(c) $\phi$: convex iff $\phi''(x) > 0$, $^\forall a < x < b$

(d) $\phi$: strictly convex iff $\phi''(x) > 0$, $^\forall a < x < b$

**Theorem 1.10.5.** (*Jensen's Inequality*). $\phi$: convex on an open interval $I$. $X$: r.v. with support $S \subset I$ and $E(X) < \infty \rightarrow \phi[E(X)] \leq E[\phi(X)]$.

(pf)Let $\xi$ is between $x$ and $\mu$, then

$$\begin{aligned}\phi(x) &= \phi(\mu)+\phi'(\mu)(x-\mu)+\frac{1}{2}\phi''(\xi)(x-\mu)^2 \\ &\geq \phi(\mu)+\phi'(\mu)(X-\mu) \Rightarrow \text{Take expectation on both sides.}\end{aligned}$$

**Example 1.10.1.** $\{a_1, \cdots, a_n\}$: set of positive numbers.
Let $X$ be a r.v. s.t. $P(X = a_i) = 1/n$, $i = 1, \cdots, n$.

(i) $E(X) = \sum_{i=1}^{n} a_i \frac{1}{n} = \bar{a}$: arithmetic mean(AM)

(ii) Since $-\log x$ is convex, we have by Jensen's ineq.,

$$\begin{aligned}-\log[E(X)] &= -\log(\bar{a}) \\ &\leq E[-\log X] \\ &= -\frac{1}{n}\sum \log a_i \\ &= -\log(a_1 \cdots a_n)^{1/n}\end{aligned}$$

i.e. $(a_1 \cdots a_n)^{1/n}$ : geometric mean(GE) $\leq \bar{a} = \frac{1}{n}\sum a_i$

(iii) Replace $a_i$ by $1/a_i$, then

$$\left(\frac{1}{a_1 \cdots a_n}\right)^{1/n} \leq \frac{1}{n}\sum \frac{1}{a_i}$$

i.e. $(a_1 \cdots a_n)^{1/n} \geq \frac{1}{\frac{1}{n}\sum \frac{1}{a_i}}$: harmonic mean(HM)

We have shown the relationship HM $\leq$ GM $\leq$ AM.

# 2 Multivariate Distributions

## 2.1 Distributions of Two Random Variables

**Definition 2.1.1.** $(X_1, X_2)$ is called *random vector* if $X_1,\ X_2$ are random variables which assign to each element $c$ of $\mathscr{C}$ one and only one ordered pair of numbers $X_1(c) = x_1,\ X_2(c) = x_2$. The space of $(X_1, X_2)$ is $\mathscr{D} = \{(x_1, x_2) : x_1 = X_1(c), x_2 = X_2(c), c \in \mathscr{C}\}$.

- will use the vector notation $\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = (X_1, X_2)'$
- The cdf of $\boldsymbol{X} = (X_1, X_2)'$ is

  $$F_{X_1,X_2}(x_1, x_2) = P(X_1 \leq x_1, X_2 \leq x_2)$$

  and can easily show

  $$P(a_1 < X_1 \leq b_1, a_2 < X_2 \leq b_2) =$$

  $$F_{X_1,X_2}(b_1, b_2) - F_{X_1,X_2}(a_1, b_2) - F_{X_1,X_2}(b_1, a_2) + F_{X_1,X_2}(a_1, a_2)$$

- The joint prob. mass function of $\boldsymbol{X} = (X_1, X_2)'$ is

  $$p_{X_1,X_2}(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$$

  if $\boldsymbol{X}$ is discrete random vector.

- For the continuous random vector, $f_{X_1,X_2}(x_1, x_2)$ satisfying

  $$F_{X_1,X_2}(x_1, x_2) = \int_{-\infty}^{x_2} \int_{-\infty}^{x_1} f_{X_1,X_2}(w_1, w_2) dw_1 dw_2$$

  is called the joint pdf, and we have

  $$\frac{\partial^2 F_{X_1,X_2}(x_1, x_2)}{\partial x_1 \partial x_2} = f_{X_1,X_2}(x_1, x_2)$$

- $F_{X_1}(x_1) = \lim_{x_2 \to \infty} F_{X_1,X_2}(x_1, x_2)$: marginal cdf of $X_1$

- $p_{X_1}(x_1) = \sum_{x_2} p_{X_1,X_2}(x_1, x_2)$: marginal pmf of $X_1$
- $f_{X_1}(x_1) = \int_{-\infty}^{\infty} f_{X_1,X_2}(x_1, x_2)dx_2$: marginal pdf of $X_1$

**Example 2.1.1.** $f(x_1, x_2) = x_1 + x_2$, $0 < x_1 < 1$, $0 < x_2 < 1$ jpdf of $X_1$ and $X_2$ compute $P(X_1 \leq 1/2)$ and $P(X_1 + X_2 \leq 1)$.

(sol)

(i) $P(X_1 \leq 1/2) = \int_0^{1/2} f_1(x_1)dx_1$

$$\begin{aligned} f_{X_1}(x_1) &= \int f(x_1, x_2)dx_2 \\ &= \int_0^1 (x_1 + x_2)dx_2 \\ &= \left[x_1x_2 + \frac{1}{2}x_2^2\right]_0^1 \\ &= x_1 + \frac{1}{2} \end{aligned}$$

$$\therefore P\left(X_1 \leq \frac{1}{2}\right) = \int_0^{1/2} (x_1 + \frac{1}{2})dx_1 = \left[\frac{x_1^2}{2} + \frac{x_1}{2}\right]_0^{1/2} = \frac{3}{8}$$

(ii)

$$\begin{aligned} P(X_1 + X_2 \leq 1) &= \int_0^1 \int_0^{1-x_2} f(x_1, x_2)dx_1dx_2 \\ &= \int_0^1 \int_0^{1-x_2} (x_1 + x_2)dx_1dx_2 \\ &= \frac{1}{3} \end{aligned}$$

- $$E[g(X_1, X_2)] = \begin{cases} \int\int g(x_1, x_2)f(x_1, x_2)dx_1dx_2 & \text{if} \int\int |g(x_1, x_2)|f(x_1, x_2)dx_1dx_2 < \infty \\ \sum_{x_1}\sum_{x_2} g(x_1x_2)p(x_1, x_2) & \text{if} \sum\sum |g(x_1, x_2)|p(x_1, x_2) < \infty \end{cases}$$

**Theorem 2.1.1.** $E[k_1 g_1(X_1, X_2) + k_2 g_2(X_1, X_2)] = k_1 E[g_1(X_1, X_2)] + k_2 E[g_2(X_1, X_2)]$: *linearity property of expectation*.

**Example 2.1.2.** $f(x_1, x_2) = 8x_1 x_2 I(0 < x_1 < x_2 < 1)$ compute $E(X_1 X_2^2)$, $E(X_2)$, and $E[7X_1 X_2^2 + 5X_2]$.

(sol)

(i) $E(X_1 X_2^2) = \int_0^1 \int_0^{x_2} x_1 x_2^2 \, 8x_1 x_2 dx_1 dx_2 = \frac{8}{21}$

(ii) $E(X_2) = \int_0^1 x_2 f_{X_2}(x_2) dx_2 = \int_0^1 x_2 \left[ \int_0^{x_2} 8x_1 x_2 dx_1 \right] dx_2 = \frac{4}{5}$

(iii) $E[7X_1 X_2^2 + 5X_2] = 7\frac{8}{21} + 5\frac{4}{5} = \frac{20}{3}$

**Definition 2.1.2.** Let $\boldsymbol{X} = (X_1, X_2)'$ be a random vector.

$$\begin{aligned} M_{\boldsymbol{X}}(\boldsymbol{t}) &= E\left[e^{\boldsymbol{t}'\boldsymbol{X}}\right], \ \boldsymbol{t} = (t_1, t_2)', \ ||\boldsymbol{t}|| < h, \ h > 0 \\ &= M_{X_1, X_2}(t_1, t_2) \\ &= E\left[e^{t_1 X_1 + t_2 X_2}\right] : \text{mgf of } \boldsymbol{X} = (X_1, X_2)' \end{aligned}$$

Note that $M_{X_1,X_2}(t_1, t_2) = \int \int e^{t_1 x_1 + t_2 x_2} f_{X_1,X_2}(x_1, x_2) dx_1 dx_2$

$$\begin{aligned} M_{X_1,X_2}(t_1, 0) &= \int \int e^{t_1 x_1} f(x_1, x_2) dx_1 dx_2 \\ &= \int \int e^{t_1 x_1} f(x_1, x_2) dx_2 dx_1 \\ &= \int e^{t_1 x_1} \left\{ \int f(x_1, x_2) dx_2 \right\} dx_1 \\ &= \int e^{t_1 x_1} f_{X_1}(x_1) dx_1 \\ &= E[e^{t_1 x_1}] \\ &= M_{X_1}(t_1) : \text{marginal mgf of } X_1 \end{aligned}$$

Similarly, $M_{X_1,X_2}(0, t_2) = M_{X_2}(t_2)$: marginal mgf of $X_2$

**Example 2.1.3.** $f(x,y) = e^{-y}I(0 < x < y < \infty)$: jpdf of $(X,Y)$

(sol) $M(t_1,t_2) = \int_0^\infty \int_x^\infty e^{t_1x+t_2y}e^{-y}dydx = \dfrac{1}{(1-t_1-t_2)(1-t_2)}$

$$M(t_1,0) = \frac{1}{1-t_1} : \text{mgf of } X,$$

$$M(0,t_2) = \frac{1}{(1-t_2)^2} : \text{mgf of } Y.$$

## 2.2 Transformations: Bivariate R.V.'s

Want to find the distribution of $Y = g(X_1,X_2)$ when jpdf of $X_1$ and $X_2$ is known. Two methods are possible. First, find the cdf of $Y$ and take derivative. Secondly, use transformation technique.

(1) discrete case

$(X_1,X_2)$: discrete random vector with jpmf $p_{X_1,X_2}(x_1,x_2)$ and support $S$
$y_1 = u_1(x_1,x_2)$ and $y_2 = u_2(x_1,x_2)$: 1-1 transformation from $S$ to $\mathscr{T}$.

$$(X_1,X_2) \xrightarrow{u_1,u_2} (Y_1,Y_2)$$

$x_1 = w_1(y_1,y_2)$, $x_2 = w_2(y_1,y_2)$: inverse function
$\Rightarrow p_{Y_1,Y_2}(y_1,y_2) = p_{X_1,X_2}(w_1(y_1,y_2),w_2(y_1,y_2)), (y_1,y_2) \in \mathscr{T}$.

**Example 2.2.1.** $p_{X_1,X_2}(x_1,x_2) = \mu_1^{x_1}\mu_2^{x_2}e^{-\mu_1-\mu_2}/x_1!x_2!, x_1 = 0,1,2,\cdots,$
$x_2 = 0,1,2,\cdots$ Find the pdf of $Y_1 = X_1 + X_2$.

(sol) need to define $Y_2$ s.t. $(x_1,x_2) \to (y_1,y_2)$ is 1-1.
Let $Y_2 = X_2$, then $y_1 = x_1 + x_2$ and $y_2 = x_2$ represent 1-1 transformation.

$$S = \{(x_1,x_2) : x_1 = 0,1,2,\cdots,\ x_2 = 0,1,2,\cdots\}$$

$$\to \mathscr{T} = \{(y_1,y_2) : y_1 = 0,1,2,\cdots,\ y_2 = 0,1,\cdots,y_1\}$$

i.e. $x_1 = y_1 - y_2$, $x_2 = y_2$. So, the jpdf of $Y_1$ and $Y_2$ is

$$p_{Y_1,Y_2}(y_1, y_2) = \frac{\mu_1^{y_1-y_2}\mu_2^{y_2}e^{-\mu_1-\mu_2}}{(y_1-y_2)!y_2!},\ (y_1, y_2) \in \mathscr{T}$$

$$\begin{aligned}
\therefore p_{Y_1}(y_1) &= \sum_{y_2=0}^{y_1} p_{Y_1,Y_2}(y_1, y_2) \\
&= \frac{e^{-\mu_1-\mu_2}}{y_1!} \sum_{y_2=0}^{y_1} \frac{y_1!}{(y_1-y_2)!y_2!}\mu_1^{y_1-y_2}\mu_2^{y_2} \\
&= \frac{(\mu_1+\mu_2)^{y_1}e^{-\mu_1-\mu_2}}{y_1!},\ y_1 = 0, 1, 2, \cdots
\end{aligned}$$

(2) continuous case

Let $X_1, X_2$ be conti. r.v.'s, and $X = (X_1, X_2)'$ be random vector with jpdf $f_{X_1,X_2}(x_1, x_2)$ and support $S$. Consider a transformation $(x_1, x_2) \to (y_1, y_2)$ s.t. $y_1 = u_1(x_1, x_2)$ and $y_2 = u_2(x_1, x_2)$ be 1-1 and let $x_1 = w_1(y_1, y_2)$ and $x_2 = w_2(y_1, y_2)$ be inverse function with the Jacobian

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix}$$

Then , the jpdf of $Y_1$ and $Y_2$ is given by

$$f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(w_1(y_1, y_2), w_2(y_1, y_2))|J|,\ (y_1, y_2) \in \mathscr{T}.$$

**Example 2.2.2.** $f_{X_1,X_2}(x_1, x_2) = I(0 < x_1 < 1,\ 0 < x_2 < 1)$ Find the pdf of $Y_1 = X_1 + X_2$

(sol) Two methods are possible

(i) cdf technique

$$F_{Y_1}(y_1) = P(Y_1 \le y_1) = P(X_1 + X_2 \le y_1)$$

$$\therefore F_{Y_1}(y_1) = \begin{cases} 0 & ,y_1 < 0 \\ \int_0^{y_1}\int_0^{y_1-x_1} dx_2dx_1 & ,0 \le y_1 < 1 \\ 1 - \int_{y_1-1}^{1}\int_{y_1-x_1}^{1} dx_2dx_1 & ,1 \le y_1 < 2 \\ 1 & ,y_1 \ge 2 \end{cases}$$

$$= \begin{cases} 0 & ,y_1 < 0 \\ y_1^2/2 & ,0 \le y_1 < 1 \\ 1-(2-y_1)^2/2 & ,1 \le y_1 < 2 \\ 1 & ,y_1 \ge 2 \end{cases}$$

$$\therefore f_{Y_1}(y) = \begin{cases} y_1 & ,0 < y_1 < 1 \\ 2-y_1 & ,1 \le y_1 < 2 \\ 0 & ,\text{o.w.} \end{cases}$$

(ii) transformation technique

Need to define $Y_2$ s.t. $(x_1, x_2) \to (y_1, y_2)$ be 1-1.
Let $Y_2 = X_2$, then $y_1 = x_1 + x_2$, $y_2 = x_2$ represent 1-1 and $x_1 = y_1 - y_2$, $x_2 = y_2$ are inverse function. Jacobian is $J = 1$.

$\therefore f_{Y_1,Y_2}(y_1, y_2) = f_{X_1,X_2}(y_1 - y_2, y_2)|J| = 1$, $0 < y_2 < 1$, $y_2 < y_1 < 1 + y_2$

$$\therefore f_{Y_1}(y) = \int f_{Y_1,Y_2}(y_1, y_2)dy_2$$

$$= \begin{cases} \int_0^{y_1} dy_2 = y_1 & ,0 < y_1 < 1 \\ \int_{y_1-1}^{1} dy_2 = 2-y_1 & ,1 \le y_1 < 2 \end{cases}$$

**Example 2.2.3.** $f_{X_1,X_2}(x_1, x_2) = \frac{1}{4}exp\left[-\frac{x_1+x_2}{2}\right]$, $0 < x_1 < \infty$, $0 < x_2 < \infty$
Find the pdf of $Y_1 = \frac{1}{2}(X_1 - X_2)$.

(sol) Let $Y_2 = X_2$, then $y_1 = \frac{1}{2}(x_1 - x_2)$, $y_2 = x_2$ is 1-1 and $x_1 = 2y_1 + y_2$, $x_2 = y_2$ are inverse function with $J = 2$.

$$\mathscr{T} = \{(y_1, y_2) : -\infty < y_1 < \infty,\ y_2 > 0,\ -2y_1 < y_2\}$$

$$
\begin{aligned}
f_{Y_1,Y_2}(y_1,y_2) &= f_{X_1,X_2}(2y_1+y_2,y_2)|J| \\
&= \frac{2}{4}exp\left[-\frac{1}{2}(2y_1+y_2)-\frac{1}{2}y_2\right] \\
&= \frac{1}{2}e^{-y_1-y_2},\ -\infty<y_1<\infty,\ y_2>0,\ -2y_1<y_2
\end{aligned}
$$

$$
\therefore f_{Y_1}(y_1) = \begin{cases} \int_{-2y_1}^{\infty}\frac{1}{2}e^{-y_1-y_2}dy_2 = \frac{1}{2}e^{y_1} & ,-\infty<y_1<0 \\ \int_{0}^{\infty}\frac{1}{2}e^{-y_1-y_2}dy_2 = \frac{1}{2}e^{-y_1} & ,y_1\geq 0 \end{cases}
$$

$$
= \frac{1}{2}e^{-|y_1|},\ -\infty<y_1<\infty : \text{double exponential or Laplace pdf}
$$

## 2.3 Conditional Distribution and Expectation

- $p_{X_2|X_1}(x_2|x_1) = \frac{p_{X_1,X_2}(x_1,x_2)}{p_{X_1}(x_1)}$: conditional pmf of $X_2$ given $X_1 = x_1$
- $f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1,X_2}(x_1,x_2)}{f_{X_1}(x_1)}$: conditional pdf of $X_2$ given $X_1 = x_1$

(e.g)

$$
P(a<X_2<b|X_1=x_1) = \int_a^b f(x_2|x_1)dx_2
$$

$E[u(X_2)|x_1] = \int_{-\infty}^{\infty} u(x_2)f(x_2|x_1)dx_2$ : conditional mean of $u(X_2)$ given $X_1 = x_1$

$Var(X_2|x_1) = E[\{X_2 - E(X_2|x_1)\}^2|x_1] = E(X_2^2|x_1) - E^2(X_2|x_1)$ : conditional var. of $X_2$ given $X_1$

**Example 2.3.1.** Find $E(X_1|x_2)$ and $Var(X_1|x_2)$ when $f(x_1,x_2) = 2I(0 < x_1 < x_2 < 1)$

(sol)

$$
f_2(x_2) = \int_0^{x_2} 2dx_1 = 2x_2 I(0<x_2<1)
$$

$$
\therefore f(x_1|x_2) = \frac{2}{2x_2} = \frac{1}{x_2}I(0<x_1<x_2<1)
$$

$$\therefore E(X_1|x_2) = \int_0^{x_2} x_1 \frac{1}{x_2} dx_1 = \frac{x_2}{2} I(0 < x_2 < 1)$$

$$\therefore Var(X_1|x_2) = \int_0^{x_2} \left(x_1 - \frac{x_2}{2}\right)^2 \frac{1}{x_2} dx_1 = \frac{x_2^2}{12} I(0 < x_2 < 1)$$

**Theorem 2.3.1.** (a) $E[E(X_2|X_1)] = E(X_2)$

(b) $Var[E(X_2|X_1)] \leq Var(X_2) = Var[E(X_2|X_1)] + E[Var(X_2|X_1)]$

(pf)

(a)

$$\begin{aligned}
E[E(X_2|X_1)] &= \int \left\{ \int x_2 f_{X_2|X_1}(x_2|x_1) dx_2 \right\} f_{X_1}(x_1) dx_1 \\
&= \int \int x_2 \frac{f(x_1, x_2)}{f_{X_1}(x_1)} f_{X_1}(x_1) dx_2 dx_1 \\
&= \int \int x_2 f(x_1, x_2) dx_1 dx_2 \\
&= \int x_2 \left\{ \int f(x_1, x_2) dx_1 \right\} dx_2 \\
&= \int x_2 f_{X_2}(x_2) dx_2 \\
&= E(X_2)
\end{aligned}$$

(b)

$$\begin{aligned}
Var(X_2) &= E[(X_2 - \mu_2)^2],\ \mu_2 = E(X_2) \\
&= E[\{X_2 - E(X_2|X_1) + E(X_2|X_1) - \mu_2\}^2] \\
&= E[\{X_2 - E(X_2|X_1)\}^2] + E[\{E(X_2|X_1) - \mu_2\}^2] + 2E[\{X_2 - E(X_2|X_1)\}\{E(X_2|
\end{aligned}$$

Now,

$$E[\{X_2 - E(X_2|X_1)\}^2] = E[E[\{X_2 - E(X_2|X_1)\}^2|X_1]] = E[Var(X_2|X_1)]$$

$$E[\{E(X_2|X_1) - \mu_2\}^2] = E[\{E(X_2|X_1) - E(E(X_2|X_1))\}^2]$$

Therefore,

$$\begin{aligned}
Var(X_2) &= E[\{X_2 - E(X_2|X_1)\}^2] + E[\{E(X_2|X_1) - \mu_2\}^2] \\
&= E[Var(X_2|X_1)] + Var[E(X_2|X_1)] \\
&\geq Var[E(X_2|X_1)]
\end{aligned}$$

## 2.4 The Correlation Coefficient

$X$: r.v. with $\mu_1 = E(X)$, $\sigma_1^2 = Var(X)$.
$Y$: r.v. with $\mu_2 = E(Y)$, $\sigma_2^2 = Var(Y)$.

$Cov(X,Y) := E[(X_1 - \mu_1)(Y - \mu_2)] = E(XY) - E(X)E(Y)$ : *covariance* between $X$ and $Y$.

$$\rho := \frac{Cov(X,Y)}{\sigma_1\sigma_2} = \frac{E[(X-\mu_1)(Y-\mu_2)]}{\sigma_1\sigma_2} \text{ : } \textit{corr. coef.} \text{ of } X \text{ and } Y.$$

**Theorem 2.4.1.** If $E(X|Y)$ is linear in $X$, then $E(Y|X) = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(X - \mu_1)$ and $E[Var(Y|X)] = \sigma_2^2(1-\rho^2)$.

(pf) Let $E(Y|X) = a + bX$, by taking expectation on both side

$$\begin{aligned} E[E(Y|X)] &= E[a+bX] \\ E(Y) &= a + bE(X) \\ \mu_2 &= a + b\mu_1 \end{aligned}$$

By multiplying $X$ on both sides of $E(Y|X) = a + bX$

$$\begin{aligned} XE(Y|X) &= aX + bX^2 \\ E[XE(Y|X)] &= E[aX + bX^2] \\ E[E(XY|X)] &= aE(X) + bE(X^2) \\ E(XY) &= a\mu_1 + b(\sigma_1^2 + \mu_1^2) \\ \rho\sigma_1\sigma_2 + \mu_1\mu_2 &= a\mu_1 + b(\sigma_1^2 + \mu_1^2) \end{aligned}$$

$$\Rightarrow\ a = \mu_2 - \rho\frac{\sigma_2}{\sigma_1}\mu_1,\ b = \rho\frac{\sigma_2}{\sigma_1}$$

$$E(Y|X) = a + bX = \mu_2 - \rho\frac{\sigma_2}{\sigma_1}\mu_1 + \rho\frac{\sigma_2}{\sigma_1}X = \mu_2 + \rho\frac{\sigma_2}{\sigma_1}(X - \mu_1)$$

$$
\begin{aligned}
E[Var(Y|X)] &= \int \left\{ \int (y - \mu_2 - \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1))^2 f_{Y|X}(y|x)dy \right\} f_X(x)dx \\
&= \int\int \left\{ y - \mu_2 - \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1) \right\}^2 f_{X,Y}(x,y)dydx \\
&= \int\int \left\{ (y-\mu_2)^2 - 2\rho(y-\mu_2)\frac{\sigma_2}{\sigma_1}(x-\mu_1) + \rho^2\frac{\sigma_2^2}{\sigma_1^2}(x-\mu_1)^2 \right\} f_{X,Y}(x,y)dydx \\
&= Var(Y) - 2\rho\frac{\sigma_2}{\sigma_1}Cov(X,Y) + \rho^2\frac{\sigma_2^2}{\sigma_1^2}Var(X) \\
&= \sigma_2^2 - 2\rho\frac{\sigma_2}{\sigma_1}\rho\sigma_1\sigma_2 + \rho^2\frac{\sigma_2^2}{\sigma_1^2}\sigma_1^2 \\
&= \sigma_2^2(1-\rho^2)
\end{aligned}
$$

Since

$$
\frac{\partial^{k+m}M(t_1,t_2)}{\partial t_1^k \partial t_2^m} = \int\int x^k y^m e^{t_1x+t_2y} f(x,y)dxdy,
$$

we have

$$
\frac{\partial^{k+m}M(t_1,t_2)}{\partial t_1^k \partial t_2^m}\Bigg]_{t_1=t_2=0} = E(X^kY^m)
$$

## 2.5 Independent Random Variables

**Definition 2.5.1.** $X$ and $Y$: *indep.* iff $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.

**Theorem 2.5.1.** $X$ and $Y$: indep. iff $f_{X,Y}(x,y) = g(x)h(y)$, where $g(x)$ is function of $x$ only and $g(y)$ is function of $y$ only.

(pf) ($\Rightarrow$) If $X$ and $Y$ are indep., then $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, so that take $g(x) = f_X(x)$, $h(y) = f_Y(y)$.

($\Leftarrow$) Assume $f_{X,Y}(x,y) = g(x)h(y)$, then

$$\begin{aligned} f_X(x) &= \int f(x,y)dy \\ &= \int g(x)h(y)dy \\ &= g(x)\int h(y)dy \\ &= c_1 g(x),\ c_1 = \int h(y)dy \end{aligned}$$

$$\begin{aligned} f_Y(y) &= \int f(x,y)dx \\ &= \int g(x)h(y)dx \\ &= h(y)\int g(x)dx \\ &= c_2 h(y),\ c_2 = \int g(x)dx \end{aligned}$$

$$\text{Also, } 1 = \int\int g(x)h(y)dxdy = c_1 c_2$$

$$\therefore f(x,y) = g(x)h(y) = c_1 g(x) c_2 h(y) = f_X(x) f_Y(y)$$

**Example 2.5.1.** $f_{X,Y}(x,y) = (x+y)I(0<x<1,\ 0<y<1)$: jpdf of $X$ and $Y$. Are $X$ and $Y$ indep.?

(sol) Note that we cannot express $f_{X,Y}(x,y)$ as a product pf $g(x)$ and $h(y)$. Hence, $X$ and $Y$ are not indep.

**Theorem 2.5.2.** $X$ and $Y$: indep. iff $F_{X,Y}(x,y) = F_X(x)F_Y(y)$.

(pf) ($\Rightarrow$)

$$\begin{aligned} F_{X,Y}(x,y) &= \int_{-\infty}^{x}\int_{-\infty}^{y} f(t,w)dwdt \\ &= \int_{-\infty}^{x}\int_{-\infty}^{y} f_X(t)f_Y(w)dwdt \\ &= \int_{-\infty}^{x} f_X(t)dt \int_{-\infty}^{y} f_Y(w)dw \\ &= F_X(x)F_Y(y) \end{aligned}$$

($\Leftarrow$)

$$\begin{aligned} f_{X,Y}(x,y) &= \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} \\ &= \frac{\partial^2 F_X(x) F_Y(y)}{\partial x \partial y} \\ &= f_X(x) f_Y(y) \end{aligned}$$

**Theorem 2.5.3.** $X$ and $Y$: indep. iff $P(a < X \leq b,\, c < Y \leq d) = P(a < X \leq b)P(c < Y \leq d)$.

(pf) ($\Rightarrow$)

$$\begin{aligned} P(a < X \leq b,\, c < Y \leq d) &= F(b,d) - F(a,d) - F(b,c) + F(a,c) \\ &= F_x(b)F_Y(d) - F_X(a)F_Y(d) - F_X(d)F_Y(c) + F_X(a)F_Y(c) \\ &= \{F_X(b) - F_X(a)\}\{F_Y(d) - F_Y(c)\} \\ &= P(a < X \leq b)P(c < Y \leq d) \end{aligned}$$

($\Leftarrow$) trivial

**Theorem 2.5.4.** $X$ and $Y$: indep. $\Rightarrow E[u(X)v(Y)] = E[u(X)]E[v(Y)]$.

(pf)

$$\begin{aligned} E[u(X)v(Y)] &= \int\int u(x)v(y) f_{X,Y}(x,y)\,dxdy \\ &= \int\int u(x)v(y) f_X(x) f_Y(y)\,dxdy \\ &= \int u(x) f_X(x)\,dx \int v(y) f_Y(y)\,dy \\ &= E[u(X)]E[v(Y)] \end{aligned}$$

**Theorem 2.5.5.** $X$ and $Y$: indep. iff $M(t_1,t_2) = M(t_1,0)M(0,t_2)$.

(pf) ($\Rightarrow$)

$$\begin{aligned} M(t_1,t_2) &= E\left[e^{t_1X+t_2Y}\right] \\ &= E\left[e^{t_1X}e^{t_2Y}\right] \\ &= E\left[e^{t_1X}\right]E\left[e^{t_2Y}\right] \\ &= M(t_1,0)M(0,t_2) \end{aligned}$$

($\Leftarrow$)

$$\begin{aligned} M(t_1,0)M(0,t_2) &= \int e^{t_1x}f_X(x)dx \int e^{t_2y}f_Y(y)dy \\ &= \int\int e^{t_1x+t_2y}f_X(x)f_Y(y)dxdy \\ &= \int\int e^{t_1x+t_2y}f_{X,Y}(x,y)dxdy \end{aligned}$$

By the uniqueness of mgf, we must have $f_{X,Y}(x,y) = f_X(x)f_Y(y)$.

## 2.6 Extension to Several Random Variables

**Definition 2.6.1.** $\boldsymbol{X} = (X_1, \cdots, X_n)'$: n-dim random vector, $X_i$'s: r.v.'s

- $F_{\boldsymbol{X}}(\boldsymbol{x}) = P(\boldsymbol{X} \le \boldsymbol{x}) = P(X_1 \le x_1, \cdots, X_n \le x_n)$: joint cdf.
- $Y = u(X_1, \cdots, X_n) \Rightarrow E(Y) = \int \cdots \int u(x_1, \cdots, x_n) f_{X_1,\cdots,X_n}(x_1, \cdots, x_n) dx_1 \cdots dx_n$

$$f_{X_1}(x_1) = \int \cdots \int f_{X_1,\cdots,X_n}(x_1, \cdots, x_n) dx_2 \cdots dx_n$$

$$f_{X_1,X_3}(x_1,x_3) = \int \cdots \int f_{X_1,\cdots,X_n}(x_1, \cdots, x_n) dx_2 dx_4 \cdots dx_n$$

$$f_{2,\cdots,n|1}(x_2, \cdots, x_n|x_1) = \frac{f_{X_1,\cdots,X_n}(x_1, \cdots, x_n)}{f_1(x_1)}$$

$$f_{1|2,\cdots,n}(x_1|x_2, \cdots, x_n) = \frac{f_{X_1,\cdots,X_n}(x_1, \cdots, x_n)}{f_{X_2,\cdots X_n}(x_2, \cdots, x_n)}$$

**Remark 2.6.1.** mutually indep. $\overset{\circ}{\underset{\times}{\rightleftharpoons}}$ pairwise indep.
(counter example)

$$f(x_1, x_2, x_3) = \frac{1}{4},\ (x_1, x_2, x_3) \in \{(1,0,0), (0,1,0), (0,0,1), (1,1,1)\}$$

$$f_{ij}(x_i, x_j) = \frac{1}{4},\ (x_i, x_j) \in \{(0,0), (1,0), (0,1), (1,1)\}$$

$$f_i(x_i) = \frac{1}{2},\ x_i = 0, 1$$

$$\therefore f_{ij}(x_i, x_j) = f_i(x_i) f_j(x_j) \text{ but } f(x_1, x_2, x_3) \neq f_1(x_1) f_2(x_2) f_3(x_3)$$

- $X_1, \cdots, X_n$ are called iid(independent and identically distributed) if $X_1, \cdots, X_n$ are mutually indep and have the same distribution.

- $E(\boldsymbol{X}) = (E(X_1), \cdots, E(X_n))'$
  $E(\boldsymbol{W}) = [E(W_{ij})]$, where $\boldsymbol{W}$ is $m \times n$ matrix of random variables.

**Theorem 2.6.1.** $\boldsymbol{W}_1$, $\boldsymbol{W}_2$: $m \times n$ matrices of r.v.'s.
$\boldsymbol{A}_1$, $\boldsymbol{A}_2$: $k \times m$ matrices of constants. $\boldsymbol{B}$: $n \times l$ matrix of constants. Then

$$E[\boldsymbol{A}_1 \boldsymbol{W}_1 + \boldsymbol{A}_2 \boldsymbol{W}_2] = \boldsymbol{A}_1 E[\boldsymbol{W}_1] + \boldsymbol{A}_2 E[\boldsymbol{W}_2]$$

$$E[\boldsymbol{A}_1 \boldsymbol{W}_1 \boldsymbol{B}] = \boldsymbol{A}_1 E[\boldsymbol{W}_1] \boldsymbol{B}$$

- $\boldsymbol{\mu} = E(\boldsymbol{X})$: mean of $\boldsymbol{X}$.

- $Cov(\boldsymbol{X}) = E[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})'] = E[\boldsymbol{X}\boldsymbol{X}'] - \boldsymbol{\mu}\boldsymbol{\mu}' = [\sigma_{ij}]$ : variance-covariance matrix.
  $Cov(\boldsymbol{A}\boldsymbol{X}) = \boldsymbol{A} Cov(\boldsymbol{X}) \boldsymbol{A}'$
  Variance-covariance matrix $Cov(\boldsymbol{X})$ is p.s.d. i.e. $\boldsymbol{a}' Cov(\boldsymbol{X}) \boldsymbol{a} \geq 0$. why?
  Let $Y = \boldsymbol{a}'\boldsymbol{X}$, then $0 \leq Var(Y) = Var(\boldsymbol{a}'\boldsymbol{X}) = \boldsymbol{a}' Cov(\boldsymbol{X}) \boldsymbol{a}$.

## 2.7 Transformation of Random Vectors

Consider transforming $n$ random variales $X_1, \cdots, X_n$ to $n$ random variables $Y_1, \cdots, Y_n$ s.t. $y_1 = u_1(x_1, \cdots, x_n), \cdots, y_n = u_n(x_1, \cdots, x_n)$.

(1) one-to-one transformation case

$S \to \mathscr{T}$ is 1-1 s.t. $x_1 = w_1(y_1, \cdots, y_n), \cdots, x_n = w_n(y_1, \cdots, y_n)$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \cdots & \frac{\partial x_1}{\partial y_n} \\ \vdots & & \vdots \\ \frac{\partial x_n}{\partial y_1} & \cdots & \frac{\partial x_n}{\partial y_n} \end{vmatrix}, \ f(x_1, \cdots, x_n) : \text{jpdf of } X_1, \cdots, X_n$$

Then, the jpdf of $Y_1, \cdots, Y_n$ is

$$g(y_1, \cdots, y_n) = |J| f(w_1(y_1, \cdots, y_n), \cdots, w_n(y_1, \cdots, y_n))$$

**Example 2.7.1.** $f(x_1, x_2, x_3) = 48 x_1 x_2 x_3 I(0 < x_1 < x_2 < x_3 < 1)$, jpdf of $Y_1 = X_1/X_2$, $Y_2 = X_2/X_3$, $Y_3 = X_3$

(sol) $x_1 = y_1 y_2 y_3$, $x_2 = y_2 y_3$, $x_3 = y_3$ (1-1 transf.)

$$J = \begin{vmatrix} y_2 y_3 & y_1 y_3 & y_1 y_2 \\ 0 & y_3 & y_2 \\ 0 & 0 & 1 \end{vmatrix} = y_2 y_3^2$$

$0 < y_1 < 1, 0 < y_2 < 1, 0 < y_3 < 1$

$\therefore g(y_1, y_2, y_3) = 48(y_1 y_2 y_3)(y_2 y_3)(y_3)|y_2 y_3^2| = 48 y_1 y_2^3 y_3^5$, $0 < y_i < 1$, $i = 1, 2, 3$

(2) many-to-one transformation case

$S \to \mathscr{T}$ is $k - 1$.

Let $A_1, \cdots, A_k$ be exhaustive sets s.t. $\bigcup_{i=1}^{k} A_i = S$ and $A_i \cap A_j = \phi$,

and $A_i \to \mathscr{T}$ is 1-1 for each $i = 1, \cdots, k$. Then, we apply the same method to each $A_i \to \mathscr{T}$. i.e.

$$g(y_1, \cdots, y_n) = \sum_{i=1}^{k} |J_i| g(w_{1i}(y_1, \cdots, y_n), \cdots, w_{ni}(y_1, \cdots, y_n))$$

**Example 2.7.2.** $f(x_1, x_2) = \frac{1}{\pi} I(0 < x_1^2 + x_2^2 < 1)$. Find the jpdf of $Y_1 = X_1^2 + X_2^2$, $Y_2 = \frac{X_1^2}{(X_1^2 + X_2^2)}$.

(sol) $y_1 y_2 = x_1^2$, $x_2^2 = y_1(1 - y_2)$, $0 < y_1 < 1$, $0 < y_2 < 1$, i.e. $x_1 = \pm\sqrt{y_1 y_2}$, $x_2 = \pm\sqrt{y_1(1 - y_2)}$.

$$A_1,\ x_1 = \sqrt{y_1 y_2},\ x_2 = \sqrt{y_1(1 - y_2)}$$

$$A_2,\ x_1 = -\sqrt{y_1 y_2},\ x_2 = \sqrt{y_1(1 - y_2)}$$

$$A_3,\ x_1 = -\sqrt{y_1 y_2},\ x_2 = -\sqrt{y_1(1 - y_2)}$$

$$A_1,\ x_1 = \sqrt{y_1 y_2},\ x_2 = -\sqrt{y_1(1 - y_2)}$$

$$J = \begin{vmatrix} \frac{1}{2}\sqrt{\frac{y_2}{y_1}} & \frac{1}{2}\sqrt{\frac{y_1}{y_2}} \\ \frac{1}{2}\sqrt{\frac{(1-y_2)}{y_1}} & -\frac{1}{2}\sqrt{\frac{y_1}{(1-y_2)}} \end{vmatrix} = -\frac{1}{4}\frac{1}{\sqrt{y_2(1 - y_2)}}$$

similarly, $J_2 = J_3 = J_4 = J_1$

$$\begin{aligned} \therefore g(y_1, y_2) &= \sum_{i=1}^{4} |J_i| f(w_{1i}(y_1, y_2), w_{2i}(y_1, y_2)) \\ &= \frac{4}{\pi}\frac{1}{4}\frac{1}{\sqrt{y_2(1 - y_2)}} \\ &= \frac{1}{\pi\sqrt{y_2(1 - y_2)}} I(0 < y_1 < 1,\ 0 < y_2 < 1) \end{aligned}$$

# 3 Some Special Distributions

## 3.1 The Binomial and Related Distributions

① binomial distribution

(i) binomial equation

$$(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} b^x a^{n-x}$$

(ii) Bernoulli trial

A seq. of experiment is called *Bernoulli trials* if each outcome is either success or failure, and each trial is indep. $X_1, \cdots, X_n$ are called Bernoulli r.v.'s if $X_1, \cdots, X_n$ are indep. and $P(X_i = 1) = p$ and $P(X_i = 0) = 1 - p$, $0 \leq p \leq 1$. We denote that $X_i \sim B(n, p)$. Note that

$$E(X_i) = \sum_{x_i=0}^{1} X_i f(x_i) = 0(1-p) + p = p$$

$$Var(X_i) = E(X_i^2) - E^2(X_i) = 0^2(1-p) + 1^2 \times p - p^2 = p - p^2 = p(1-p)$$

(iii) pmf of binomial distriution

Let $X_1, \cdots, X_n$ be bernoulli trials with prob. of success $p$. i.e. $X_i \sim B(1, p)$, $X_i$'s are indep. Then, $X = \sum_{i=1}^{n} X_i$ is the number of success out of $n$ trials, and $X$ is called to have *binomial distribution* with pmf

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x},\ x = 0, 1, \cdots, n$$

(iv) mgf

$$
\begin{aligned}
M(t) &= E[e^{tX}] \\
&= \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x (1-p)^{n-x} \\
&= [(1-p) + pe^t]^n
\end{aligned}
$$

(v) $\mu$ and $\sigma^2$

$$
\begin{aligned}
M'(t) &= n[(1-p) + pe^t]^{n-1} pe^t \\
M''(t) &= n(n-1)[(1-p) + pe^t]^{n-2} p^2 e^{2t} + n[(1-p) + pe^t]^{n-1} pe^t
\end{aligned}
$$

$\therefore \mu = M'(0) = np$, $\sigma^2 = M''(0) - M'(0)^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p)$

(vi) $X_i \sim B(n_i, p)$, $i = 1, \cdots, m$. $X_i$'s are indep. $\Rightarrow Y = \sum_{i=1}^{m} X_i \sim B\left(\sum_{i=1}^{m} n_i, p\right)$

(pf) Use the uniqueness of mgf, i.e. 1-1 correspondence between pdf and mgf

$$
\begin{aligned}
M_Y(t) &= E[e^{tY}] \\
&= E[e^{t\sum_{x=1}^{m} X_i}] \\
&= E[e^{tX_1 + \cdots + tX_m}] \\
&= E[e^{tX_1} \cdots e^{tX_m}] \\
&= \prod_{i=1}^{m} E[e^{tX_i}] \\
&= \prod_{i=1}^{m} M_{X_i}(t) \\
&= \prod_{i=1}^{m} [(1-p) + pe^t]^{n_i} \\
&= [(1-p) + pe^t]^{\sum_i^m n_i} : \text{mgf of } B\left(\sum_{i=1}^{m} n_i, p\right)
\end{aligned}
$$

**Example 3.1.1.** (*WLLN: Weak Law of Large Numbers*)
$Y \sim B(n,p) \Rightarrow P\left(\left|\frac{Y}{n} - p\right| \geq \varepsilon\right) \to 0$ as $n \to \infty$

(pf) Will use Tchebyshev's inequality

$$\begin{aligned} P\left(\left|\frac{Y}{n} - p\right| \geq \varepsilon\right) &= P(|Y - np| \geq n\varepsilon),\ E(Y) = np,\ Var(Y) = np(1-p) = \sigma^2 \\ &= P\left(|Y - \mu| \geq \varepsilon\sqrt{\frac{n}{p(1-p)}}\sqrt{np(1-p)}\right) \\ &= P\left(|Y - \mu| \geq \varepsilon\sqrt{\frac{n}{p(1-p)}}\sigma\right) \\ &= P(|Y - \mu| \geq k\sigma),\ k = \varepsilon\sqrt{\frac{n}{p(1-p)}} \\ &\leq \frac{1}{k^2} = \frac{p(1-p)}{n\varepsilon^2} \to 0 \text{ as } n \to \infty \end{aligned}$$

**Example 3.1.2.** $X_1, X_2, X_3$: indep. with the same pdf $f(x)$ and cdf $F(x)$. Find the pdf of $Y = mid(X_1, X_2, X_3)$.

(sol) First, find the cdf of $Y$

$$\begin{aligned} G(y) &= P(Y \leq y) \\ &= P(mid(X_1, X_2, X_3) \leq y) \\ &= P(\text{at least two of } X_1, X_2, X_3 \text{ are } \leq y) \end{aligned}$$

Let $\{X_i \leq y\}$ be success, and $Y$ be the number of successes out of 3.
i.e. $Y \sim B(3,p)$, $p = P(X_i \leq y) = F(y)$.
Now,

$$\begin{aligned} G(y) &= P(Y \geq 2) \\ &= P(Y = 2) + P(Y = 3) \\ &= \binom{3}{2}[F(y)]^2[1 - F(y)] + [F(y)]^3 \end{aligned}$$

$$\therefore g(y) = G'(y) = 6F(y)[1 - F(y)]f(y)$$

② negative binomial distribution

(i) definition

Consider a seq. of indep. Bernoulli trials $B(1,p)$. Let $Y$ be the number of failures before the $r$-th success, then $Y$ is called to have the *negative binomial distribution*. The pmf of $Y$ is

$$p(y) = \binom{y+r-1}{r-1} p^r (1-p)^y,\ y = 0,1,2,\cdots$$

and denoted by $Y \sim NB(r,p)$.

(ii) mgf

$$\begin{aligned} M_Y(t) &= E[e^{tY}] \\ &= \sum_{y=0}^{\infty} e^{ty} \binom{y+r-1}{r-1} p^r (1-p)^y \\ &= p^r \sum_{y=0}^{\infty} \binom{y+r-1}{r-1} [(1-p)e^t]^y \\ &= p^r \left\{ 1x^0 + rx + \frac{r(r+1)}{2}x^2 + \cdots \right\},\ (1-p)e^t = x \end{aligned}$$

Now, we consider

$$g(x) = (1-x)^{-r}$$

By using the Taylor expansion of $g(x)$ w.r.t. $x = 0$,

$$\begin{aligned} g(x) &= g(0) + (x-0)g'(0) + \frac{1}{2}(x-0)^2 g''(0) + \cdots \\ g(0) &= (1-0)^{-r} = 1 \\ g'(x) &= -r(1-x)^{-r-1}(-1) \Rightarrow g'(0) = r \\ g''(x) &= -r(-r-1)(1-x)^{-r-2} \Rightarrow g''(0) = r(r+1) \end{aligned}$$

$$\begin{aligned} \therefore g(x) &= (1-x)^{-r} \\ &= 1 + rx + \frac{r(r+1)}{2}x^2 + \cdots \end{aligned}$$

$$\therefore M_Y(t) = p^r(1-x)^{-r} = p^r[1-(1-p)e^t]^{-r}$$

(iii) $\mu$ and $\sigma^2$.

$$\begin{aligned} M'(t) &= p^r(-r)[1-(1-p)e^t]^{-r-1}[-(1-p)e^t] \\ M''(t) &= p^r[-r(r+1)][1-(1-p)e^t]^{-r-2}[-(1-p)^2e^{2t}] \\ &\quad +p^r(-r)[1-(1-p)e^t]^{-r-1}[-(1-p)e^t] \end{aligned}$$

Therefore,

$$\begin{aligned} \mu &= M'(0) \\ &= p^r r p^{-r-1}(1-p) \\ &= \frac{r(1-p)}{p} \end{aligned}$$

$$\begin{aligned} \sigma^2 &= M''(0) - M'(0)^2 \\ &= \frac{p^r r(r+1)p^{-r-2}(1-p)^2 + p^r p^{-r-1} r(1-p) - r^2(1-p)^2}{p^2} \\ &= \frac{r(1-p)}{p^2} \end{aligned}$$

(iv) geometric distribution

$Y \sim NB(1,p)$ is called the geometric distribution, i.e.

$$\begin{aligned} p(y) &= \binom{y+r-1}{r-1} p^r(1-p)^y,\ y = 0,1,2,\cdots \\ &= \binom{y}{0} p^1(1-p)^y,\ y = 0,1,2,\cdots \\ &= p(1-p)^y,\ y = 0,1,2,\cdots \end{aligned}$$

which is called *geometric distribution*.

③ trinomial distribution

(i) pmf

The jpdf of the random vector $(X, Y)$ is

$$f(x,y) = \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y p_3^{n-x-y}$$

which is called *trinomial* pmf and denoted by $(X, Y) \sim T(n, p_1, p_2)$.

(ii) mgf

$$\begin{aligned}
M_{X,Y}(t_1,t_2) &= E[e^{t_1X+t_2Y}] \\
&= \sum_{x=0}^{n}\sum_{y=0}^{n-x} e^{t_1x+t_2y}\frac{n!}{x!y!(n-x-y)!}p_1^x p_2^y p_3^{n-x-y} \\
&= \sum_{x=0}^{n} e^{t_1x}\frac{n!}{x!(n-x)!}p_1^x\left\{\sum_{y=0}^{n-x}\frac{(n-x)!e^{t_2y}}{y!(n-x-y)!}p_2^y p_3^{n-x-y}\right\} \\
&= \sum_{x=0}^{n}\binom{n}{x}(p_1e^{t_1})^x\sum_{y=0}^{n-x}\binom{n-x}{y}(p_2e^{t_2})^y p_3^{n-x-y} \\
&= \sum_{x=0}^{n}\binom{n}{x}(p_1e^{t_1})^x(p_2e^{t_2}+p_3)^{n-x} \\
&= (p_1e^{t_1}+p_2e^{t_2}+p_3)^n
\end{aligned}$$

(iii) marginal pmf

The marginal pmf of $X$ is

$$\begin{aligned} f_X(x) &= \sum_{y=0}^{n-x} f_{X,Y}(x,y) \\ &= \sum_{y=0}^{n-x} \frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y (1-p_1-p_2)^{n-x-y} \\ &= \frac{n!}{x!(n-x)!} p_1^x \sum_{y=0}^{n-x} \frac{(n-x)!}{y!(n-x-y)!} p_2^y (1-p_1-p_2)^{n-x-y} \\ &= \frac{n!}{x!(n-x)!} p_1^x (1-p_1)^{n-x} : \text{pmf of } B(n,p_1) \end{aligned}$$

i.e. $X \sim B(n, p_1)$.
Similarly, the marginal pmf of $Y$ is $B(n, p_2)$.

(iv) conditional pmf

The conditional pmf of $Y$ given $X$ is

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f_{X,Y}(x,y)}{f_X(x)} \\ &= \frac{\frac{n!}{x!y!(n-x-y)!} p_1^x p_2^y (1-p_1-p_2)^{n-x-y}}{\frac{n!}{x!(n-x)!} p_1^x (1-p_1)^y} \\ &= \frac{(n-x)!}{y!(n-x-y)!} \frac{p_2^y (1-p_1-p_2)^{n-x-y}}{(1-p_1)^{n-x-y+y}} \\ &= \binom{n-x}{y} \left(\frac{p_2}{1-p_1}\right)^y \left(\frac{1-p_1-p_2}{1-p_1}\right)^{n-x-y} \\ &= \binom{n-x}{y} \left(\frac{p_2}{1-p_1}\right)^y \left(1-\frac{p_2}{1-p_1}\right)^{n-x-y} \sim B\left(n-x, \frac{p_2}{1-p_1}\right) \end{aligned}$$

i.e. conditional pmf of $Y$ given $X = x$ is $B\left(n-x, \frac{p_2}{1-p_1}\right)$.
Can easily show $X|Y \sim B\left(n-y, \frac{p_1}{1-p_2}\right)$.

④multinomial distribution

(i) pmf

The pmf of random vector $\boldsymbol{X} = (X_1, \cdots, X_{k-1})$ is

$$f(x_1, \cdots, x_{k-1}) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

where

$x_k = n - x_1 - \cdots - x_{k-1}$, $p_k = 1 - p_1 - \cdots - k_{k-1}$, $0 \leq x_1 + \cdots + x_{k-1} \leq n$

and denoted by $\boldsymbol{X} \sim \mathcal{M}(n, p_1, \cdots, p_{k-1})$.

(ii) mgf

$$M(t_1, \cdots, t_{k-1}) = (p_1 e^{t_1} + p_2 e^{t_2} + \cdots + p_{k-1} e^{t_{k-1}} + p_k)^n$$

(iii) Each one-variable marginal pmf is binomial, each two-variables marginal pmf is trinomial, and so on.

## 3.2 The Poisson Distribution

① pmf

motivation: Consier a Taylor expansion of $g(m) = e^m$ about $m = 0$, i.e.

$$\begin{aligned} g(m) &= g(0) + \frac{g'(0)}{1!}(m-0)' + \frac{g''(0)}{2!}(m-0)^2 + \cdots \\ &= 1 + m + \frac{m^2}{2!} + \cdots \\ &= \sum_{x=0}^{\infty} \frac{m^x}{x!} = e^m \end{aligned}$$

A r.v. is said to have *Poisson distriution* with parameter $m$ if its pmf is given by

$$p(x) = \frac{e^{-m} m^x}{x!},\ x = 0, 1, 2, \cdots$$

and it is denoted by $X \sim \mathcal{P}(m)$.

② mgf

$$\begin{aligned} M(t) &= E[e^{tX}] \\ &= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-m} m^x}{x!} \\ &= e^{-m} \sum_{x=0}^{\infty} \frac{(me^t)^x}{x!} \\ &= e^{-m} e^{me^t} = exp[m(e^t - 1)] \end{aligned}$$

$$\begin{aligned} M'(t) &= me^t exp[m(e^t - 1)] \\ M''(t) &= me^t e^{m(e^t-1)} + me^t me^t e^{m(e^t-1)} \end{aligned}$$

$$\begin{aligned} \mu &= M'(0) = me^0 e^{m(e^0-1)} = m \\ \sigma^2 &= M''(0) - M'(0)^2 = (m + m^2) - m^2 = m \end{aligned}$$

③ property

$X_i \sim \mathcal{P}(m_i)$, $X_i$'s are indep. $\Rightarrow Y = \sum_{i=1}^{n} X_i \sim \mathcal{P}\left(\sum_{i=1}^{n} m_i\right)$.

(pf)

$$\begin{aligned} M_Y(t) &= E[e^{tY}] \\ &= E[e^{t\sum X_i}] \\ &= E[e^{tX_1} e^{tX_2} \cdots e^{tX_n}] \\ &= \prod_{i=1}^{n} E[e^{tX_i}] \\ &= \prod_{i=1}^{n} M_{X_i}(t) \\ &= \prod_{i=1}^{n} exp[m_i(e^t - 1)] \\ &= exp\left[\sum m_i(e^t - 1)\right] \end{aligned}$$

## 3.3 The $\Gamma$, $\chi^2$, and $\beta$ Distributions

① gamma distribution

(i) *gamma function*

$$\Gamma(\alpha) = \int_0^\infty y^{\alpha-1}e^{-y}dy,\ \alpha > 0$$

(ii) properties of gamma function

a. For $\alpha > 1$, $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$

(pf)

$$\begin{aligned}\Gamma(\alpha) &= \int_0^\infty y^{\alpha-1}e^{-y}dy \\ &= [y^{\alpha-1}(-e^{-y})]_0^\infty - \int_0^\infty (\alpha-1)y^{\alpha-2}(-e^{-y})dy \\ &= 0 + (\alpha-1)\int_0^\infty y^{\alpha-2}e^{-y}dy \\ &= (\alpha-1)\Gamma(\alpha-1)\end{aligned}$$

b. If $\alpha$ is positive integer, then $\Gamma(\alpha) = (\alpha-1)!$

(pf)

$$\begin{aligned}\Gamma(\alpha) &= (\alpha-1)\Gamma(\alpha-1) \\ &\vdots \\ &= (\alpha-1)(\alpha-2)\cdots 1\Gamma(1)\end{aligned}$$

Now,

$$\Gamma(1) = \int_0^\infty y^{1-1}e^{-y}dy = 1$$
$$\therefore \Gamma(1) = (\alpha-1)!$$

c. $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

(pf)

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty y^{\frac{1}{2}} e^{-y} dy$$

Let $y = \frac{x^2}{2}$, $x > 0$, then

$$\begin{aligned}
\Gamma\left(\frac{1}{2}\right) &= \int_0^\infty \left(\frac{x^2}{2}\right)^{-\frac{1}{2}} e^{-\frac{x^2}{2}} x dx \\
&= \sqrt{2} \int_0^\infty e^{-\frac{x^2}{2}} dx \\
&= \sqrt{2} \frac{\sqrt{2\pi}}{2} \\
&= \sqrt{\pi}
\end{aligned}$$

For example, $\Gamma\left(\frac{7}{2}\right) = \frac{5}{2}\Gamma\left(\frac{5}{2}\right) = \frac{5}{2}\frac{3}{2}\frac{1}{2}\sqrt{\pi} = \frac{15}{8}\sqrt{\pi}$

(iii) pdf

The continuous r.v. $X$ is called to have the *gamma distribution* with parameters $\alpha > 0$ and $\beta > 0$ if its pdf is given by

$$f(x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha} I(x > 0)$$

and denoted by $X \sim \Gamma(\alpha, \beta)$.

(idea) By letting $y = \dfrac{x}{\beta}$ in $\Gamma(\alpha)$, we have

$$\Gamma(\alpha) = \int_0^\infty \left(\frac{x}{\beta}\right)^{\alpha-1} e^{-x/\beta} \frac{1}{\beta} dx$$

$$\therefore 1 = \int_0^\infty \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha} dx$$

(iv) mgf

$$\begin{aligned}
M(t) &= E[e^{tX}] \\
&= \int_0^\infty e^{tx} \frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha} dx \\
&= \int_0^\infty \frac{x^{\alpha-1}e^{-x\left(-t+\frac{1}{\beta}\right)}}{\Gamma(\alpha)\beta^\alpha} dx \\
&= \int_0^\infty \frac{\Gamma(\alpha)\left(\frac{\beta}{1-\beta t}\right)^\alpha}{\Gamma(\alpha)\beta^\alpha} \frac{x^{\alpha-1}e^{-x/\left(\frac{\beta}{1-\beta t}\right)}}{\Gamma(\alpha)\left(\frac{\beta}{1-\beta t}\right)^\alpha} dx \\
&= \frac{\frac{\beta^\alpha}{(1-\beta t)^\alpha}}{\beta^\alpha} \\
&= (1-\beta t)^{-\alpha}
\end{aligned}$$

$$\begin{aligned}
M'(t) &= (-\alpha)(1-\beta t)^{-\alpha-1}(-\beta) = \alpha\beta(1-\beta t)^{-\alpha-1} \\
M''(t) &= \alpha\beta(-\alpha-1)(1-\beta t)^{-\alpha-2}(-\beta) = \alpha\beta^2(1-\beta t)^{-\alpha-2}(\alpha+1)
\end{aligned}$$

$$\begin{aligned}
\mu &= M'(0) = \alpha\beta \\
\sigma^2 &= M''(0) - M'(0)^2 = \alpha^2\beta^2 + \alpha\beta^2 - (\alpha\beta)^2 = \alpha\beta^2
\end{aligned}$$

(v) sum of indep. gamma

$X_i \sim \Gamma(\alpha_i, \beta)$, $X_i$'s are indep. $\Rightarrow Y = \sum_{i=1}^n X_i \sim \Gamma\left(\sum \alpha_i, \beta\right)$

(pf)

$$\begin{aligned}
M_Y(t) &= E[e^{t\sum X_i}] \\
&= E[e^{tX_1} \cdots e^{tX_n}] \\
&= \prod_{i=1}^n E[e^{tX_i}] \\
&= \prod (1-\beta t)^{-\alpha_i} \\
&= (1-\beta t)^{-\sum \alpha_i}
\end{aligned}$$

(vi) relationship with Poisson distribution

$W$: time needed to obtain $k$ changes(or deaths)

$$G(w) = P(W \leq w) = 1 - P(W > w)$$

Now, $\{W > w\}$ is equivalent to "less than $k$ changes in an interval of length $w$". i.e.

$$P(W > w) = \sum_{x=0}^{k-1} P(X = x) = \sum_{x=0}^{k-1} \frac{(\lambda w)^x e^{-\lambda w}}{x!},$$

where, $X$: number of changes in an interval of length $w$. Now, it can be shown

$$\sum_{x=0}^{k-1} \frac{(\lambda w)^x e^{-\lambda w}}{x!} = \int_{\lambda w}^{\infty} \frac{z^{k-1} e^{-z}}{(k-1)!} dz$$

$$\therefore G(w) = \int_0^{\lambda w} \frac{z^{k-1} e^{-z}}{\Gamma(k)} dz.$$

Let $z = \lambda y$, then

$$G(w) = \int_0^w \frac{\lambda^k y^{k-1} e^{-\lambda y}}{\Gamma(k)} dy \Rightarrow g(w) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{\Gamma(k)} \sim \Gamma\left(k, \frac{1}{\lambda}\right)$$

②$\chi^2$-distribution

(i) definition

If $X \sim \Gamma\left(\frac{r}{2}, 2\right)$, then $X$ is called to have *chi-square distribution* with d.f. $r$, and denoted by $X \sim \chi^2(r)$.

(ii) pdf

$$f(x) = \frac{x^{\frac{r}{2}-1} e^{-\frac{x}{2}}}{\Gamma\left(\frac{r}{2}\right) 2^{\frac{r}{2}}},\ x > 0$$

(iii) mgf

$$M(t) = (1-2t)^{-r/2}$$

(iv) $\mu$ and $\sigma^2$.

$$\mu = r,\ \sigma^2 = 2r$$

(v) property
$X_i \sim \chi^2(r_i)$, $i = 1, \cdots, n$. $X_i$'s: indep. $\Rightarrow Y = \sum X_i \sim \chi^2(\sum r_i)$.

③ beta distribution

(i) pdf

A r.v. $X$ is said to have *beta distribution* with parameters $\alpha$ and $\beta$ if its pdf is given by

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1},\ 0 < x < 1$$

and denoted by $X \sim Beta(\alpha, \beta)$.

(ii) mgf

$$M_X(t) = E[e^{tX}] = \int_0^1 e^{tx} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} dx$$

In fact, $M_X(t)$ does not have a closed(analytic) form.
Hence, to compute mean and variance, use the definition of expectation, i.e.

$$\begin{aligned} E(X) &= \int_0^1 x \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} dx \\ &= \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+1+\beta)} \frac{\Gamma(\alpha+1+\beta)}{\Gamma(\alpha+1)\Gamma(\beta)} x^{\alpha+1-1}(1-x)^{\beta-1} dx \\ &= \frac{\Gamma(\alpha+1)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+1)} \\ &= \frac{\alpha}{\alpha+\beta} \end{aligned}$$

Similarly, can easily show

$$Var(X) = E(X^2) - E^2(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

(iii) derivation from gamma distribution

$X_1 \sim \Gamma(\alpha, 1)$, $X_2 \sim \Gamma(\beta, 1)$, $X_1$ and $X_2$ are indep. $\Rightarrow \frac{X_1}{X_1+X_2} \sim Beta(\alpha, \beta)$.
(pf) Let

$$Y_1 = X_1 + X_2,\ Y_2 = \frac{X_1}{X_1+X_2}$$

then it is 1-1 transformation, and the inverse function is

$$x_1 = y_1y_2,\ x_2 = y_1 - y_1y_2 = y_1(1-y_2)$$

Also,

$$S = \{(x_1, x_2) : 0 < x_1 < \infty, 0 < x_2 < \infty\}$$

and

$$\mathscr{T} = \{(y_1, y_2) : 0 < y_1 < \infty, 0 < y_2 < 1\}$$

$$J = \begin{vmatrix} y_2 & y_1 \\ 1-y_2 & -y_1 \end{vmatrix} = -y_1y_2 - y_1(1-y_2) = -y_1$$

Therefore, the jpdf pf $(Y_1, Y_2)$ is

$$g(y_1, y_2) = f(y_1y_2, y_1(1-y_2))|J|$$

where

$$\begin{aligned} f(x_1, x_2) &= \frac{x_1^{\alpha-1}e^{-x_1/1}}{\Gamma(\alpha)1^\alpha}\frac{x_2^{\beta-1}e^{-x_2/1}}{\Gamma(\beta)1^\beta} \\ &= \frac{x_1^{\alpha-1}x_2^{\beta-1}e^{-x_1-x_2}}{\Gamma(\alpha)\Gamma(\beta)} \end{aligned}$$

$$\begin{aligned} \therefore g(y_1, y_2) &= \frac{(y_1y_2)^{\alpha-1}\{y_1(1-y_2)\}^{\beta-1}e^{-y_1y_2-y_1(1-y_2)}}{\Gamma(\alpha)\Gamma(\beta)}|-y_1| \\ &= \frac{y^{\alpha+\beta-2+1}y_2^{\alpha-1}(1-y_2)^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)}e^{-y_1} \end{aligned}$$

Finally, the pdf of $Y_2 = \dfrac{X_1}{X_1 + X_2}$ is

$$\begin{aligned} g_2(y_2) &= \int g(y_1, y_2) dy_1 \\ &= \int_0^\infty \frac{y_2^{\alpha-1}(1-y_2)^{\beta-1}}{\Gamma(\alpha)\Gamma(\beta)} \Gamma(\alpha+\beta) \frac{1}{\Gamma(\alpha+\beta)} y_1^{\alpha+\beta-1} e^{-y_1} dy_1 \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} y_2^{\alpha-1}(1-y_2)^{\beta-1} I(0 < y_2 < 1) \\ &\sim Beta(\alpha, \beta) \end{aligned}$$

④ *Dirichlet distribution*

$X_i \sim \Gamma(\alpha_i, 1)$, $X_i$'s are indep. $i = 1, 2, \cdots, k+1$.
$Y_i = \dfrac{X_i}{(X_1 + \cdots + X_{k+1})}$, $i = 1, \cdots, k$ and $Y_{k+1} = X_1 + \cdots + X_{k+1}$.
Then, the jpf of $Y_1, \cdots, Y_k$ is called the Dirichlet distribution with pdf

$$g(y_1, \cdots, y_k) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_{k+1})}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_{k+1})} y_1^{\alpha_1 - 1} y_2^{\alpha_2 - 1} \cdots y_k^{\alpha_k - 1} (1 - y_1 - \cdots - y_k)^{\alpha_{k+1} - 1}$$

(pf) cf. If $k = 1$, the Dirichlet distribution reduces to the Beta distriution.
i.e., the Dirichlet distriution is a multivariate extension of Beta distribution.

$$y_1 = \frac{x_1}{\sum_{i=1}^{k+1} x_i}, \; y_2 = \frac{x_2}{\sum_{i=1}^{k+1} x_i}, \cdots, y_k = \frac{x_k}{\sum_{i=1}^{k+1} x_i}, \; y_{k+1} = \sum_{i=1}^{k+1} x_i$$

which is 1-1 from $(x_1, \cdots, x_{k+1})$ to $(y_1, \cdots, y_{k+1})$.
Inverse functions are

$$x_1 = y_1 y_{k+1}, \; x_2 = y_2 y_{k+1}, \cdots, x_k = y_k y_{k+1}, \; x_{k+1} = y_{k+1}(1 - y_1 - y_2 - \cdots - y_k)$$

$$S = \{(x_1, \cdots, x_{k+1}) : 0 < x_i < \infty, \, i = 1, \cdots, k+1\}$$
$$\mathscr{T} = \{(y_1, \cdots, y_{k+1}) : 0 < y_i < 1, \, i = 1, \cdots, k; \, 0 < y_{k+1} < \infty\}$$

$$J = \begin{vmatrix} y_{k+1} & 0 & 0 & \cdots & 0 & y_1 \\ 0 & y_{k+1} & 0 & \cdots & 0 & y_2 \\ 0 & 0 & y_{k+1} & \cdots & 0 & y_3 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & y_{k+1} & y_k \\ -y_{k+1} & -y_{k+1} & -y_{k+1} & \cdots & -y_{k+1} & 1 - \sum_{i=1}^{k} y_i \end{vmatrix} = y_{k+1}^k$$

Also, the jpdf of $(X_1, \cdots, X_{k+1})$ is

$$\begin{aligned} f(x_1, \cdots, x_{k+1}) &= \frac{x_1^{\alpha_1 - 1} e^{-x_1/1}}{\Gamma(\alpha_1) 1^{\alpha_1}} \cdots \frac{x_{k+1}^{\alpha_{k+1} - 1} e^{-x_{k+1}/1}}{\Gamma(\alpha_{k+1}) 1^{\alpha_{k+1}}} \\ &= \frac{x_1^{\alpha_1 - 1} \cdots x_{k+1}^{\alpha_{k+1} - 1} e^{-(x_1 + \cdots + x_{k+1})}}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_{k+1})} \end{aligned}$$

Hence, the jpdf of $(Y_1, \cdots, Y_{k+1})$ is

$$\begin{aligned} g(y_1, \cdots, y_{k+1}) &= f(y_1 y_{k+1}, \cdots, y_k y_{k+1}, y_{k+1}(1 - y_1 - \cdots - y_k)) |y_{k+1}^k| \\ &= \frac{(y_1 y_{k+1})^{\alpha_1 - 1} \cdots (y_k y_{k+1})^{\alpha_k - 1} \{y_{k+1}(1 - y_1 - \cdots - y_k)\}^{\alpha_{k+1} - 1}}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_{k+1})} e^{-y_{k+1}} y_{k+1}^k \end{aligned}$$

Finally, the jpdf of $(Y_1, \cdots, Y_k)$ is

$$\begin{aligned} g(y_1, \cdots, y_k) &= \int_0^\infty g(y_1, \cdots, y_{k+1}) dy_{k+1} \\ &= \frac{\Gamma(\alpha_1 + \cdots + \alpha_{k+1})}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_{k+1})} y_1^{\alpha_1 - 1} \cdots y_k^{\alpha_k - 1} (1 - y_1 - \cdots y_k)^{\alpha_{k+1} - 1} \end{aligned}$$

## 3.4 The Normal Distribution

① derivation

First, want to compute

$$I = \int_{-\infty}^{\infty} e^{-y^2/2} dy$$

consider

$$\left( \sum_{i=1}^{n} a_i \right)^2 = \left( \sum_{i=1}^{n} a_i \right) \left( \sum_{j=1}^{n} a_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j$$

Similarly, consider

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-y^2/2}dy\right)^2 = \left(\int_{-\infty}^{\infty} e^{-y^2/2}dy\right)\left(\int_{-\infty}^{\infty} e^{-z^2/2}dz\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(y^2+z^2)/2}dydz$$

use polar coordinate system, i.e. $y = rcos\theta, z = rsin\theta$

$\{(y,z) : -\infty < y < \infty, -\infty < z < \infty\} \rightarrow \{(r,\theta) : 0 < r < \infty, 0 < \theta < 2\pi\}$

1-1 correspondence between $(y,z)$ and $(r,\theta)$

$$J = \begin{vmatrix} \frac{dy}{dr} & \frac{dy}{d\theta} \\ \frac{dz}{dr} & \frac{dz}{d\theta} \end{vmatrix} = \begin{vmatrix} cos\theta & -rsin\theta \\ sin\theta & rcos\theta \end{vmatrix} = rcos^2\theta - (-r)sin^2\theta = r$$

Therefore,

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(y^2+z^2)/2}dydz \\ &= \int\int e^{-r^2/2}rdrd\theta \\ &= \int_0^{2\pi}\left[-e^{-r^2/2}\right]_0^{\infty}d\theta \\ &= \int_0^{2\pi} 1d\theta \\ &= 2\pi \end{aligned}$$

$$\therefore I = \sqrt{2\pi}$$

i.e.

$$1 = \int \frac{1}{\sqrt{2\pi}}e^{-y^2/2}dy$$

Also, let

$$y = \frac{x-\mu}{\sigma},$$

then

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}e^{-\left(\frac{x-\mu}{\sigma}\right)^2/2}\frac{1}{\sigma}dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma}exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]dx \end{aligned}$$

② pdf

The continuous r.v. $X$ is said to have *normal distribution* with mean $\mu$ and variance $\sigma^2$ if its pdf is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \infty < x < \infty$$

and denoted by $X \sim N(\mu, \sigma^2)$.

As a special case, if $\mu = 0$ and $\sigma^2 = 1$, then it is called standard normal(Gaussian) distriution with mean 0, variance 1

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

③ mgf

$$\begin{aligned}
M(t) &= E[e^{tX}] \\
&= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[\frac{1}{2\sigma^2}\{-2\sigma^2 tx + x^2 - 2\mu x + \mu^2\}\right] dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[\frac{1}{2\sigma^2}\{x^2 - 2(\sigma^2 t + \mu)x + (\sigma^2 t + \mu)^2 - (\sigma^2 t + \mu)^2 + \mu^2\}\right] dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[\frac{1}{2\sigma^2}\{(x - (\sigma^2 t + \mu))^2 - \sigma^4 t^2 - 2\sigma^2 t\mu\}\right] dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{\{x - (\sigma^2 t + \mu)\}^2}{2\sigma^2} + \left(t\mu + \frac{\sigma^2 t^2}{2}\right)\right] dx \\
&= exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} exp\left[-\frac{\{x - (\sigma^2 t + \mu)\}^2}{2\sigma^2}\right] dx \\
&= exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)
\end{aligned}$$

$$
\begin{aligned}
M'(t) &= (\mu + \sigma^2 t) exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right) \\
M''(t) &= \sigma^2 exp\left[\mu t + \frac{\sigma^2 t^2}{2}\right] + (\mu + \sigma^2 t^2) exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)
\end{aligned}
$$

$$
\begin{aligned}
\mu &= M'(0) = \mu \\
\sigma^2 &= M''(0) - M'(0)^2 = \sigma^2 + \mu^2 - \mu^2 = \sigma^2
\end{aligned}
$$

④ higher order moments

Let $X \sim N(\mu, \sigma^2)$, then $Z = \dfrac{X-\mu}{\sigma^2} \sim N(0,1)$ and $M_Z(t) = e^{t^2/2}$.
Using the mgf of $Z$, we can get $E(X^k)$, $k = 1, 2, \cdots$
Recall that

$$
M_Z(t) = 1 + E(Z)t + \frac{E(Z^2)}{2!}t^2 + \frac{E(Z^3)}{3!}t^3 + \cdots
$$

also,

$$
\begin{aligned}
e^{t^2/2} &= 1 + \frac{t^2}{2} + \frac{1}{2!}\left(\frac{t^2}{2}\right)^2 + \frac{1}{3}\left(\frac{t^2}{2}\right)^3 + \cdots \\
&= 1 + \frac{t^2}{2!} + \frac{3 \times 1}{4!}t^4 + \cdots + \frac{(2k-1)\cdots(3)(1)}{(2k)!}t^{2k} + \cdots
\end{aligned}
$$

Therefore,

$$
\begin{cases}
E(Z^{2k}) = (2k-1)\cdots(3)(1) = \dfrac{(2k)!}{2^k k!} \\
E(Z^{2k+1}) = 0
\end{cases}
$$

Now,

$$
\begin{aligned}
E(X^k) &= E[(\mu + \sigma z)^k] \\
&= E\left[\sum_{j=0}^{k}\binom{k}{j}(\sigma z)^j \mu^{k-j}\right] : \text{binomial eq.} \\
&= \sum_{j=0}^{k}\binom{k}{j}\sigma^j E(Z^j)\mu^{k-j}
\end{aligned}
$$

⑤ properties

(i) $Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2(1)$

(pf) Let $V = Z^2$, then the cdf of $V$ is

$$\begin{aligned} F(v) &= P(V \le v) \\ &= P(Z^2 \le v) \\ &= P(-\sqrt{v} \le Z \le \sqrt{v}),\ v > 0 \\ &= 2P(0 \le Z \le \sqrt{v}) \\ &= 2\int_0^{\sqrt{v}} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz \end{aligned}$$

Let $y = z^2$, then $dy = 2zdz$, i.e. $dz = \dfrac{1}{2\sqrt{y}} dy$

$$\therefore F(v) = 2\int_o^v \frac{1}{\sqrt{2\pi}} e^{-y/2} \frac{1}{2\sqrt{y}} dy$$

Therefore, the pdf of $V$ is

$$\begin{aligned} f(v) &= F'(v) \\ &= \frac{1}{\sqrt{2\pi}} e^{-v/2} \frac{1}{\sqrt{v}} \\ &= \frac{v^{-1/2} e^{-v/2}}{\sqrt{\pi}\sqrt{2}} \\ &= \frac{v^{\frac{1}{2}-1} e^{-v/2}}{\Gamma\left(\frac{1}{2}\right) 2^{1/2}} : \text{pdf of } \chi^2(1) \end{aligned}$$

(ii) $X_i \sim N(\mu_i, \sigma_i^2)$, $X_i$'s are indep. $\Rightarrow Y = \sum_{i=1}^n a_i X_i \sim N\left(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2\right)$

(pf)

$$\begin{aligned}
M_Y(t) &= E[e^{tY}] \\
&= E[exp(t\sum a_i X_i)] \\
&= E[e^{ta_1X_1}e^{ta_2X_2}\cdots e^{ta_nX_n}] \\
&= \prod_{i=1}^{n} E[e^{ta_iX_i}] \\
&= \prod_{i=1}^{n} M_{X_i}(ta_i) \\
&= \prod_{i=1}^{n} exp\left(\mu t a_i + \frac{1}{2}\sigma_i^2 t^2 a_i^2\right) \\
&= exp\left[t\sum a_i\mu_i + \frac{1}{2}t^2\sum a_i^2\sigma_i^2\right] : \text{mgf of } N\left(\sum a_i\mu_i, \sum a_i^2\sigma_i^2\right)
\end{aligned}$$

⑥ contaminated normals

$Z \sim N(0,1)$, $I_\varepsilon \sim B(1, 1-\varepsilon)$, $Z$ and $I_\varepsilon$ are indep.
Want to find the pdf of

$$W = I_\varepsilon Z + (1 - I_\varepsilon)\sigma_c Z$$

$$\begin{aligned}
F_W(w) &= P(W \le w) \\
&= P(W \le w, I_\varepsilon = 1) + P(W \le w, I_\varepsilon = 0) \\
&= P(W \le w | I_\varepsilon = 1)P(I_\varepsilon = 1) + P(W \le w | I_\varepsilon = 0)P(I_\varepsilon = 0) \\
&= (1-\varepsilon)P(Z \le w) + \varepsilon P(Z \le w/\sigma_c) \\
&= (1-\varepsilon)\Phi(w) + \varepsilon\Phi\left(\frac{w}{\sigma_c}\right)
\end{aligned}$$

$$\therefore f_W(w) = (1-\varepsilon)\phi(w) + \frac{\varepsilon}{\sigma_c}\phi\left(\frac{w}{\sigma_c}\right)$$

## 3.5 The Multivariate Normal Distribution

① derivation

(i) standard normal case

$\boldsymbol{Z} = (Z_1, \cdots, Z_n)'$, $Z_i \sim N(0,1)$, $Z_i$'s are indep.
Then, the pdf of $\boldsymbol{Z}$ is

$$\begin{aligned} f(\boldsymbol{z}) &= \prod_{i=1}^{n} f(Z_i) \\ &= \prod_{i=1}^{n} (2\pi)^{-1/2} exp\left[-\frac{1}{2} z_i^2\right] \\ &= (2\pi)^{-n/2} exp\left(-\frac{1}{2}\sum_{i=1}^{n} z_i^2\right) \\ &= (2\pi)^{-n/2} exp\left(-\frac{1}{2}\boldsymbol{z'z}\right) \end{aligned}$$

which is called the standard multivariate normal distribution with $E(\boldsymbol{Z}) = \boldsymbol{0}$, $Cov(\boldsymbol{Z}) = I_n$ and denoted by $\boldsymbol{Z} \sim N_n(\boldsymbol{0}, I_n)$.
Now, the mgf of $\boldsymbol{Z}$ is

$$\begin{aligned} M_{\boldsymbol{Z}}(\boldsymbol{t}) &= E[e^{\boldsymbol{t'Z}}] \\ &= E[e^{t_1 Z_1} \cdots e^{t_n Z_n}] \\ &= \prod_{i=1}^{n} E[e^{t_i Z_i}] \\ &= \prod_{i=1}^{n} M_{Z_i}(t_i) \\ &= \prod_{i=1}^{n} exp\left[0 t_i + \frac{1}{2} 1^2 t_i^2\right] \\ &= \prod_{i=1}^{n} exp\left(\frac{1}{2} t_i^2\right) \\ &= exp\left(\frac{1}{2}\boldsymbol{t't}\right) \end{aligned}$$

(ii) spectral decomposition

**Theorem 3.5.1.** spectral decomposition theorem
Let $A$ be $n \times n$ symmetric matrix, then $\exists$ an orthogonal matrix $\Gamma$ s.t. $A = \Gamma'\Lambda\Gamma$, then $\Lambda = diag(\lambda_1, \cdots, \lambda_n)$ and $\lambda_i$'s are eigenvalues of $A$ and corresponding eigenvectors are column vectors of $\Gamma$.

(iii) general normal case

Let $\Sigma$ be $n \times n$ symmetric and positive definite matrix. By spectral decomposition,

$$\Sigma = \Gamma'\Lambda\Gamma$$

where $\lambda = diag(\lambda_1, \cdots, \lambda_n)$ s.t. $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n > 0$.
Then, $\exists\, \Lambda^{1/2} = diag(\sqrt{\lambda_1}, \cdots, \sqrt{\lambda_n})$ and

$$\begin{aligned} \Sigma &= \Gamma'\Lambda\Gamma \\ &= \Gamma'\Lambda^{1/2}\Lambda^{1/2}\Gamma \\ &= \Gamma'\Lambda^{1/2}\Gamma\Gamma'\Lambda^{1/2}\Gamma \\ &= \Sigma^{1/2}\Sigma^{1/2} \end{aligned}$$

where $\Sigma^{1/2} = \Gamma'\Lambda^{1/2}\Gamma$

$$\begin{aligned} \therefore (\Sigma^{1/2})^{-1} &= (\Gamma'\Lambda^{1/2}\Gamma)^{-1} \\ &= (\Gamma)^{-1}(\Lambda^{1/2})^{-1}(\Gamma')^{-1} \\ &= \Gamma'\Lambda^{-1/2}\Gamma \end{aligned}$$

Let $\boldsymbol{Z} \sim N_n(\boldsymbol{0}, I_n)$ and let

$$\boldsymbol{X} = \Sigma^{1/2}\boldsymbol{Z} + \boldsymbol{\mu}$$

i.e. $\Sigma^{1/2}\boldsymbol{Z} = \boldsymbol{X} - \boldsymbol{\mu} \Rightarrow \boldsymbol{Z} = \Sigma^{-1/2}(\boldsymbol{X} - \boldsymbol{\mu})$

$$J = \left|\frac{dz}{dx}\right| = |\Sigma^{-1/2}| = |\Sigma|^{-1/2}$$

$$g(z) = (2\pi)^{-n/2} exp\left[-\frac{1}{2}\boldsymbol{z'z}\right] : \text{pdf of } N_n(\boldsymbol{0}, I_n)$$

Hence, the pdf of $\boldsymbol{X}$ is

$$\begin{aligned} f(\boldsymbol{x}) &= g(\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu}))|J| \\ &= (2\pi)^{-n/2} exp\left[-\frac{1}{2}\{\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\}'\{\Sigma^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})\}\right]|\Sigma|^{-1/2} \\ &= (2\pi)^{-n/2}|\Sigma|^{-1/2} exp\left[-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu})'\Sigma^{-1}(\boldsymbol{X}-\boldsymbol{\mu})\right] \end{aligned}$$

which is called the multivariate normal pdf with mean $\mu$ and variance-covariance matrix $\Sigma$, and denoted by $\boldsymbol{X} \sim N_n(\boldsymbol{\mu}, \Sigma)$.
Also, the mgf of $\boldsymbol{X}$ is

$$\begin{aligned} M_{\boldsymbol{X}}(\boldsymbol{t}) &= E[e^{\boldsymbol{t'X}}] \\ &= E[exp\{\boldsymbol{t}'(\Sigma^{1/2}\boldsymbol{z}+\boldsymbol{\mu})\}] \\ &= E[exp(\boldsymbol{t}'\Sigma^{1/2}\boldsymbol{z}+\boldsymbol{t'\mu})] \\ &= exp(\boldsymbol{t'\mu})E[exp\{(\Sigma^{1/2}\boldsymbol{t})'\boldsymbol{z}\}] \\ &= exp(\boldsymbol{t'\mu})M_{\boldsymbol{Z}}(\Sigma^{1/2}t) \\ &= exp(\boldsymbol{t'\mu})exp\left[\frac{1}{2}(\Sigma^{1/2}\boldsymbol{t})'(\Sigma^{1/2}\boldsymbol{t})\right] \\ &= exp\left[\boldsymbol{t'\mu}+\frac{1}{2}\boldsymbol{t}'\Sigma\boldsymbol{t}\right] \end{aligned}$$

② bivariate normal distribution

$$\boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

$$\begin{aligned} f(x,y) &= (2\pi)^{-1}|\Sigma|^{-1/2} exp\left[-\frac{1}{2}(\boldsymbol{X}-\boldsymbol{\mu})'\Sigma^{-1}(\boldsymbol{X}-\boldsymbol{\mu})\right] \\ &= \frac{1}{2\pi\sigma_2\sigma_2\sqrt{1-\rho^2}} exp[-\frac{1}{2(1-\rho^2)}\{\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) \\ &\quad + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\}] \end{aligned}$$

③ linear transformation

$\boldsymbol{X} \sim N_n(\boldsymbol{\mu}, \Sigma) \Rightarrow \boldsymbol{Y} = A\boldsymbol{X} + \boldsymbol{b} \sim N_m(A\boldsymbol{\mu} + \boldsymbol{b}, A\Sigma A')$

(pf) First, recall that

$$M_{\boldsymbol{X}}(\boldsymbol{t}) = exp\left[\boldsymbol{t'}\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t'}\Sigma\boldsymbol{t}\right]$$

Next, compute mgf of $\boldsymbol{Y}$

$$\begin{aligned} M_{\boldsymbol{Y}}(\boldsymbol{t}) &= E[e^{\boldsymbol{t'}\boldsymbol{Y}}] \\ &= E[exp\{\boldsymbol{t}'(A\boldsymbol{X} + \boldsymbol{b})\}] \\ &= E[exp\{(A'\boldsymbol{t})'\boldsymbol{X}\}e^{\boldsymbol{t'}\boldsymbol{b}}] \\ &= e^{\boldsymbol{t'}\boldsymbol{b}}exp\left[(A'\boldsymbol{t})'\boldsymbol{\mu} + \frac{1}{2}(A'\boldsymbol{t})'\Sigma(A'\boldsymbol{t})\right] \\ &= exp\left(\boldsymbol{t'}\boldsymbol{b} + \boldsymbol{t}'A\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}'A\Sigma A'\boldsymbol{t}\right) \\ &= exp\left(\boldsymbol{t'}(A\boldsymbol{\mu} + \boldsymbol{b}) + \frac{1}{2}\boldsymbol{t}'A\Sigma A'\boldsymbol{t}\right) \end{aligned}$$

which is mgf of $N_m(A\boldsymbol{\mu} + \boldsymbol{b}, A\Sigma A')$

④ marginal and conditional distribution

Assume that

$$\boldsymbol{X} \sim N_n(\boldsymbol{\mu}, \Sigma)$$

Decompose $\boldsymbol{X}$ s.t.

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \end{pmatrix}, \boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

(i) marginal distribution

Let $A = (I_m : O)$, then

$$A\boldsymbol{X} = (I_m : O) \begin{pmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \end{pmatrix} = \boldsymbol{X}_1$$

i.e. $\boldsymbol{X}_1$ is linear transformation of $\boldsymbol{X}$

$$\begin{aligned} E(\boldsymbol{X}_1) &= E(A\boldsymbol{X}) \\ &= A\boldsymbol{\mu} \\ &= (I_m : O) \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix} \\ &= \boldsymbol{\mu}_1 \end{aligned}$$

$$\begin{aligned} Cov(\boldsymbol{X}_1) &= Cov(A\boldsymbol{X}) \\ &= ACov(\boldsymbol{X})A' \\ &= (I_m : O) \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I_m \\ O \end{pmatrix} \\ &= \Sigma_{11} \end{aligned}$$

$$\therefore \boldsymbol{X}_1 \sim N_n(\boldsymbol{\mu}_1, \Sigma_{11})$$

Similarly,

$$\boldsymbol{X}_2 \sim N_n(\boldsymbol{\mu}_2, \Sigma_{22})$$

(ii) independence of $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$

$\boldsymbol{X} \sim N_m(\boldsymbol{\mu}, \Sigma)$, $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are indep. iff $\Sigma_{12} = O$.

(pf) The joint mgf of $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ is

$$\begin{aligned} M_{\boldsymbol{X}_1,\boldsymbol{X}_2}(\boldsymbol{t}_1,\boldsymbol{t}_2) &= E[exp(\boldsymbol{t}_1'\boldsymbol{X}_1 + \boldsymbol{t}_2'\boldsymbol{X}_2)] \\ &= E[e^{\boldsymbol{t}'\boldsymbol{X}}] \\ &= exp\left(\boldsymbol{t}'\boldsymbol{\mu} + \frac{1}{2}\boldsymbol{t}'\Sigma\boldsymbol{t}\right) \\ &= exp\left[(\boldsymbol{t}_1'\boldsymbol{t}_2')\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix} + \frac{1}{2}(\boldsymbol{t}_1'\boldsymbol{t}_2')\begin{pmatrix}\Sigma_{11} & \Sigma_{12}\\ \Sigma_{21} & \Sigma_{22}\end{pmatrix}\begin{pmatrix}\boldsymbol{t}_1\\ \boldsymbol{t}_2\end{pmatrix}\right] \\ &= exp\left[\boldsymbol{t}_1'\boldsymbol{\mu}_1 + \boldsymbol{t}_2'\boldsymbol{\mu}_2 + \frac{1}{2}\boldsymbol{t}_1'\Sigma_{11}\boldsymbol{t}_1 + \frac{1}{2}\boldsymbol{t}_1'\Sigma_{12}\boldsymbol{t}_2 + \frac{1}{2}\boldsymbol{t}_2'\Sigma_{21}\boldsymbol{t}_1 + \frac{1}{2}\boldsymbol{t}_2'\Sigma_{22}\boldsymbol{t}_2\right] \end{aligned}$$

Now,

$$M_{\boldsymbol{X}_1}(\boldsymbol{t}_1)M_{\boldsymbol{X}_2}(\boldsymbol{t}_2) = exp\left[\boldsymbol{t}_1'\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{t}_1'\Sigma_{11}\boldsymbol{t}_1\right] exp\left[\boldsymbol{t}_2'\boldsymbol{\mu}_2 + \frac{1}{2}\boldsymbol{t}_2'\Sigma_{22}\boldsymbol{t}_2\right]$$

Hence, $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are indep. iff $M_{\boldsymbol{X}_1,\boldsymbol{X}_2}(\boldsymbol{t}_1,\boldsymbol{t}_2) = M_{\boldsymbol{X}_1}(\boldsymbol{t}_1)M_{\boldsymbol{X}_2}(\boldsymbol{t}_2)$
iff $\Sigma_{12} = O$.

(iii) conditional distribution of $\boldsymbol{X}_1$ given $\boldsymbol{X}_2$

$\boldsymbol{X}_1|\boldsymbol{X}_2 \sim N_m(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\boldsymbol{X}_2 - \boldsymbol{\mu}), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$

(pf) Let

$$\boldsymbol{W} = \boldsymbol{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{X}_2$$

and consider jpdf of $\boldsymbol{W}$ and $\boldsymbol{X}_2$

$$\begin{pmatrix}\boldsymbol{W}\\ \boldsymbol{X}_2\end{pmatrix} = \begin{pmatrix}I_m & -\Sigma_{12}\Sigma_{22}^{-1}\\ O & I_(n-m)\end{pmatrix}\begin{pmatrix}\boldsymbol{X}_1\\ \boldsymbol{X}_2\end{pmatrix} = A\boldsymbol{X}$$

i.e.

$$\begin{pmatrix}\boldsymbol{W}\\ \boldsymbol{X}_2\end{pmatrix} = A\boldsymbol{X} \sim N_n(A\mu, A\Sigma A')$$

where

$$\begin{aligned} A\mu &= \begin{pmatrix}I & -\Sigma_{12}\Sigma_{22}^{-1}\\ O & I\end{pmatrix}\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix} \\ &= \begin{pmatrix}\boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{\mu}_2\\ \boldsymbol{\mu}_2\end{pmatrix} \end{aligned}$$

$$
\begin{aligned}
A\Sigma A' &= \begin{pmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \\ O & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} I & O \\ -\Sigma_{22}^{-1}\Sigma_{21} & I \end{pmatrix} \\
&= \begin{pmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} & O \\ O & \Sigma_{22} \end{pmatrix}
\end{aligned}
$$

Hence, $\boldsymbol{W}$ and $\boldsymbol{X}_2$ are indep.
Therefore,

$$
\boldsymbol{W} \sim \boldsymbol{W}|\boldsymbol{X}_2 \sim N_m(\boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{\mu}_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})
$$

Now,

$$
\begin{aligned}
\boldsymbol{X}_1|\boldsymbol{X}_2 &\sim \boldsymbol{W} + \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{X}_2 \\
&\sim N_m(\boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{\mu}_2 + \Sigma_{12}\Sigma_{22}^{-1}\boldsymbol{X}_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma 21) \\
&\sim N_m(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\boldsymbol{X}_2 - \boldsymbol{\mu}), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})
\end{aligned}
$$

⑤ relationship with $\chi^2$-distribution

$$
\boldsymbol{X} \sim N_n(\boldsymbol{\mu}, \Sigma) \;\Rightarrow\; \boldsymbol{W} = (\boldsymbol{X} - \boldsymbol{\mu})'\Sigma^{-1}(\boldsymbol{X} - \boldsymbol{\mu}) \sim \chi^2(n)
$$

(pf)

$$
\begin{aligned}
\boldsymbol{Z} &= \Sigma^{-1/2}(\boldsymbol{X} - \boldsymbol{\mu}) \sim N_n(\boldsymbol{0}, I_n) \\
\boldsymbol{W} &= \boldsymbol{Z}'\boldsymbol{Z} = \sum z_i^2 \sim \chi^2(n)
\end{aligned}
$$

## 3.6 $t$- and $F$-Distributions

① $t$-distribution

(i) definition

$W \sim N(0,1)$, $V \sim \chi^2(r)$, $W$ and $V$ are indep. Define

$$
T = \frac{W}{\sqrt{V/r}}
$$

then the r.v. $T$ is called to have a *t-distribution* with degree of freedom $r$, and denoted by $T \sim t(r)$.

(ii) derivation of pdf

Consider a transformation

$$(w, v) \rightarrow (t, u)$$

where

$$t = \frac{w}{\sqrt{v/r}},\ u = v$$

then it is 1-1 transformation with inverse function

$$w = t\sqrt{u}/\sqrt{r},\ v = u$$

and

$$J = \begin{vmatrix} \frac{dw}{dt} & \frac{dw}{du} \\ \frac{dv}{dt} & \frac{dv}{du} \end{vmatrix} = \begin{vmatrix} \sqrt{u}/\sqrt{r} & t/2\sqrt{ur} \\ 0 & 1 \end{vmatrix} = \sqrt{u}/\sqrt{r}$$

Now, the joint pdf of $(w, v)$ is

$$f(w, v) = \frac{1}{\sqrt{2\pi}} e^{-w^2/2} \frac{v^{\frac{r}{2}-1} e^{-v/2}}{\Gamma\left(\frac{r}{2}\right) 2^{r/2}} = \frac{e^{-w^2/2} v^{\frac{r}{2}-1} e^{-v/2}}{\sqrt{2\pi}\Gamma\left(\frac{r}{2}\right) 2^{r/2}}$$

Then, the joint pdf of $(t, u)$ is

$$\begin{aligned} g(t, u) &= f\left(\frac{t\sqrt{u}}{\sqrt{r}}, u\right) |J| \\ &= \frac{1}{\sqrt{2\pi}\Gamma\left(\frac{r}{2}\right) 2^{r/2}} exp\left(-\frac{t^2 u}{2r}\right) u^{\frac{r}{2}-1} e^{-u/2} \frac{u^{1/2}}{r^{1/2}} \\ &= \frac{1}{\sqrt{2\pi}\Gamma\left(\frac{r}{2}\right) 2^{r/2}} u^{\frac{r+1}{2}-1} exp\left\{-\frac{1}{2}\left(1 + \frac{t^2}{r}\right) u\right\} \frac{1}{r^{1/2}} \end{aligned}$$

Hence, the pdf of $t$ is

$$\begin{aligned} g(t) &= \int_0^\infty g(t, u) du \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi}\Gamma\left(\frac{r}{2}\right) 2^{r/2}\sqrt{r}} \Gamma\left(\frac{r+1}{2}\right) \left(\frac{2}{1 + \frac{t^2}{r}}\right)^{(r+1)/2} \frac{u^{\frac{r+1}{2}-1} e^{-u/\left(\frac{2}{1+\frac{t^2}{r}}\right)}}{\Gamma\left(\frac{r+1}{2}\right) \left(\frac{2}{1+\frac{t^2}{r}}\right)^{(r+1)/2}} du \\ &= \frac{\Gamma\left(\frac{r+1}{2}\right)}{\sqrt{\pi r}\Gamma\left(\frac{r}{2}\right)} \left(1 + \frac{t^2}{r}\right)^{-(r+1)/2}, -\infty < t < \infty \end{aligned}$$

(iii) $\mu$ and $\sigma^2$

$$\begin{aligned} E(T) &= E\left[\frac{W}{\sqrt{V/r}}\right] \\ &= E\left[W\frac{1}{\sqrt{V/r}}\right] \\ &= E(W)E\left(\frac{1}{\sqrt{V/r}}\right) \\ &= 0 \end{aligned}$$

$$\begin{aligned} Var(T) &= E(T^2) - E^2(T) \\ &= E(T^2) \\ &= E\left[\frac{W^2}{V/r}\right] \\ &= E(W^2)E\left(\frac{r}{V}\right) \end{aligned}$$

Now, $E(W^2) = Var(W) + E^2(W) = 1 + 0^2 = 1$

$$\begin{aligned} E\left(\frac{r}{V}\right) &= rE(V^{-1}) \\ &= r\int_0^\infty v^{-1}\frac{v^{\frac{r}{2}-1}e^{-v/2}}{\Gamma\left(\frac{r}{2}\right)2^{r/2}}dv \\ &= r\int_0^\infty v^{-1}\frac{\Gamma\left(\frac{r-2}{2}\right)2^{\frac{r-2}{2}}}{\Gamma\left(\frac{r}{2}\right)2^{r/2}}\frac{v^{\frac{r-2}{2}-1}e^{-v/2}}{\Gamma\left(\frac{r-2}{2}\right)2^{\frac{r-2}{2}}}dv \\ &= r\frac{\Gamma\left(\frac{r}{2}-1\right)}{\Gamma\left(\frac{r}{2}\right)2} \\ &= r\frac{\Gamma\left(\frac{r}{2}-1\right)}{\left(\frac{r}{2}-1\right)\Gamma\left(\frac{r}{2}-1\right)} \\ &= \frac{r}{r-2} \end{aligned}$$

② $F$-distribution

(i) definition

$U \sim \chi^2(r_1)$, $V \sim \chi^2(r_2)$, $U$ and $V$ are indep. Then

$$W = \frac{U/r_1}{V/r_2}$$

is called to have *F-distribution* with degrees of freedom $r_1$ and $r_2$, and denoted by $W \sim F(r_1, r_2)$.

(ii) derivation of pdf

Consider a transformation

$$(u, v) \rightarrow (w, z)$$

where

$$w = \frac{u/r_1}{v/r_2},\ z = v$$

then it is 1-1 transformation with inverse function

$$u = \frac{r_1}{r_2} wz,\ v = z$$

and

$$J = \begin{vmatrix} \frac{du}{dw} & \frac{du}{dz} \\ \frac{dv}{dw} & \frac{dv}{dz} \end{vmatrix} = \begin{vmatrix} \frac{r_1}{r_2} z & \frac{r_1}{r_2} w \\ 0 & 1 \end{vmatrix} = \frac{r_1}{r_2} z$$

Now, the joint pdf of $(U, V)$ is

$$\begin{aligned} f(u, v) &= f_U(u) f_V(v) : U \text{ and } V \text{ are indep.} \\ &= \frac{u^{\frac{r_1}{2}-1} e^{-u/2}}{\Gamma\left(\frac{r_1}{2}\right) 2^{r_1/2}} \frac{v^{\frac{r_1}{2}-1} e^{-v/2}}{\Gamma\left(\frac{r_2}{2}\right) 2^{r_2/2}} \\ &= \frac{u^{\frac{r_1}{2}-1} v^{\frac{r_1}{2}-1} e^{-(u+v)/2}}{\Gamma\left(\frac{r_1}{2}\right) \Gamma\left(\frac{r_2}{2}\right) 2^{(r_1+r_2)/2}} \end{aligned}$$

Then, the joint pdf of $(W, Z)$ is

$$\begin{aligned} g(w,z) &= f\left(\frac{r_1}{r_2}wz, z\right)|J| \\ &= \frac{\left(\frac{r_1}{r_2}wz\right)^{\frac{r_1}{2}-1} z^{\frac{r_2}{2}-1} e^{-\left(\frac{r_1}{r_2}wz+z\right)/2}}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)2^{(r_1+r_2)/2}}\frac{r_1}{r_2}z \\ &= \frac{\left(\frac{r_1}{r_2}\right)^{\frac{r_1}{2}-1+1} w^{\frac{r_1}{2}-1} z^{\frac{r_1}{2}-1+\frac{r_2}{2}-1+1} e^{-z\left(\frac{r_1}{r_2}w+1\right)/2}}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)2^{(r_1+r_2)/2}} \\ &= \frac{\left(\frac{r_1}{r_2}\right)^{\frac{r_1}{2}} w^{\frac{r_1}{2}-1} z^{\frac{r_1+r_2}{2}-1}}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)2^{(r_1+r_2)/2}} exp\left[-z/\left(\frac{2}{\frac{r_1}{r_2}w+1}\right)\right] \end{aligned}$$

Hence, the pdf of $W$ is

$$\begin{aligned} g(w) &= \int_0^\infty g(w,z)dz \\ &= \frac{\left(\frac{r_1}{r_2}\right)^{\frac{r_1}{2}} w^{\frac{r_1}{2}-1}}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)2^{(r_1+r_2)/2}} \int_0^\infty \Gamma\left(\frac{r_1+r_2}{2}\right)\left(\frac{2}{\frac{r_1}{r_2}w+1}\right)^{(r_1+r_2)/2} \\ &\quad \frac{z^{\frac{r_1+r_2}{2}-1} e^{-z/\left(\frac{2}{\frac{r_1}{r_2}w+1}\right)}}{\Gamma\left(\frac{r_1+r_2}{2}\right)\left(\frac{2}{\frac{r_1}{r_2}w+1}\right)^{(r_1+r_2)/2}}dz \\ &= \frac{\Gamma\left(\frac{r_1+r_2}{2}\right)\left(\frac{r_1}{r_2}\right)^{\frac{r_1}{2}} w^{\frac{r_1}{2}-1}}{\Gamma\left(\frac{r_1}{2}\right)\Gamma\left(\frac{r_2}{2}\right)\left(\frac{r_1}{r_2}w+1\right)^{(r_1+r_2)/2}} \end{aligned}$$

(iii) $\mu$ and $\sigma^2$

$$F = \frac{U/r_1}{V/r_2} \sim F(r_1, r_2)$$

$$
\begin{aligned}
E(F) &= \frac{r_2}{r_1} E\left(\frac{U}{V}\right) \\
&= \frac{r_2}{r_1} E(U) E\left(\frac{1}{V}\right) \\
&= \frac{r_2}{r_1} E(U) E(V^{-1}) \\
&= \frac{r_2}{r_1} r_1 \frac{1}{r_2 - 2} \\
&= \frac{r_2}{r_2 - 2}
\end{aligned}
$$

$$
\begin{aligned}
Var(F) &= E(F^2) - E^2(F) \\
&= \frac{r_2^2}{r_1^2} E\left(\frac{U^2}{V^2}\right) - \left\{\frac{r_2}{r_1} E(U) E(V^{-1})\right\}^2 \\
&= \frac{r_2^2}{r_1^2} E(U^2) E(V^{-2}) - \left\{\frac{r_2}{r_1} E(U) E(V^{-1})\right\}^2 \\
&= r_2^2 \frac{2(r_1 + r_2 - 2)}{r_1 (r_2 - 2)^2 (r_2 - 4)}
\end{aligned}
$$

In general,

$$
\begin{aligned}
E(V^{-k}) &= \int_0^\infty v^{-k} \frac{v^{\frac{r_2}{2} - 1} e^{-v/2}}{\Gamma\left(\frac{r_2}{2}\right) 2^{r_2/2}} dv \\
&= \int_0^\infty \frac{\Gamma\left(\frac{r_2}{2} - k\right) 2^{\frac{r_2}{2} - k}}{\Gamma\left(\frac{r_2}{2}\right) 2^{r_2/2}} \frac{v^{\left(\frac{r_2}{2} - k\right) - 1} e^{-v/2}}{\Gamma\left(\frac{r_2}{2} - k\right) 2^{\frac{r_2}{2} - k}} dv \\
&= \frac{\Gamma\left(\frac{r_2}{2} - k\right)}{\Gamma\left(\frac{r_2}{2}\right) 2^k}, \; k = 1, 2, 3, \cdots
\end{aligned}
$$

③ Student's theorem

**Theorem 3.6.1.** $X_1, \cdots, X_n$: iid $N(\mu, \sigma^2)$, $\bar{X} = \frac{1}{n} \sum X_i$, $s^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ Then

(a) $\bar{X} \sim N(\mu, \sigma^2/n)$

(b) $\bar{X}$ and $s$ are indep.

(c) $\dfrac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$

(d) $\dfrac{\bar{X}-\mu}{s/\sqrt{n}} \sim t(n-1)$

(pf)

(a) Let

$$\boldsymbol{X} = (X_1, \cdots, X_n)', \mathbf{1} = (1, \cdots, 1)'$$

Now, let

$$\boldsymbol{a} = \frac{1}{n}\mathbf{1} = \left(\frac{1}{n}, \cdots, \frac{1}{n}\right)$$

then

$$\boldsymbol{a'X} \sim N(\boldsymbol{a}'(\mu\mathbf{1}), \boldsymbol{a}'Cov(\boldsymbol{X})\boldsymbol{a})$$

because

$$\boldsymbol{X} \sim N_n(\mu\mathbf{1}, \sigma^2 I)$$

i.e.

$$\bar{X} = \boldsymbol{a'X} \sim N(\mu, \sigma^2/n)$$

(b) Let

$$\boldsymbol{Y} = (X_1 - \bar{X}, \cdots, X_n - \bar{X})'$$

Consider

$$\boldsymbol{W} = \begin{pmatrix} \bar{X} \\ \boldsymbol{Y} \end{pmatrix} = \begin{pmatrix} \frac{1}{n}\mathbf{1} \\ I - \frac{1}{n}\mathbf{11'} \end{pmatrix} \boldsymbol{X} = A\boldsymbol{X}$$

First, will show $\bar{X}$ and $\boldsymbol{Y}$ are indep.
Recall that if both $\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are normally distributed then,

$$Cov(\boldsymbol{X_1}, \boldsymbol{X_2}) = 0$$

implies that $\boldsymbol{X_1}$ and $\boldsymbol{X_2}$ are indep.
Since $\bar{X}$ and $\boldsymbol{Y}$ are normal to show independence between $\bar{X}$ and $\boldsymbol{Y}$ we need to show $Cov(\bar{X}, \boldsymbol{Y}) = \mathbf{0}$

$$\begin{aligned}
Cov(\boldsymbol{W}) &= Cov\begin{pmatrix} \bar{X} \\ \boldsymbol{Y} \end{pmatrix} \\
&= \begin{pmatrix} Var(\bar{X}) & Cov(\bar{X}, \boldsymbol{Y}) \\ Cov(\boldsymbol{Y}, \bar{X}) & Cov(\boldsymbol{Y}) \end{pmatrix} \\
&= Cov(A\boldsymbol{X}) \\
&= ACov(\boldsymbol{X})A' \\
&= \begin{pmatrix} \frac{1}{n}\mathbf{1} \\ I - \frac{1}{n}\mathbf{11'} \end{pmatrix} \boldsymbol{X}\sigma^2 I \begin{pmatrix} \frac{1}{n}\mathbf{1} & I - \frac{1}{n}\mathbf{11'} \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} \frac{1}{n^2}\mathbf{1'1} & \frac{1}{n}\mathbf{1}'\left(I - \frac{1}{n}\mathbf{11'}\right) \\ \left(I - \frac{1}{n}\mathbf{11'}\right)\frac{1}{n}\mathbf{1} & \left(I - \frac{1}{n}\mathbf{11'}\right)^2 \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} \frac{1}{n} & \frac{1}{n}\left(\mathbf{1}' - \frac{1}{n}\mathbf{1'11'}\right) \\ \frac{1}{n}\left(\mathbf{1}' - \frac{1}{n}\mathbf{11'1}\right) & I - \frac{1}{n}\mathbf{11'} \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} \frac{1}{n} & \mathbf{0} \\ \mathbf{0} & I - \frac{1}{n}\mathbf{11'} \end{pmatrix}
\end{aligned}$$

Therefore,

$$Cov(\bar{X}, \boldsymbol{Y}) = \mathbf{0}$$

i.e. $\bar{X}$ and $\boldsymbol{Y}$ are indep.
Finally, note that

$$s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2 = \frac{1}{n-1}\boldsymbol{Y'Y},\ \boldsymbol{Y} = (X_1 - \bar{X}, \cdots, X_n - \bar{X})^2$$

i.e. $s^2$ is function of $\boldsymbol{Y}$. Therefore, $\bar{X}$ and $\boldsymbol{Y}$ are indep.

(c) Note that

$$\begin{aligned}\sum_{i=1}^{n}(X_i-\mu)^2 &= \sum_{i=1}^{n}(X_i-\bar{X}+\bar{X}-\mu)^2\\ &= \sum_{i=1}^{n}(X_i-\bar{X})^2+2\sum_{i=1}^{n}(X_i-\bar{X})(\bar{X}-\mu)+n(\bar{X}-\mu)^2\\ &= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\mu)^2\end{aligned}$$

Therefore,

$$\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2=\frac{\sum(X_i-\bar{X})^2}{\sigma^2}+\frac{n(\bar{X}-\mu)^2}{\sigma^2}$$

Now,

$$\frac{X_i-\mu}{\sigma}\sim N(0,1)\ \Rightarrow\ \left(\frac{X_i-\mu}{\sigma}\right)^2\sim\chi^2(1)\ \Rightarrow\ \sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2\sim\chi^2(n)$$

$$\frac{n(\bar{X}-\mu)}{\sigma^2}=\left(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right)^2\sim\chi^2(1)$$

Apply mgf technique on both sides, i.e.

$$\begin{aligned}M_A(t) &= E[e^{tA}]\\ &= E[e^{t(B+C)}]\\ &= E[e^{tB}e^{tC}]\\ &= E[e^{tB}]E[e^{tC}]\end{aligned}$$

$$(1-2t)^{-n/2}=E[e^{tB}](1-2t)^{-1/2}$$

$\therefore E(t^{tB})=(1-2t)^{-n/2}(1-2t)^{-1/2}=(1-2t)^{-(n-1)/2}$ : mgf of $\chi^2(n-1)$

(d)

$$\frac{\bar{X}-\mu}{s/\sqrt{n}}=\frac{\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2}{\sigma^2}/(n-1)}}=\frac{Z}{\sqrt{V/(n-1)}}\sim t(n-1)$$

where, $Z\sim N(0,1)$, $V\sim\chi^2(n-1)$, $Z$ and $V$ are indep.

# 4 Unbiasedness, Consistency and Limiting Distribution

## 4.1 Expectation of Functions

① definitions

(i) $X_1, X_2, \cdots, X_n$ are called *random sample* (r.s.) if they are iid.

(ii) T is called a *statistic* if T is a fuction of random sample only.

(iii) $\bar{X} = \sum \frac{X_i}{n}$ : *sample mean*, $\quad s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ : *sample variance*

② expectations

$\boldsymbol{x} = (X_1, \cdots, X_n)', \quad \boldsymbol{y} = (Y_1, \cdots, Y_n)'$ : ramdom vector

$\boldsymbol{a} = (a_1, \cdots, a_n)', \quad \boldsymbol{b} = (b_1, \cdots, b_n)'$ : constant vector

Let $T = \boldsymbol{a}'\boldsymbol{x}, \quad W = \boldsymbol{b}'\boldsymbol{y}$ be statistics

(i) $E(T) = \boldsymbol{a}'E(\boldsymbol{x})$

(ii) $Var(T) = \boldsymbol{a}'Cov(\boldsymbol{x})\boldsymbol{a}$

(iii) $Cov(T, W) = Cov(\boldsymbol{a}'\boldsymbol{x}, \boldsymbol{b}'\boldsymbol{y}) = \boldsymbol{a}'Cov(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{b}$

③ unbiasedness

$X_1, \cdots, X_n$ : random sample from $f(\boldsymbol{x} : \theta)$, $\quad \theta \in \Omega$ is parameter

$T = T(X_1, \cdots, X_n)$ : statistic, $\quad$ T is called *unbiased* if $E(T) = \theta$, $\quad {}^{\forall}\theta \in \Omega$.

## 4.2 Convergence in Probability

① definitions

**Definition 4.2.1.** Let $\{X_n\}$ be a seq.of r.v.'s and $X$ be a r.v.
We say $X_n$ *converges in probability* to $X$ if $\forall \varepsilon > 0, \quad P(|X_n - X| > \varepsilon) \to 0$ as $n \to \infty$, and denoted by $X_n \xrightarrow{P} X$.

② properties

(i) WLLN

**Theorem 4.2.1.** $\{X_n\}$ : seq. of *iid* r.v.'s with mean $\mu$, variance $\sigma^2 < \infty$.
Then, $\bar{X_n} \xrightarrow{P} \mu$.

(pf) Can be shown easily by Chebyshev's ineq.

(ii)

**Theorem 4.2.2.** $X_n \xrightarrow{P} X, \quad Y_n \xrightarrow{P} Y \;\Rightarrow\; X_n + Y_n \xrightarrow{P} X + Y$

(pf) By triangular ineq.
we have $|X_n - X| + |Y_n - Y| \geq |(X_n + Y_n) - (X + Y)|$

$$\begin{aligned} \therefore P(|(X_n + Y_n) - (X + Y) \geq \varepsilon) &\leq P(\{|X_n - X| + |Y_n - Y|\} \geq \varepsilon) \\ &\leq P(|(X_n - X) \geq \varepsilon/2) + P(|Y_n - Y| \geq \varepsilon/2) \end{aligned}$$

(iii)

**Theorem 4.2.3.** $X_n \xrightarrow{P} X, \quad , a$ : const. $\Rightarrow aX_n \xrightarrow{P} aX$

(pf) $P(|aX_n - aX| \geq \varepsilon) = P(|a||X_n - X| \geq \varepsilon) = P(|X_n - X| \geq \varepsilon/|a|)$

(iv)

**Theorem 4.2.4.** $X_n \xrightarrow{P} a$, $g$ : conti. function at $a \Rightarrow g(X_n) \xrightarrow{P} g(a)$

(pf) Let $\varepsilon > 0$. Since g is conti. at a, $\exists\, \delta > 0$ s.t. if $|x-a| < \delta$, then $|g(x) - g(a)| < \varepsilon$. Thus, $|g(x) - g(a)| \geq \varepsilon$ implies $|x-a| > \delta$. Therefore, $P(|g(X_n) - g(a)| \geq \varepsilon) \leq P(|X_n - a| \geq \delta) \to 0$

(v) (Remark)

**Theorem 4.2.5.** $X_n \xrightarrow{P} X$, $g$ : conti. $\Rightarrow g(X_n) \xrightarrow{P} g(X)$

③ consistency

(i) definition : a statistic $T_n$ is called *consistent* est. of $\theta$ if $T_n \xrightarrow{P} \theta$.

(ii)

**Example 4.2.1.** $X_1, \cdots, X_n$ : r.s. from $dist^n$ with mean $\mu$, variance $\sigma^2$. Then, $s^2 = \sum(X_i - \bar{X})^2/(n-1)$ is consistent est. of $\sigma^2$

(pf) $s^2 = \frac{n}{n-1}(\frac{1}{n}\sum X_i^2 - \bar{X}^2) \xrightarrow{P} 1 \cdot [E(X_1^2) - \mu^2] = \sigma^2$

## 4.3 Convergence in Distribution

① convergence in distributions

(i) def

**Definition 4.3.1.** $\{X_n\}$ : seq of r.v.'s with cdf $F_{X_n}$, $X$ : r.v. with cdf $F_X$. We say $X_n$ *converges in distribution* to $X$ if $\lim F_{X_n}(x) = F_X(x)$, $^\forall x$ in which $F_X$ is conti., and denoted by $X_n \xrightarrow{D} X$.
$X$ is also called *limiting distribution* of $X_n$ or asymptotic distribution of $X_n$.

(ii)

**Example 4.3.1.** Let $X_n$ have the cdf

$$\begin{aligned} F_{X_n}(x) &= \int_{-\infty}^{x} \frac{1}{\sqrt{1/n}\sqrt{2\pi}} e^{-nw^2/2} dw \\ &= \int_{-\infty}^{\sqrt{n}x} \frac{1}{2\pi} e^{-v^2/2} dv \end{aligned}$$

by changing variable $v = \sqrt{n}w$.

$$\therefore \lim_{x\to\infty} F_{X_n}(x) = \begin{cases} 0, & x < 0 \\ 1/2, & x = 1 \\ 1, & x > 1. \end{cases}$$

So, take

$$F(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

We call $X$ is degenerate at $x = 0$.

(iii)

**Example 4.3.2.** Let $X_1, \cdots, X_n$ be r.s. from $u(0, \theta)$, and let $Z_n = n(\theta - Y_n)$, where $Y_n = max(X_1, \cdots, X_n)$. Find the limiting distribution of $Z_n$.

(sol) The cdf of $Z_n$ is

$$\begin{aligned} F_{Z_n}(t) &= P(Z_n \leq t) = P(Y_n \geq \theta - \frac{t}{n}) = 1 - P(Y_n \leq \theta - \frac{t}{n}) \\ &= 1 - (\frac{\theta - t/n}{\theta})^n = 1 - (1 - \frac{t/\theta}{n})^n \Rightarrow 1 - e^{-t/\theta} \\ &: \quad cdf \;\; of \;\; \varepsilon(\theta), \qquad i.e. \quad Z_n \xrightarrow{D} \varepsilon(\theta). \end{aligned}$$

(iv)

**Theorem 4.3.1.** $X_n \xrightarrow{P} X \Rightarrow X_n \xrightarrow{D} X$

(pf) Let $x$ be a continuous point in $F_X(x)$ and $\varepsilon > 0$.

$$\begin{aligned} F_{X_n}(x) &= P(X_n \le x) = P(\{X_n \le x\} \cap \{|X_n - X| < \varepsilon\}) + P(\{X_n \le x\} \cap \{|X_n - X| \ge \varepsilon\} \\ &\le P(X \le x + \varepsilon) + P(|X_n - X| \ge \varepsilon) \end{aligned}$$

$$\therefore \ \overline{\lim}\ F_n(x) \le F_X(x + \varepsilon)$$

now,

$$\begin{aligned} P(X_n > x) &= P(\{X_n > x\} \cap \{|X_n - X| < \varepsilon\}) + P(\{X_n > x\} \cap \{|X_n - X| \ge \varepsilon\}) \\ &\le P(X > x - \varepsilon) + P(|X_n - X| \ge \varepsilon) \end{aligned}$$

$$\text{i.e.} \quad 1 - P(X_n > x) \ge 1 - P(X \ge x - \varepsilon) - P(|X_n - X| \ge \varepsilon)$$

$$\text{i.e.} \quad F_{X_n}(x) \ge P(X \le x - \varepsilon) - P(|X_n - X| \ge \varepsilon)$$

$$\therefore \ \underline{\lim}\ F_{X_n}(x) \ge F_X(x - \varepsilon)$$

Conclusively, $F_X(x - \varepsilon) \le \underline{\lim}\ F_{X_n}(x) \le \overline{\lim}\ F_{X_n}(x) \le F_X(x + \varepsilon)$
By letting $\varepsilon \to 0$, we have $\lim F_{X_n}(x) = F_X(x)$.

(v) Remark : The converse of Thm 4.3.1 does not hold.

i.e. $X_n \xrightarrow{D} X \not\Rightarrow X_n \xrightarrow{P} X$. However, if $X$ is degenerate at $c$, then it is true.

i.e. $X_n \xrightarrow{D} c \Rightarrow X_n \xrightarrow{P} c$.

(pf) $\lim P(|X_n - c| \le \varepsilon) = \lim\{F_{X_n}(c + \varepsilon) - F_{X_n}(c - \varepsilon)\} = F_X(c+) - F_X(c-) = 1 - 0 = 1$.

(vi)

**Theorem 4.3.2.** $X_n \xrightarrow{D} X,\ Y_n \xrightarrow{P} 0 \ \Rightarrow X_n + Y_n \xrightarrow{D} X$

(vii)

**Theorem 4.3.3.** $X_n \xrightarrow{D} X,\ \mathrm{g}$ : conti. ftn. $\Rightarrow g(X_n) \xrightarrow{D} g(X)$

(viii)

**Theorem 4.3.4.** (Slutzky Thm) $X_n,\ X,\ Y_n,\ Z_n$ : r.v's $k_1,\ k_2$ : const.
s.t. $X_n \xrightarrow{D} X,\ Y_n \xrightarrow{P} k_1,\ Z_n \xrightarrow{P} k_2 \ \Rightarrow\ Y_n + Z_n X_n \xrightarrow{D} k_1 + k_2 X.$

② bounded in probability

(i) Landau's Big Oh and little oh

When we write $x_n \to 0$ as $n \to \infty$, what is the rate of convergence?

Let $\{r_n\} \subset (0, \infty)$ be the rate of convergence (*e.g.* $r_n = n^{-P},\ P > 0$)

- $x_n = o(r_n)$ iff $\dfrac{x_n}{r_n} \to 0$ as $n \to \infty$
- $x_n = O(r_n)$ iff $\underline{\lim} \dfrac{x_n}{r_n} < \infty$

  iff $\exists\ M \in (0, \infty),\ \exists\ N$ s.t. $^{\forall} n \geq N,\ |\frac{x_n}{r_n}| \leq M.$

We can extend this notation when $X_n$ is r.v.

- $X_n = o_p(r_n)$ iff $\dfrac{X_n}{r_n} \xrightarrow{P} 0$ iff $^{\forall}\varepsilon > 0,\ P(|\frac{X_n}{r_n}| > \varepsilon) \to 0$
- $X_n = O_p(r_n)$ iff $^{\forall}\varepsilon > 0, \exists\ M$ s.t. $P(|\frac{X_n}{r_n}| > M) < \varepsilon$

  if $X_n = O_p(1)$, then $\{X_n\}$ is called bounded in prob.

(ii) Taylor expansion

If $g(x)$ is $k$-times differentiable at $x = x_0$,
we have $g(x) = \sum_{j=0}^{k} \frac{1}{j!}\ g^{(j)}(x_0) \cdot (x - x_0)^j + o(|x - x_0|^k)$ as $|x - x_0| \to 0$

(iii)

**Theorem 4.3.5.** $X_n \xrightarrow{D} X \;\Rightarrow\; X_n = Op(1)$

(pf) Let $\eta$ be continuous point in $F_X(x)$, then

$$P(|X_n| \le \eta) = F_{X_n}(\eta) - F_{X_n}(-\eta^-) \to F_X(\eta) - F_X(-\eta) \quad \cdots (*)$$

now, can choose $\eta_1$ and $\eta_2$ s.t. for a given $\varepsilon > 0$,

$$F_X(x) < \frac{\varepsilon}{2} \;\; for \;\; x \le \eta_1 \;\; \& \;\; F_X(x) > 1 - \frac{\varepsilon}{2} \;\; for \;\; x \ge \eta_2$$

Take $\eta = max(|\eta_1|, |\eta_2|)$, then

$$P(|X| \le \eta) = F_X(\eta) - F_X(-\eta^-) \ge 1 - \frac{\varepsilon}{2} - \frac{\varepsilon}{2} = 1 - \varepsilon$$

By taking limit in (*), we have $\lim P(|X_n| \le \eta) \ge 1 - \varepsilon$.

(iv)

**Theorem 4.3.6.** $X_n = Op(1),\; Y_n \xrightarrow{P} 0 \;\Rightarrow\; X_n Y_n \xrightarrow{P} 0$

(pf)

$$P(|X_n Y_n| \ge \varepsilon) = P(|X_n Y_n| \ge \varepsilon,\; |X_n| \le M) + P(|X_n Y_n| \ge \varepsilon,\; |X_n| > M)$$

$$\begin{aligned} \therefore \lim P(|X_n Y_n| \ge \varepsilon) &= \lim P(|X_n Y_n \ge \varepsilon,\; |X_n| \le M) \\ &\le \lim P(|Y_n| \ge \varepsilon / M) = 0 \end{aligned}$$

③ Δ-method

(i)

**Theorem 4.3.7.** $Y_n = Op(1),\; X_n = o_p(Y_n) \Rightarrow\; X_n \xrightarrow{P} 0$

(pf)

$$\begin{aligned} P(|X_n| \ge \varepsilon) &= P(|X_n| \ge \varepsilon,\; |Y_n| \le M) + P(|X_n| \ge \varepsilon,\; |Y_n| > M) \\ &\le P(|\frac{X_n}{Y_n}| \ge \frac{\varepsilon}{M}) + P(|Y_n| > M) \end{aligned}$$

Take limit on both sides.

$$\begin{aligned}\lim P(|X_n| \geq \varepsilon) &\leq \lim P(|\frac{X_n}{Y_n}| \geq \frac{\varepsilon}{M}) + \lim P(|Y_n| > M) \\ &= 0\end{aligned}$$

(ii) $\Delta$-method

**Theorem 4.3.8.** Assume $\sqrt{n}(X_n - \theta) \xrightarrow{D} N(0, \sigma^2)$ and $g(x)$ is diff. at $x = \theta$, $g'(\theta) \neq 0$. Then, $\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{D} N(0, \sigma^2 g'(\theta)^2)$

(pf) By Taylor expansion,

$$g(X_n) = g(\theta) + g'(\theta)(X_n - \theta) + o_p(|X_n - \theta|)$$

$$\textit{i.e.} \quad \sqrt{n}(g(X_n) - g(\theta)) = g'(\theta) \cdot \sqrt{n}(X_n - \theta) + o_p(\sqrt{n}|X_n - \theta|)$$

now, $\sqrt{n}(X_n - \theta) = O_p(1)$, so that by Thm 4.3.8., $o_p(\sqrt{n}|X_n - \theta|) = o_p(|O_p(1)|) \xrightarrow{P} 0$.

(iii) Example

Assume $\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2)$.

Find the limiting distribution of $\sqrt{n}(\bar{X}^2 - \mu^2)$.

(sol) $\quad g(x) = x^2, \quad g'(x) = 2x \qquad \therefore g'(\mu)^2 = 4\mu^2$

$$\therefore \quad \sqrt{n}\ (\bar{X}^2 - \mu^2) \xrightarrow{D} N(0,\ 4\mu^2\sigma^2)$$

④ mgf technique

(i)

**Theorem 4.3.9.** $\{X_n\}$ : seq. of r.v's with mgf $M_{X_n}(t)$.
$X$ : r.v. with mgf $M_X(t)$. If $\lim M_{X_n}(t) = M_X(t)$, then $X_n \xrightarrow{D} X$.

(ii) useful result for limit

If $\lim \psi(n) = 0$, then $\lim(1 + \frac{b}{n} + \frac{\psi(n)}{n})^{cn} = e^{bc}$

(iii)

**Example 4.3.3.** $Z \sim \chi^2(n)$. Show that $Y = \frac{Z-n}{\sqrt{2n}} \xrightarrow{D} N(0,\ 1)$

(pf)

$$\begin{aligned} M_Y(t) &= E\left[\exp\left\{t \cdot (\frac{Z-n}{\sqrt{2n}})\right\}\right] = e^{-tn/\sqrt{2n}} \cdot E\left[e^{tZ/\sqrt{2n}}\right] \\ &= \exp\left[-(t\sqrt{\frac{2}{n}})\frac{n}{2}\right] \cdot (1 - 2 \cdot \frac{t}{\sqrt{2n}})^{-n/2} \\ &= (e^{-t/\sqrt{2/n}} - t\sqrt{\frac{2}{n}}\, e^{\,t\sqrt{2/n}})^{-\frac{n}{2}} \end{aligned}$$

now, $e^{t\sqrt{2/n}} = 1 + t\sqrt{\frac{2}{n}} + \frac{1}{2}(t\sqrt{\frac{2}{n}})^2 + \frac{1}{6}(t\sqrt{\frac{2}{n}})^3 + o(n^{-3/2})$

$$\begin{aligned} \therefore\ e^{\,t\sqrt{2/n}} - t\sqrt{\frac{2}{n}}\, e^{t\sqrt{2/n}} &= (1 + t\sqrt{\frac{2}{n}} + \frac{t^2}{n} + \frac{\sqrt{2}t^3}{3n^{3/2}} + o(n^{-3/2}) - t\sqrt{\frac{2}{n}} - \frac{2t^2}{n} - \frac{\sqrt{2}t^3}{n^{3/2}} + \\ &= (1 - \frac{t^2}{n} + \frac{\psi(n)}{n})\,, \qquad \psi(n) = -\frac{2\sqrt{2}t^3}{3n^{1/2}} + O(n^{-1/2}) \end{aligned}$$

$$\therefore\ M_Y(t) \to e^{t^2/2} : mgf\ \ of\ \ N(0,\ 1)$$

## 4.4 Central Limit Theorem

**Theorem 4.4.1.** $X_1, \cdots, X_n$ : r.s. from a distribution with mean $\mu$, variance $\sigma^2$. Then, $\sqrt{n}(\bar{X} - \mu)/\sigma \xrightarrow{D} N(0,\ 1)$

(pf) Let $m(t) = E[e^{t(x-\mu)}]$ : mgf of $Y = X - \mu$.

$$\begin{aligned} m(t) &= m(o) + m'(o)t + \frac{m''(o)}{2}t^2 + \frac{m'''(o)}{6}t^3 + \cdots \\ &= 1 + \frac{\sigma^2}{2}t^2 + \frac{m'''(o)}{6}t^3 + \cdots \end{aligned}$$

Now, consider mgf of $Z = \sqrt{n}(\bar{X} - \mu)/\sigma$

$$\begin{aligned} M_Z(t) &= E\left[\exp(t \cdot \frac{\sum X_i - n\mu}{\sigma\sqrt{n}})\right] = E\left[\exp(t \cdot \frac{X_1 - \mu}{\sigma\sqrt{n}}) \cdots \exp(t \cdot \frac{X_n - \mu}{\sigma\sqrt{n}})\right] \\ &= [E\{\exp(t \cdot \frac{X_1 - \mu}{\sigma\sqrt{n}})\}]^n = \{m(\frac{t}{\sigma\sqrt{n}})\}^n \\ &= [1 + \frac{\sigma^2}{2} \cdot \frac{t^2}{\sigma^2 n} + \frac{m'''(o)}{6} \cdot \frac{t^3}{\sigma^2 n^{3/2}} + \cdots]^n = (1 + \frac{t^2}{2n} + \frac{\psi(n)}{n})^n \to e^{t^{2/2}} \end{aligned}$$

**Example 4.4.1.** $X_1, \cdots, X_n$ : r.s. from $B(1, , p)$. Then, by CLT,
$\sqrt{n}(\bar{X} - p)/\sqrt{p(1-p)} \xrightarrow{D} N(0, 1)$

**Example 4.4.2.** Find $h$ s.t. $\sqrt{n}(h(\bar{X}) - h(p)) \to N(0, c^2)$, c : const.

(sol)

$\sqrt{n}(\bar{X} - p) \to N(0, p(1-p)) \Rightarrow \sqrt{n}(h(\bar{X}) - h(p)) \to N(0, h'(p)^2 p(1-p))$

$$\therefore h'(p)^2 p(1-p) = c^2 \Rightarrow h'(p) = \sqrt{c^2/p(1-p)}$$
$$h(p) = (2c) \cdot arc\ sin(\sqrt{p})$$

This kind of transformation called the variance stabilizing transformation.

## 4.5 Asymptotics for Multivariate Distributions

① Euclidean norm

(i) definition : $\boldsymbol{v} = (v_1, \cdots, v_p)' \in R^p$, $||\boldsymbol{v}|| = (\sum v_i^2)^{1/2}$ : Euclidean norm.

(ii) properties

(a) $||\boldsymbol{v}|| \leq 0$. Equality holds when $\boldsymbol{v} = \boldsymbol{0}$

(b) $^{\forall}a \in R'$, $||a\boldsymbol{v}|| = |a||\boldsymbol{v}|$

(c) $||\boldsymbol{u}+\boldsymbol{v}|| \le ||\boldsymbol{u}|| + ||\boldsymbol{v}||$ : triangular inequality.

(iii) basis

$\boldsymbol{e}_i = (0, \cdots, 0, 1, 0, \cdots, 0) \quad \boldsymbol{e}_1, \cdots, \boldsymbol{e}_p$ : basis for $R_p$

$\boldsymbol{v} = \sum_{i=1}^{p} v_i \boldsymbol{e}_i$

(iv)

**Lemma 4.5.1.** $|v_j| \le ||v|| \le \sum_{i=1}^{p} |v_i|, \quad j = 1, \cdots, p$

(pf) $v_j^2 \le \sum_{i=1}^{p} v_i^2 = ||\boldsymbol{v}||^2 \Rightarrow |v_j| \le ||\boldsymbol{v}||$

also, $||\boldsymbol{v}|| = ||\sum v_i \boldsymbol{e}_i|| \le \sum |v_i| ||e_i|| = \sum |v_i|$.

② Convergence in probability

(i) definition : $\{\boldsymbol{X}_n\}$ converges in prob. to $\boldsymbol{X}$ if $P(||\boldsymbol{X}_n - \boldsymbol{X}|| \ge \varepsilon) \to 0$, and denoted by $\boldsymbol{X}_n \xrightarrow{P} \boldsymbol{X}$.

(ii)

**Theorem 4.5.1.** $\boldsymbol{X}_n \xrightarrow{P} \boldsymbol{X}$ iff $X_{n_j} \xrightarrow{P} X_j, \ j = 1, \cdots, p$

(pf)

($\Rightarrow$) By Lemma 4.5.1, $|X_{n_j} - X_j| \le ||\boldsymbol{X}_n - \boldsymbol{X}||$

($\Leftarrow$) By Lemma 4.5.1, $\sum_{i=1}^{p} |X_{n_j} - X_j| \ge ||\boldsymbol{X}_n - \boldsymbol{X}||$

$$\therefore P(||\boldsymbol{X}_n - \boldsymbol{X}|| \ge \varepsilon) \le P(\sum |X_{n_j} - X_j| \ge \varepsilon) \le \sum_{i=1}^{p} P(|X_{n_j} - X_j| \ge \varepsilon/p)$$

(iii) Examples

(i) $X_1, \cdots, X_n$ : r.s. from a distribution with mean $\mu$ and variance $\sigma$.
We know that $\bar{X} \xrightarrow{P} \mu, \ s^2 \xrightarrow{P} \sigma^2$, by Thm 4.5.1, $(\bar{X}, \ s^2) \xrightarrow{P} (\mu, \ \sigma^2)$.

(ii) $\boldsymbol{X}_1, \cdots, \boldsymbol{X}_n$ : r.s. from a distribution with mean $\boldsymbol{\mu}$ and var-cov. $\boldsymbol{\Sigma}$.
We know that $\bar{X}_j \xrightarrow{P} \mu_j$, $j = 1, \cdots, p$, then $\bar{\boldsymbol{X}} \xrightarrow{P} \boldsymbol{\mu}$
also, $s_j^2 = \frac{1}{n-1}\sum_{i=1}^n (X_{ij} - \bar{X}_j)^2 \xrightarrow{P} \sigma_j^2$ and $s_{jk} = \frac{1}{n-1}\sum_{i=1}^n (X_{ij} - \bar{X}_j)(X_{ik} - \bar{X}_j) \xrightarrow{P} \sigma_{jk}$, we have $S \xrightarrow{P} \Sigma$.

③ Convergence in distribution

(i) definition : $\{X_n\}$ converges in distribution to $\boldsymbol{X}$ if $F_{X_n}(\boldsymbol{X}) \to F_X(\boldsymbol{X})$ for all points $\boldsymbol{x}$ at which $F_X(\boldsymbol{x})$ is conti, and denoted by $\boldsymbol{X}_n \xrightarrow{D} \boldsymbol{X}$.

(ii)

**Theorem 4.5.2.** $\boldsymbol{X} \xrightarrow{D} \boldsymbol{X}$, g: conti. $\Rightarrow$ $g(\boldsymbol{X}_n) \xrightarrow{D} g(\boldsymbol{X})$

(iii)

**Theorem 4.5.3.** $\boldsymbol{X}_n \xrightarrow{D} \boldsymbol{X}$ iff $M_n(\boldsymbol{t}) \to M(\boldsymbol{t})$

④ CLT

(i) multivariate CLT

**Theorem 4.5.4.** $\{\boldsymbol{X}_n\}$ : seq. of iid random vectors with mean $\mu$, var-cov. $\Sigma$ $\Rightarrow$ $\boldsymbol{Y}_n = \frac{1}{\sqrt{n}}\sum_{i=1}^n (\boldsymbol{X}_i - \boldsymbol{\mu}) = \sqrt{n}\,(\bar{\boldsymbol{X}} - \boldsymbol{\mu}) \xrightarrow{D} N_p(\boldsymbol{0},\ \Sigma)$

(pf)

$$\begin{aligned} M_n(\boldsymbol{t}) &= E\,[\exp\{t' \cdot \frac{1}{\sqrt{n}}\sum(X_i - \mu)\}] = E\,[\exp\{\frac{1}{\sqrt{n}}\sum t'(X_i - \mu)\}] \\ &= E\,[\exp\{\frac{1}{\sqrt{n}}\sum W_i\}],\ \ W_i = t'(X_i - \mu) \end{aligned}$$

now, $W_1, \cdots, W_n$ are iid with mean 0, variance $t'\Sigma t$.
Then, by CLT, $Z = \frac{1}{\sqrt{n}}\sum W_i \xrightarrow{D} N(0,\ t'\Sigma t)$
now, $M_n(t) = E\,[\exp\{\frac{1}{\sqrt{n}}\sum W_i\}] = E\,[e^{1 \cdot Z}]$, i.e. mgf evaluated at

$t = 1$.

Therefore, $M_n(t) \to \exp(0 \cdot 1 + \frac{1}{2} t'\Sigma t \cdot 1^2) = \exp(t'\Sigma t/2)$ which is mgf of

$N_p(\mathbf{0},\ \Sigma)$.

(ii)

**Theorem 4.5.5.** $X_n \xrightarrow{D} N_p(\boldsymbol{\mu},\ \Sigma)$. A : m×p, $\boldsymbol{b}$ : m×1 $\Rightarrow$

$A\boldsymbol{X}_n + \boldsymbol{b} \xrightarrow{D} N_m(A\boldsymbol{\mu} + \boldsymbol{b},\ A\Sigma A')$

(iii)

**Theorem 4.5.6.** $\{\boldsymbol{X}_n\}$ : seq. of $p$-dim random vector.

$\sqrt{n}\ (X_n - \mu) \xrightarrow{D} N_p(\mathbf{0},\ \Sigma)$. $g(\boldsymbol{X}) = (g_1(X), \cdots, g_k(X))' : R^p \to R^k$

$B = (\dfrac{\partial g_i}{\partial \mu_j})$ : $k \times p$ matrix. Then, $\sqrt{n}(g(\boldsymbol{X}_n) - g(\boldsymbol{\mu})) \xrightarrow{D} N_k(\mathbf{0},\ B\Sigma B')$

# CHAPTER 5. Some Elementary Statistical Inference

## 5.1 Sampling and Statistic

- sampling with (without) replacement.
- random sample, statistic

## 5.2 Order Statistic

① definition

$X_1, \cdots, X_n$: r.s. from a pdf $f(x)$ and cdf $F(x)$.
Let $Y_1$ be the smallest of $X_i$'s, $Y_2$ be the 2nd smallest of $X_i$'s, $\cdots$, and $Y_n$ be the largest of $X_i$'s. Then, $Y_1 < Y_2 < \cdots < Y_n$ is called the order statistics of $X_1, \cdots, X_n$.
② pdf

(i) joint pdf of $Y_1, \cdots, Y_n$ (Thm 5.2.1)

$$g(y_1, \cdots, y_n) = n!\, f(y_1) \cdots f(y_n), \qquad y_1 < y_2 < \cdots < y_n$$

(pf) Consider transformation $(x_1, \cdots, x_n) \to (y_1, \cdots, y_n)$.
Then, there are $n!$ methods, and Jacobian is $\pm 1$.
Therefore, $g(y_1, \cdots, y_n) = \sum_{i=1}^{n!} |J_i|\, f(y_1) \cdots f(y_n) = n! \prod_{i=1}^{n} f(y_i)$.

(ii) marginal pdf of $Y_k$

$$g_k(y_k) = \frac{n!}{(k-1)!(n-k)!} (F(y_k))^{k-1} (1 - F(y_k))^{n-k} f(y_k)$$

(iii) joint pdf of $Y_i$ and $Y_j$ $(i < j)$

$$g_{ij}(y_i,\, y_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} (F(y_i))^{i-1} (F(y_j) - F(y_i))^{j-i-1}$$

$$(1 - F(y_j))^{n-j} f(y_i) f(y_j)$$

(iv) **Example 5.2.3.**

$X_1, X_2, X_3$ : r.s. from $u(0,\ 1)$, $Y_1 < Y_2 < Y_3$ : order stat. Find the pdf of the sample range $Z_1 = Y_3 - Y_1$.

(sol) $z_1 = y_3 - y_1,\ z_2 = y_3 \longrightarrow y_1 = z_2 - z_1,\ y_3 = z_2,\ \ |J| = 1.$
The jpdf of $Y_1$ and $Y_3$ is

$$g_{13}(y_1, y_3) = 6(y_3 - y_1)\ ,\ \ 0 < y_1 < y_3 < 1.$$

$$\therefore\ h(z_1,\ z_2) = 6z_1\ ,\ \ 0 < z_1 < z_2 < 1.$$

$$\therefore\ h_1(z_1) = \int_{z_1}^{1} 6z_1\ dz_2 = 6z_1(1 - z_1),\ \ 0 < z_1 < 1.$$

③ quantiles

(i) definition

$X$ : r.v. with conti. cdf $F(x)$. $\xi_p = F^{-1}(p)$ : $p$ th quantile of $X$

(ii) estimator of $\xi_p$.

Let $Y_1 < \cdots < Y_n$ be order statistic, and consider $Y_k$, where $k = [p(n+1)]$, as an estimator of $\xi_p$. For $Y_k$ to be a good estimator of $\xi_p$.

$$\begin{aligned} E[F(Y_k)] &= \int F(y_k) g_k(y_k)\ dy_k \\ &= \int F(y_k) \frac{n!}{(k-1)!(n-k)!} (F(y_k))^{k-1} (1 - F(y_k))^{n-k} f(y_k)\ dy_k \end{aligned}$$

Let $z = F(y_k)$, then $dz = f(y_k) dy_k$, so that

$$E[F(Y_k)] = \int \frac{n!}{(k-1)!(n-k)!} z^k (1-z)^{n-k} dz = \frac{k}{n+1} \simeq p.$$

$Y_k$ is called the $p$ th sample quantile or $100p - th$ percentile.

(iii) five number summary (J. Tukey)

$$Y_1,\ \ Y_{[.25(n+1)]},\ Y_{[.5(n+1)]},\ Y_{[.75(n+1)]},\ Y_n$$

(iv) boxplot

Boxplot is based on the five number summary.
To exhibit potential outliers, define the lower and upper fence (LF, UF),
LF= $Q_1 - h$, UF= $Q_3 + h$, $h = 1.5(Q_3 - Q_1)$.
Points that lie outside the fence (LF, UF) are called potential outliers.

(v) **Example 5.2.4.**

data : 56, 70, 89, 94, 96, $\cdots$ , 110, 113, 116 $(n = 15)$

$Y_1 = 56,\ Q_1 = y_4 = 94,\ Q_2 = y_8 = 102,\ Q_3 = y_{12} = 108,\ y_{15} = 116,$
$h = 1.5(108 - 94) = 21 \rightarrow$ (LF, UF)= $(73,\ 129)$
$\therefore$ 56, 70 are potential outliers.

(vi) q-q plot

$X$ : r.v. from a location-scale family with cdf $F(\frac{x-a}{b})$, where $F(x)$ is known, but a and b are unknown. Let $\xi_{z,p}$ be the $p$th quantile of $z = \frac{x-a}{b}$ now,

$$p = P(X \le \xi_{x,p}) = P(Z \le \frac{\xi_{x,p} - a}{b}) = P(Z \le \xi_{z,p})$$

$\therefore\ \xi_{x,p} = b\,\xi_{z,p} + a$ ( $\xi_{z,p}$ : known, $\xi_{x,p}$ : unknown )
Now, $Y_k$ is estimator of $\xi_{x,p_k}$, where $p_k = k/(n+1)$. The plot of $Y_k$ vs $\xi_{z,p_k}$ is called q-q plot. If $X$ is distributed as $F = (\frac{x-a}{b})$, then the q-q plot should be linear.

④ confidence intervals of quantiles

$Y_k$ : point est. of $\xi_p$, where $k = [(n+1)p]$.
As a C.I. for $\xi_p$, consider $(Y_i,\ Y_j)$ s.t. $i < [(n+1)p] < j$.
When we say $(Y_i,\ Y_j)$ as $100\gamma\%$ C.I., what is $\gamma$?
Need to compute $\gamma = P(Y_i < \xi_p < Y_j)$.
$\{Y_i < \xi_p\} \Leftrightarrow \{$at least $i$ of $X$ values are less than $\xi_p\}$
$\{Y_j > \xi_p\} \Leftrightarrow \{$fewer than $j$ of $X$ values are less than $\xi_p\}$

now, consider this problem as Bernoulli trial, i.e., if $X < \xi_p$ , then success, and if $X \geq \xi_p$, then failure. Also, the prob. of success is $P(X < \xi_p) = F(\xi_p) = p$.

$\therefore \quad P(Y_i < \xi_p < Y_j) = \sum_{k=i}^{j-1} \binom{n}{k} p^k (1-p)^{n-k} \equiv \gamma.$
We call $(Y_i,\ Y_j)$ as $100\gamma\%$ C.I. for $\xi_p$.

## 5.3 Tolerance Limits for Distributions

① definition

$X_1, \cdots, X_n$ : r.s. from a $dist^n$ with cdf $F(x)$, $Y_1 < \cdots < Y_n$ is order stat. Then, $(y_i,\ y_j)$ s.t. $\gamma = P[F(Y_j) - F(Y_i) \geq p]$ is called $100\gamma\%$ tolerance limits for $100p\%$ of the prob. for the $dist^n$ of $X$.

② computation of $\gamma$

(i) jpdf of $Z_1 = F(Y_n), \cdots, Z_n = F(Y_n)$

Note that $Z = F(X) \sim u(0,1)$ because $G(z) = P(Z \leq z) = P(F(X) \leq z) = P(X \leq F^{-1}(z)) = F(F^{-1}(z)) = \delta$. Hence, $Z_1, \cdots, Z_n$ are order stat. from $u(0,1)$, so that the jpdf of $Z_1, \cdots, Z_n$ is

$$h(\delta_1, \cdots, \delta_n) = n!\ I(0 < Z_1 < \cdots < Z_n < 1)$$

(ii) computation of $\gamma$

To compute $\gamma$, note that

$$\gamma = P(Z_j - Z_i \geq p) = \int_0^{1-p} \int_{p+z_i}^1 h_{ij}(z_i,\ z_j)\ dz_j dz_i$$

where

$$h(Z_i, Z_j) = \frac{n!}{(i-1)!(j-i-1)!(n-j)!} Z_i^{i-1} (Z_j - Z_i)^{j-i-1} (1 - Z_j)^{n-j}$$

This computation is quite tedious. So, we use an alternative way to compute $\gamma$. Consider transforming

$$W_1 = Z_1, W_2 = Z_2 - Z_1, W_3 = Z_3 - Z_2, \cdots, W_n = Z_n - Z_{n-1}$$

Then $Z_i = \sum_{j=1}^{i} W_j$ and $|J| = 1$. Therefore, the jpdf of $W_1, \cdots, W_n$ is

$$k(w_1, \cdots, w_n) = n!,\ 0 < w_i,\ i = 1, \cdots, n,\ w_1 + \cdots + w_n < 1$$

Now, the jpdf of $W_1, \cdots, W_n$ is symmetric in $w_1, \cdots, w_n$, and hence, the pdf of $W_{i+1} + W_{i+2} + \cdots + W_j$ is the same as that of $W_1 + W_2 + \cdots + W_{j-i}$. Note that $W_{i+1} + \cdots + W_j = Z_j - Z_i$ and $W_1 + \cdots + W_{j-i} = Z_{j-i}$. Therefore,

$$\gamma = P(Z_j - Z_i \geq p) = P(Z_{j-i} \geq p)$$

where the pdf of $Z_k$ is

$$h_k(v) = \frac{n!}{(k-1)!(n-k)!} v^{k-1}(1-v)^{n-k}$$

Hence,

$$\gamma = \int_p^1 h_{j-i}(v)dv$$

(iii) **Example 5.3.1**

Let $Y_1 < \cdots < Y_6$ be the order statistic from conti. distribution. Compute $\gamma$ when we use $(y_1, y_6)$ as a tolerance limit for 80% of the distribution.

$$\gamma = P(F(Y_6) - F(Y_1) \geq 0.8) = 1 - \int_0^{0.8} 30v^4(1-v)dv = 0.34$$

## 5.4 More on Confidence Intervals

① approximate C.I.

When $Z = (X - \mu)/\sigma \sim N(0,1)$, then $100(1-\alpha)\%$ C.I. for $\mu$ is $(X - Z_{\alpha/2}\sigma,\ X + Z_{\alpha/2}\sigma)$ if $\sigma$ is known. When we known the asymptotic distribution only by CLT such as $\sqrt{n}(T - \theta) \to N(0, \sigma^2)$ then $(T - Z_{\alpha/2}\sigma/\sqrt{n},\ T + Z_{\alpha/2}\sigma/\sqrt{n})$ is called approximate $100(1-\alpha)\%$ C.I. for $\theta$.

② examples

(i) C.I. for $\mu$

$X_1, \cdots, X_n$: r.s. from a dist. with mean $\mu$ and $\sigma^2$ (both unknown)
Find approximate $100(1-\alpha)\%$ C.I. for $\mu$.

(sol) $\sqrt{n}(\bar{X}-\mu)/s \xrightarrow{\mathscr{D}} N(0,1)$

$$\therefore \bar{x} \pm Z_{\alpha/2}\frac{s}{\sqrt{n}}$$

(ii) C.I. for $p$

$X_1, \cdots, X_n$: r.s. from $B(1,p)$. By CLT, $\sqrt{n}(\hat{p}-p) \to N(0, p(1-p))$
also, $\sqrt{n}(\hat{p}-p)/\sqrt{\hat{p}(1-\hat{p})} \to N(0,1)$

$$\therefore \hat{p} \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

(iii) C.I. for $\mu$ under normality

$X_1, \cdots, X_n$: r.s. from $N(\mu, \sigma^2)$. $\sqrt{n}(\bar{X}-\mu)/s \sim t(n-1)$

$$\therefore \bar{X} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$$

(iv) C.I. for $\mu_1 - \mu_2$

$X_1, \cdots, X_{n_1}$: r.s. from a dist. with mean $\mu_1$, variance $\sigma_1^2$ and
$Y_1, \cdots, Y_{n_2}$: r.s. from a dist. with mean $\mu_2$, variance $\sigma_2^2$ are indep.
Want to obtain approximate C.I. for $\mu_1 - \mu_2$.
Let $n = n_1 + n_2$, and assume $\frac{n_1}{n} \to \lambda_1$, $\frac{n_2}{n} = \lambda_2$. By CLT,
$\sqrt{n_1}(\bar{X}-\mu)/\sigma_1 \to N(0,1)$.

$$\therefore \sqrt{n}(\bar{X}-\mu)/\sigma_1 = \sqrt{\frac{n}{n_1}}\sqrt{n_1}(\bar{X}-\mu)/\sigma_1$$

$$\to \sqrt{\frac{1}{\lambda_1}}N(0,1) = N(0, 1/\lambda_1).$$

Similarly, $\sqrt{n}(\bar{Y} - \mu_2)/\sigma_2 \to N(0, 1/\lambda_2)$.

$$\therefore \sqrt{n}[(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)] \to N\left(0, \frac{\sigma_1^2}{\lambda_1} + \frac{\sigma_2^2}{\lambda_2}\right),$$

Also,

$$n\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right) \xrightarrow{P} \frac{\sigma_1^2}{\lambda_1} + \frac{\sigma_2^2}{\lambda_2}$$

$$\therefore \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \to N(0, 1)$$

$$\Rightarrow (\bar{X} - \bar{Y}) \pm Z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

(v) C.I. for $\mu_1 - \mu_2$ under normality

$X_1, \cdots, X_{n_1}$: r.s. from $N(\mu_1, \sigma^2)$ and
$Y_1, \cdots, Y_{n_2}$: r.s. from $N(\mu_2, \sigma^2)$ are indep.(common variances)
$\bar{X} \sim N(\mu_1, \sigma^2/n_1)$, $\bar{Y} \sim N(\mu_2, \sigma^2/n_2)$

$$\frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim N(0, 1)$$

Now, $(n_1 - 1)s_1^2/\sigma^2 \sim \chi^2(n_1 - 1)$ and $(n_2 - 1)s_2^2/\sigma^2 \sim \chi^2(n_2 - 1)$

$$\Rightarrow \frac{1}{\sigma^2}\{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2\} \sim \chi^2(n_1 + n_2 - 2)$$

$$\therefore T = \frac{Z}{\sqrt{V/(n_1 + n_2 - 2)}} \sim t(n_1 + n_2 - 2),\ s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

(vi) C.I. for $p_1 - p_2$

$X_1, \cdots, X_{n_1}$: r.s. from $B(1, p_1)$ and
$Y_1, \cdots, Y_{n_2}$: r.s. from $B(1, p_2)$ are indep.
By the same argument as in (iv),

$$(\hat{p}_1 - \hat{p}_2) \pm Z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

## 5.5 Introduction to Hypothesis Testing

① definitions

(i) null and alternative hypothesis

$X_1, \cdots, X_n$: r.s. from a dist. with pdf $f(x:\theta), \theta \in \Omega$.
statistical hypothesis: $H_0 : \theta \in \omega_0$ vs $H_1 : \theta \in \omega_1$, $\omega_0 \cup \omega_1 = \Omega$, $\omega_0 \cap \omega_1 = \phi$

$H_0$ : null hypothesis, $H_1$ : alternative hypothesis

(ii) two type of errors

test statistic: a statistic $T = T(X_1, \cdots, X_n)$ for testing $H_0$ vs $H_1$

rejection region(critical region): a set $C$ where $H_0$ is rejected

type I error: an error caused by rejecting $H_0$ even when $H_0$ is true

type II error: an error caused by accepting $H_0$ when $H_1$ is true

a critical region $C$ is of size $\alpha$ if $\alpha = max_{\theta \in \omega_0} P[(X_1, \cdots, X_n) \in C]$

a power of a test is $P_\theta[(X_1, \cdots, X_n) \in C]$, $\theta \in \omega_1$, i.e. power is $1 - P_\theta[\text{type II error}]$, $\theta \in \omega_1$

the power function of a critical region $C$ is

$$\gamma_c(\theta) = P_\theta[(X_1, \cdots, X_n) \in C], \theta \in \omega_1$$

② examples

(i) test for $p$

$X_1, \cdots, X_n$: r.s. from $B(1, p)$. Want to make a test of size $\alpha$ for testing $H_0 : p = p_0$ vs $H_1 : p < p_0$.
test statistic: $S = \sum_{i=1}^{n} X_i$: # of successes
rejection region: reject $H_0$ if $S \leq k$ s.t. $\alpha = P_{H_0}(S \leq k)$
assume $n = 20$, $P_0 = 0.7$, $\alpha = 0.15$, $P_{H_0}(S \leq 11) = .1133$, $P_{H_0}(S \leq 12) = .2277$
Hence, a test of size .15 rejects $H_0$ if $S \leq 11$.
Compare power function for $S \leq 11$ and $S \leq 12$.(Fig. 5.5.1)

(ii) large sample test for $\mu$

$X_1, \cdots, X_n$: r.s. from a dist. with mean $\mu$, variance $\sigma^2$.
test of size $\alpha$ for testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu > \mu_0$
test stat: $\bar{X}$, rejection region : $\bar{X} \geq k$
By using $\sqrt{n}(\bar{X} - \mu_0)/s \to N(0,1)$ under $H_0$, reject $H_0$ if
$\sqrt{n}(\bar{X} - \mu_0)/s \geq Z_\alpha$
Now, compute approximate power function

$$\begin{aligned}\gamma(\mu) &= P_\mu(Z \geq Z_\alpha) = P_\mu(\bar{X} \geq \mu_0 + Z_\alpha s/\sqrt{n}) \\ &= P_\mu\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} \geq \frac{\mu_0 - \mu}{s/\sqrt{n}} + Z_\alpha\right) \simeq 1 - \Phi\left(\frac{\mu_0 - \mu}{s/\sqrt{n}} + Z_\alpha\right)\end{aligned}$$

(iii) test for $\mu$ under normality

$X_1, \cdots, X_n$: r.s. from $N(\mu, \sigma^2)$
test of size $\alpha$ for testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu > \mu_0$
test stat.: $t = \dfrac{\bar{X} - \mu_0}{s/\sqrt{n}} > t_\alpha(n-1)$

## 5.6 Additional Comments about Statistical Tests

(i) Large sample two-sided test for $\mu$

$X_1, \cdots, X_n$: r.s. from a dist with mean $\mu$, variance $\sigma^2$
test of size $\alpha$ for testing $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$
Intuitively, we reject $H_0$ if $\bar{X} \leq h$ or $\bar{X} \geq k$ s.t.
$\alpha = P_{H_0}(\bar{X} \leq h \text{ or } \bar{X} \geq k) = P_{H_0}(\bar{X} \leq h) + P_{H_0}(\bar{X} \geq k)$
Now, it is reasonable to set $P_{H_0}(\bar{X} \leq h) = \alpha/2$ and $P_{H_0}(\bar{X} \geq k) = \alpha/2$
i.e. Reject $H_0$ if $\left|\dfrac{\bar{X} - \mu_0}{s/\sqrt{n}}\right| \geq Z_{\alpha/2}$

(ii) Randomized test

$X_1, \cdots, X_{10}$ r.s. from $P(\theta)$.
test of size $\alpha = .05$ for testing $H_0 : \theta = 0.1$ vs $H_1 : \theta > 0.1$
test stat: $Y = \sum_{i=1}^{10} X_i$, critical region: $Y \geq k$.

Note that $Y \sim P(1)$, therefore $P(Y \geq 3) = .080$, $P(Y \geq 4) = .019$
Hence, size $\alpha = .05$ test is rejecting $H_0$ if $Y \geq 4$
This test is called a non-randomized test. To achieve $.05 = P_{H_0}(\text{reject } H_0)$, we need a randomized test.
Let $W$ be a Bernoulli trial with prob. of succes
$P(W = 1) = \dfrac{0.050 - 0.019}{0.080 - 0.019} = \dfrac{31}{61}$, and let the rejection region be $\{\sum_{i=1}^{10} X_i \geq 4\}$ or $\{\sum X_i = 3 \text{ and } W = 1\}$, then

$$P(Y \geq 4) + P(Y = 3)\frac{31}{61} = .019 + (.080 - .019)\frac{.050 - .019}{.080 - .019} = .050$$

(iii) p-value(observed significance level)

$Y = u(X_1, \cdots, X_n)$: test stat.
rejection region: $Y \leq c$
If the observed test stat. is $d$, then $P_{H_0}(Y \leq d)$ is called the p-value of $d$. In general, p-value is defined as the minimum of prob. of type I error to reject $H_0$ for a given value of test stat.

## 5.7 Chi-square Tests

3 types of chi-square test:
goodness-of-fit(GOF) test, homogeneity test, independence test

① goodness-of-fit test

(i) derivation

$X_1 \sim B(n, p_1)$, $X_2 = n - X_1$, $p_2 = 1 - p_1$

$$\begin{aligned} Q_1 &= \frac{(X_1 - np_1)^2}{np_1(1 - p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_1 - np_1)^2}{n(1 - p_1)} \\ &= \frac{(X_1 - np_1)^2}{np_1} + \frac{(X_2 - np_2)^2}{np_2} \xrightarrow{\mathscr{D}} \chi^2(1) \end{aligned}$$

In general, let $\boldsymbol{X} = (X_1, \cdots, X_{k-1})' \sim \mathscr{M}(n, p_1, \cdots, p_{k-1})$ and $X_k = n - (X_1 + \cdots + X_{k-1})$, $p_k = 1 - (p_1 + \cdots + p_{k-1})$, then

$$Q_{k-1} = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \xrightarrow{\mathscr{D}} \chi^2(k-1)$$

(ii) **Example.5.7.1**

Want to test a die is fair by tossing 60 times.
$H_0 : p_{10} = \cdots = p_{60} = \frac{1}{6}$, $p_{i0}$ =prob. of obtaining face $i$
data: $13, 19, 11, 8, 5, 4 \therefore np_i = 60 \cdot \frac{1}{6} = 10$

$$Q_5 = \frac{(13-10)^2}{10} + \cdots + \frac{(4-10)^2}{10} = 15.6 > \chi^2_{.05}(5) = 11.1$$

(iii) computation of degree of freedom

$H_0 : p_1 = p_{10}, \cdots, p_k = p_{k0}$
Where, $p_{i0}$ is not completely specified, for example

$$p_i = \int_{A_i} \frac{1}{\sqrt{2\pi}\sigma} exp[-(y-\mu)^2/2\sigma^2]dy,\ i = 1, \cdots, k$$

In this case, $Q_{k-1} = \sum_{i=1}^{k} \frac{(X_i - np_i)^2}{np_i} \xrightarrow{\mathscr{D}} \chi^2(k-1-2)$
i.e. 2 d.f. are lost to estimate $\mu$ and $\sigma^2$

② homogeneity test

Consider two indep. multinomial dist.
$\boldsymbol{X}_1 = (X_{11}, \cdots, X_{k1}) \sim \mathscr{M}(n_1, p_{11}, \cdots, p_{k1})$,
$\boldsymbol{X}_2 = (X_{12}, \cdots, X_{k2}) \sim \mathscr{M}(n_2, p_{12}, \cdots, p_{k2})$
$H_0 : p_{11} = p_{12}, \cdots, p_{k1} = p_{k2}$
test stat: $Q = \sum_{j=1}^{2} \sum_{i=1}^{k} \frac{(X_{ij} - n_j\hat{p}_{ij})^2}{n_j\hat{p}_{ij}}$, $\hat{p}_{ij} = \frac{X_{i1} + X_{i2}}{n_1 + n_2}$, $^\forall j = 1, 2$
d.f.: $(k-1) + (k-1) - (k-1) = k - 1$
In general, for the $r \times c$ contingency table, d.f. is $(r-1) \times (c-1)$

③ independence test

Consider two categorical variales $A$ and $B$. $A$ has $a$ categories $A_1, \cdots, A_a$ and $B$ has $b$ categories $B_1, \cdots, B_b$.
Let $P_{ij} = P(A_i \cap B_j)$, $i = 1, \cdots, a$, $j = 1, \cdots, b$.
$H_0$: two variables are indep.
test stat.: $Q = \sum_{j=1}^{b} \sum_{i=1}^{a} \frac{(X_{ij} - n\hat{p}_{ij})^2}{n\hat{p}_{ij}}$, $n = \sum_j \sum_i X_{ij}$
$\hat{p}_{ij} = \hat{p}_{i.}\hat{p}_{.j} = \frac{X_{i.}}{n}\frac{X_{.j}}{n}$, $X_{i.} = \sum_j X_{ij}$, $X_{.j} = \sum_i X_{ij}$
d.f.: $(ab-1) - \{(a-1) + (b-1)\} = ab - a - b + 1 = (a-1)(b-1)$

## 5.8 The Method of Monte Carlo

① random number generation

(i) **Thm.5.8.1**
$U \sim u(0,1)$, $F$: conti. d.f. $\Rightarrow$ $X = F^{-1}(U) \sim F$
(pf) $P(X \le x) = P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$

(ii) Ex.(generation of $\mathscr{E}(1)$, i.e. $F(x) = 1 - e^{-x}$)

$\therefore F^{-1}(u) = -\log(1-u)$, $0 < u < 1$, then $X = -\log(1-U) \sim \mathscr{E}(1)$

(iii) Ex.(estimation of $\pi$)

$U_1$, $U_2$: i.i.d. $u(0,1)$

$$X = \begin{cases} 1, & U_1^2 + U_2^2 < 1 \\ 0, & O.W. \end{cases}$$

$$\therefore E(X) = \pi/4 \Rightarrow \pi = 4E(X)$$

$$\therefore \text{ Monte Carlo estimation of } \pi \text{ is } \hat{\pi} = 4 \cdot \frac{1}{n} \sum_{i=1}^{n} X_i$$

② Monte Carlo integration

(i) Monte Carlo integration

$$\int_a^b g(x)dx = (b-a)\int_a^b g(x)\frac{1}{b-a}dx = (b-a)E[g(X)],\ X \sim u(a,b)$$

Therefore, the Monte Carlo integration of $g(x)$ is

$$\int_a^b g(x)dx \simeq (b-a)\frac{1}{n}\sum_{i=1}^{n} g(X_i),\ X_i \sim u(a,b)$$

(ii) estimation of $\pi$ by Monte Carlo integration

Let $g(x) = 4\sqrt{1-x^2}$, $0 < x < 1$,
then $\pi = \int_0^1 g(x)dx = E[g(X)],\ X \sim u(0,1)$

$$\therefore \hat{\pi} \simeq \frac{1}{n}\sum_{i=1}^{n} g(X_i)$$

③ Box-Muller transformation

$Y_1,\ Y_2$: i.i.d. $u(0,1)$, $X_1 = (-2\log y_1)^{1/2}\cos(2\pi y_2)$, $X_2 = (-2\log y_1)^{1/2}\sin(2\pi y_2)$
$\therefore y_1 = exp[-(x_1^2+x_2^2)/2]$, $y_2 = \frac{1}{2}\arctan(\frac{x_2}{x_1})$

$$J = \begin{vmatrix} (-x_1)exp[-(x_1^2+x_2^2)/2] & (-x_2)exp[-(x_1^2+x_2^2)/2] \\ \frac{-x_2/x_1^2}{(2\pi)(1+x_2^2/x_1^2)} & \frac{1/x_1}{(2\pi)(1+x_2^2/x_1^2)} \end{vmatrix} = \frac{-1}{2\pi}exp\left[-\frac{x_1^2+x_2^2}{2}\right]$$

$\therefore X_1,\ X_2$: indep $N(0,1)$

④ accept-reject generation algorithm

(i) algorithm: $Y$: r.v. with pdf $g(y)$. $U \sim u(0,1)$, $Y$, $U$: indep.
$f(x)$: pdf s.t. $f(x)/g(x) \leq M$. Then, the following alforithm generate r.v. $X$ with pdf $f(x)$.

(1) generate $Y$ and $U$
(2) If $U \le f(y)/Mg(y)$, then take $X = Y$. Else return to (1)
(pf)

$$\begin{aligned} P(X \le x) &= P[Y \le x | U \le f(y)/Mg(y)] \\ &= \frac{P[Y \le x, U \le f(y)/Mg(y)]}{P(U \le f(y)/Mg(y)]} \\ &= \frac{\int_{-\infty}^{x} \{\int_0^{f(y)/Mg(y)} du\} g(y) dy}{\int_{-\infty}^{\infty} \{\int_0^{f(y)/Mg(y)} du\} g(y) dy} \\ &= \int_{-\infty}^{x} f(y) dy \end{aligned}$$

(ii) example: $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, $g(x) = \pi^{-1}(1+x^2)^{-1}$

easy to generate since its inverse cdf is known.
$\frac{f(x)}{g(x)} = \sqrt{\frac{\pi}{2}} e^{-x^2/2}(1+x^2)$ is maximized at $x = \pm 1$

$$\therefore M = 1.52$$

## 5.9 Bootstrap Procedures

① definition

$X_1, \cdots, X_n$: r.s. from a dist. with pdf $f(x:\theta)$, $\theta \in \Omega$.
$\hat{\theta} = \hat{\theta}(\boldsymbol{X})$, $\boldsymbol{X} = (X_1, \cdots, X_n)$
$\boldsymbol{X}^* = (X_1^*, \cdots, X_n^*)$ is called bootstrap sample if $X_j^*$, $j = 1, \cdots, n$ is drawn with replacement from $(X_1, \cdots, X_n)$, i.e. $X_i$ is selected with prob. $1/n$.

② percentile bootstrap confidence interval

$\hat{\theta}_j^* = \hat{\theta}(\boldsymbol{X}_j^*)$, $\boldsymbol{X}_j^*$: $j$-th bootstrap sample, $j = 1, \cdots, B$, where $B$ is bootstrap size which is usually larger than 3000.
$\hat{\theta}_{(1)}^* \le \hat{\theta}_{(2)}^* \le \cdots \le \hat{\theta}_{(B)}^*$: order stat. for $\hat{\theta}_1^*, \cdots, \hat{\theta}_B^*$.
Then $100(1-\alpha)\%$ percentile bootstrap C.I. for $\theta$ is $(\hat{\theta}_{(m)}^*, \hat{\theta}_{(B+1-m)}^*)$, where $m = \left[\frac{\alpha}{2} B\right]$.

③ bootstrap testing

$X_1, \cdots, X_{n_1}$: r.s. from a dist. with cdf $F(x)$
$Y_1, \cdots, Y_{n_2}$: r.s. from a dist. with cdf $F(x - \Delta)$
$H_0 : \Delta = 0$ vs $H_1 : \Delta > 0$
$\boldsymbol{X}^* = (x_1^*, \cdots, x_{n_1}^*)$, $\boldsymbol{Y}^* = (y_1^*, \cdots, y_{n_2}^*)$: bootstrap sample.
Want to obtain bootstrap p-value.
Let $\bar{x} = \frac{1}{n_1}\sum x_i$, $\bar{y} = \frac{1}{n_2}\sum y_i$, and $\bar{x}_j^*$, $\bar{y}_j^*$ be sample mean of the $j$-th bootstrap samples, then the p-value of $H_0 : \Delta = 0$ is

$$\frac{1}{B}\sum_{j=1}^{B} I(\bar{y}_j^* - \bar{x}_j^* \geq \bar{y} - \bar{x})$$